

ORACLE METHODSM

PROJECT MANAGEMENT

PROCESS AND TASK REFERENCE

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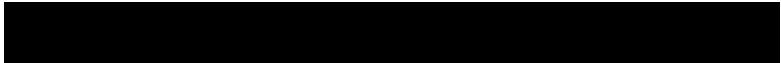
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Preface

The *PJM Process and Task Reference* provides detailed descriptions of the tasks and deliverables of the Project Management Method (PJM). PJM defines a set of organized and flexible process steps that guide Oracle consultants through the project management processes.

The material in this reference includes a description of PJM tasks and deliverables. In addition, we provide guidelines on the prerequisites, approach and techniques, roles and responsibilities, deliverable components, audience, distribution, and usage, completion criteria, and deliverable templates. This reference is provided to assist in expediting and standardizing all of the tasks executed and deliverables produced in a project using PJM.

This reference, and the Project Management Method itself, are part of Oracle MethodSM--Oracle's integrated approach to solution delivery.

Audience

This *Project Management Process and Task Reference* is written for the project management team, consisting of the project manager, team leaders, and project support staff members. Project managers and team leaders use this reference to better understand the nature of project management tasks for which they are responsible. Project support staff use this reference as a detailed guide to executing tasks in PJM.

How The Manual is Organized

This reference consists of an introduction chapter followed by a chapter on each of the PJM processes.

Introduction: The introduction presents a brief overview of tasks, deliverables, and processes, followed by an overview of the organization of PJM. It also provides conventions used within the reference.

Process Chapters: Each process chapter consists of two main sections, an overview and task definitions. The overview section provides the following information for each process:

- **Process Flow** - depicts the flow of tasks for this process
- **Approach** - describes the approach for this process
- **Tasks and Deliverables** - lists the tasks executed and the deliverables produced. This table also provides the responsible role and the type of task. Task type definitions can be found in the Glossary at the end of the PJM Method Handbook.
- **Objectives** - describes the objectives of the process
- **Key Deliverables** - lists and defines the key deliverables for this process
- **Key Responsibilities** - lists and defines the key roles for the process
- **Critical Success Factors** - lists the success factors for the process

- **References and Publications** - lists any references and publications for the process

The task detail section provides the following:

- **Deliverable** - identifies and describes the deliverable produced by the task
- **Prerequisites** - lists the prerequisites for the task and the source
- **Task Steps** - lists the steps of the task
- **Approach and Techniques** - describes the approach and suggested techniques for the task
- **Roles and Responsibilities** - provides the responsible roles for the task
- **Deliverable Components or Outputs** - lists the sections of the deliverable document or the outputs produced by an ongoing task
- **Component Descriptions or Output Descriptions** - describes each section of the deliverable document or each output
- **Audience, Distribution, and Usage** - provides guidance on who should receive the deliverable document and how it should be used
- **Quality Criteria** - provides a checklist of questions for use by team members in reviewing the deliverable document

Appendix A: This appendix indexes PJM tasks by deliverable name and by task name.

Appendix B: Appendix B lists the roles referenced in PJM and descriptions of each role.

How to Use This Manual

The *Project Management Method Process and Task Reference* provides the detailed task and deliverable information that makes up the Project Management Method. Each task produces a deliverable and these deliverables are described in this reference. Each task has an identifier

(task ID) which makes it easy to locate. The task ID corresponds directly to the deliverable ID of the task that creates the deliverable.

All users of this reference should read the introductory chapter to better understand tasks, deliverables, and processes.

Oracle Services recommends that users of all of the PJM handbooks, and the Project Management Method itself, take advantage of PJM training courses provided by Oracle's Education Services. In addition to the PJM handbooks and training, Oracle Services also provides experienced PJM consultants, automated work management tools customized for PJM, and PJM deliverable templates.

Conventions Used in this Manual

We use several notational conventions to make this reference easy to read.

Attention

We sometimes highlight especially important information or considerations to save you time or simplify the task at hand. We mark such information with an attention graphic, as follows:



Attention: It is essential that you agree on the acceptance criteria with the client at the beginning of the project.

For More Information

Throughout the reference we alert you to additional information you may want to review. This may be a task section, appendix, manual reference, or web site. We highlight these references with an easy-to-notice graphic. Here is an example:



Reference: For more information about managing the Project Library, refer to the Document Control ongoing task.



Web site: You can find further information on Oracle's Home Web Page <http://www.oracle.com/>

Suggestions

We provide you with helpful suggestions throughout the reference to help you get the most out of the method. We highlight these suggestions with an illuminated light bulb. Here is an example of a suggestion:



Suggestion: Where possible, use available tools and easy to collect measurements.

Related Publications

Books available in the PJM suite include:

- *PJM Method Handbook*
- *PJM Process and Task Reference* (this book)

PJM tasks and supporting materials are tightly integrated with the deployment methods of Oracle Method. For more information on these, refer:

- *Custom Development Method Handbook*
- *Application Implementation Method Handbook*
- *Enterprise Architecture Method Handbook*

Your Comments are Welcome

Oracle Corporation values and appreciates your comments as an Oracle PJM user and reader of the reference. As we write, revise, and evaluate our documentation, your comments are the most valuable input we receive. If you would like to contact us regarding comments on this or other Oracle Method manuals, please use the following address:

email: pjminfo@us.oracle.com

Contents

INTRODUCTION	Introduction	xi
	What is a Task?	xii
	What is a Deliverable?.....	xv
	What is a Process?	xx
	What is a Project?	xxv
	PJM Relationship to Industry Standards	xxviii
CHAPTER 1	Control and Reporting (CR).....	1-1
	CR.010 - Establish Scope, Objectives, and Approach.....	1-8
	CR.020 - Define Control and Reporting Strategies, Standards, and Procedures	1-20
	CR.030 - Establish Management Plans.....	1-26
	CR.040 - Issue/Risk Management.....	1-41
	CR.050 - Problem Management.....	1-49
	CR.060 - Change Control	1-57
	CR.070 - Status Monitoring and Reporting	1-65

CR.080 - Secure Client Acceptance	1-73
---	------

CHAPTER 2 Work Management (WM)..... 2-1

WM.010 - Define Work Management Strategies, Standards, and Procedures.....	2-8
---	-----

WM.020 - Establish Workplan.....	2-14
----------------------------------	------

WM.030 - Establish Finance Plan	2-22
---------------------------------------	------

WM.040 - Workplan Control.....	2-36
--------------------------------	------

WM.050 - Financial Control.....	2-43
---------------------------------	------

CHAPTER 3 Resource Management (RM) 3-1

RM.010 - Define Resource Management Strategies, Standards, and Procedures.....	3-6
---	-----

RM.020 - Establish Staffing and Organization Plan.....	3-11
--	------

RM.025 - Create Project Orientation Guide.....	3-17
--	------

RM.030 - Implement Organization	3-25
---------------------------------------	------

RM.040 - Establish Physical Resource Plan	3-29
---	------

RM.050 - Establish Infrastructure	3-35
---	------

RM.060 - Staff Control	3-42
------------------------------	------

RM.070 - Physical Resource Control.....	3-46
---	------

RM.080 - Release Staff.....	3-50
-----------------------------	------

RM.090 - Release Physical Resources	3-54
---	------

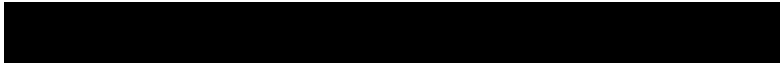
CHAPTER 4	Quality Management (QM).....	4-1
	QM.010 - Define Quality Management Strategies, Standards, and Procedures.....	4-8
	QM.020 - Quality Review.....	4-14
	QM.030 - Quality Audit	4-24
	QM.040 - Quality Measurement.....	4-34
	QM.045 - Support Healthcheck	4-39
	QM.050 - Perform Quality Assessment	4-46
CHAPTER 5	Configuration Management (CM).....	5-1
	CM.010 - Define Configuration Management Strategies, Standards, and Procedures.....	5-8
	CM.020 - Document Control	5-15
	CM.030 - Configuration Control	5-25
	CM.035 - Knowledge Management.....	5-32
	CM.040 - Release Management	5-40
	CM.050 - Configuration Status Accounting	5-46
	CM.060 - Audit Key Deliverables	5-53
	CM.070 - Conclude Configuration Management.....	5-57
APPENDIX A	Tasks & Deliverables.....	A-1
	Index by Deliverable Name.....	A-2
	Index by Task Name	A-4

APPENDIX B

PJM Roles

Role Descriptions

B-1
B-2



Introduction

This section begins with a general description of processes, tasks, and deliverables. The second part of this section gives an overview of projects, the organization of PJM, and PJM's relationship to industry standards.

Processes, tasks, and deliverables are the basis of PJM. They are the building blocks from which project managers construct PJM projects. The *PJM Process and Task Reference* gives the details of every process that plays a part in PJM, and every task included in each process. As an introduction, this section gives a brief overview of the concepts of *process*, *task*, and *deliverable*, and describes the information in this reference that is given for each.

Project management has been defined as the application of knowledge, skills, tools, and techniques in order to meet or exceed stakeholder expectations. It is a relatively young discipline, and demands a unique blend of management, technical, and organizational talents. The use of PJM improves project quality by institutionalizing a consistent project management approach and culture.

What is a Task?

A **task** is a unit of work that results in the output of a single deliverable. Tasks are the most elementary unit of work that one would put into a project plan -- they provide the basis of the work breakdown structure. A task produces a single, measurable deliverable and is usually assigned to be the responsibility of a single team member (although many others may play contribution, review, and approval roles). Project progress is usually measured by the successful completion of tasks.

Task IDs and Names

Each task in PJM has a unique ID. That ID is composed of the alphabetical ID of the process in which the task plays a part (more on this later), followed by a three digit sequence number, that usually increments by 10. The sequence number generally reflects the order in which tasks are to be completed. For example, you can be reasonably sure that task WM.020 will be completed well in advance of WM.050 within the Work Management (WM) process.

Note that the task ID does not reflect the strict order or phase in which the task occurs within a project. A project manager may combine tasks and processes in different ways to meet the needs of different development approaches. Therefore tasks may have different sequences, relative to each other, in different types of PJM projects.

Each task also has a unique name. A task name always consists of a verb (such as *create...*, *determine...*, *design...*), followed by an object. In most cases the object is the name of the deliverable that the task produces. You will find that the text always refers to task names with title case letters, for example, Establish Scope, Objectives, and Approach.

Finding Information About Tasks

Each task in PJM has a task guideline. If you know a task's ID, it is easy to find the guideline for that task in the *PJM Process and Task Reference*.

Locate the process chapter by the first part of the ID, and then locate the task within the chapter by using the numerical sequence number part of the task ID.

If you only know the name of a task, you can use Appendix A to find the ID. Appendix A contains an alphabetical listing of tasks by task name. It also contains an alphabetical listing of tasks by deliverable name.

Prerequisites

Each task assumes that certain things (such as information, programs, hardware platforms, etc.) have been previously produced or compiled, and are available for use. In most cases, these **prerequisites** of the task are specific deliverables of previous tasks. In some cases, they are expected from the client business organization.

Each task guideline lists that task's prerequisites. Under each prerequisite you will find an indication of which components or specific information within the prerequisite is used in the task. The text also indicates how you will use that component or information when carrying out the task.

Task Steps

Many tasks may be broken down into smaller units of work called **task steps**. In some cases, a task guideline may indicate a suggested set of task steps to follow. Many times, the team member responsible for the task (the “owner” of the task) will want to specify the task steps. The task owner will want to base those steps upon techniques that are appropriate to the overall development approach and the tools and resources that are available to the project. Any set of task steps that reaches the deliverable is acceptable as long as it includes adequate quality assurance steps.

Roles and Responsibilities

In addition to the task owner, many other project team members may spend time on a task. Each of these team members will be fulfilling a particular **role**. Their **responsibilities** may be to contribute information, analyze, document, set up, review, administer, or control. The effort they spend on the task may be significant or nominal.

Each task guideline gives a suggested list of roles that may play a part in completing the task. Next to each role is a description of the responsibilities of that role during the task, and an indication of the percentage of the total task effort that that role may contribute. These are suggestions that can be used in planning -- actual role contributions will of course depend upon the task steps that the task owner assigns.

What is a Deliverable?

PJM is a deliverable-based method. This means that each task produces a single **deliverable**, or output, whose quality you can measure. Deliverables can have many formats, such as documents, schedules, program code, or test results.

Each deliverable in PJM is **recognizable** (it has a specific name and ID) and **measurable**. Each deliverable has the same ID as its corresponding task. Each deliverable also has a unique name, which the text always refers to using title case letters. An example is the Project Management Plan. If you know the name of a deliverable, you can find its ID, and the name of its corresponding task, by using Appendix A.

A PJM deliverable can be further broken down into smaller units called **deliverable components**. For example, the deliverable Project Management Plan contains the following deliverable components:

- Introduction
- Scope
- Objectives
- Approach
- Control and Reporting
- Work Management
- Resource Management
- Quality Management
- Configuration Management

PJM employs **ongoing tasks** which execute continuously during a period of time. Ongoing tasks produce repetitive results rather than a single deliverable. In PJM, these results are identified as **outputs** and are described in the same way as deliverable components. For example, the Issue/Risk Management ongoing task produces the following outputs:

- Resolved Issues
- Risk Containment Measures

- Change Requests

A **deliverable definition** is a specification of the content, organization, and usage of the product from a PJM task. This reference attempts to specify a standard set of deliverable definitions for PJM.

Dependencies between PJM tasks are based on PJM deliverables. Each task requires specific deliverables from prior tasks before it can commence. A single task may impact many subsequent tasks in the current and future phases, based on the deliverable it produces.

The *PJM Process and Task Reference* lists the prerequisite deliverables for each task in the task guidelines, and also indicates this information on the process flow diagram at the start of each chapter. The *PJM Method Handbook* does the same in the diagram at the start of each phase chapter. This makes it easy for a project team member or manager to verify which deliverables must be collected as input to a particular task, before that task can be started.

Project Deliverables

You identify your deliverables using the project workplan. Each task in the project workplan should produce a single unique deliverable. As you tailor the tasks in your workplan, you will need to tailor your deliverables as well.

When you begin preparing the project workplan using standard PJM routes (work breakdown structures), each PJM task will initially refer to the name of its corresponding deliverable. As you create or revise the tasks in your workplan, make sure that your deliverable names are unique and meaningful. For example, if you create a separate instance of a PJM task for multiple project teams, you would append a qualifier, such as the team name, to the deliverable name for each new task as well.

Project tasks and dependencies can also be tailored based on the prior availability of project deliverables. If a deliverable is already available prior to a project or phase, the task that normally produces it can be reduced to a “review” task. In some cases it can be eliminated.

Reviewing Deliverables

The production of deliverables is a way to measure the progress of a project. You verify successful completion of a deliverable by performing a quality review. The quality review should be based on the quality criteria specified for the PJM deliverable definition in this reference. You can also establish alternate or additional criteria, such as those required by the business management. Whatever the case, make sure that the completion criteria for each deliverable are clearly understood by the entire project staff.

Key Deliverables

The characteristics of your project will determine the relative importance of each deliverable to the project. For instance, in a project that involves the use of the latest, unfamiliar technology, the Project Management Plan deliverable may be critical. On the other hand, in a project where the hardware and software platforms are well known and are specified by the IS organization, this deliverable may be minor.

A deliverable that is particularly important is called a **key deliverable**. Key deliverables are usually reviewed and signed off with the client or business, and then controlled following sign-off. Other deliverables may be produced only as project internal deliverables, combined with other deliverables, or not produced at all.

Each PJM process and phase gives a standard list of key deliverables, but this is only a suggestion based on experience. You should tailor this list for your project when you revise the project workplan. It is a good practice to designate key deliverables for all of your key project milestones, whether they are signed off by your client or not.

Documenting Deliverables

Project deliverables can take many formats. Paper and electronic formats are the most common, but other formats include computer hardware and software (for example, System Test Environment), and even human beings (for example, Trained Users). In many cases, we

will want to produce not only the project deliverable itself, but also a record or representation of that deliverable that may be easier to review, record and sign-off. For example, in addition to producing the actual Trained Users deliverable, we will also want to document the training events that were actually attended by each student, including an indication of which users were not trained as expected.

You should keep in mind that in many cases, the document only **represents** the project deliverable, or only documents the parts or aspects of the deliverable that are most relevant to communicate. Much more information can often be required to actually meet the quality criteria of the deliverable. In some cases you may not need to produce a document at all. The production of the document alone should not be the goal.

Controlling Deliverables

You should determine the level of control of each project deliverable as either **controlled** or **uncontrolled**. You control a deliverable and its corresponding documentation in order to protect its integrity and manage its approval and revision.

As a rule, every key deliverable of the project should be controlled. You control the content of the deliverable using configuration control procedures, to restrict access and changes to the deliverable to only those authorized. You also track each version of the deliverable over time and reconstruct any previous version as needed.

You control documentation for a deliverable using document control procedures, to define how documents are prepared, approved and distributed. A controlled document is assigned a version number and its date and distribution list is clearly indicated. You may also want to number each copy of the document with a copy number. As authorized changes are made to the contents of the document, new versions are periodically created and sent out to the original distribution list, at a minimum. You also include a log of changes within each version. If you had numbered copies, you may also want to request that superseded copies be returned.

A deliverable may be uncontrolled because it will not be updated or will be shortly replaced by a controlled deliverable. Changes to an

uncontrolled deliverable would not go through the project's configuration control process. However, you should still review the deliverable for quality and retain a copy of the deliverable in the project's repository.

In the case of uncontrolled documents such as meeting minutes or memos, document control requirements can be reduced. You would not need to associate a version number with an uncontrolled document. You may not even need to formally distribute it, although a copy should always be retained in the project library.

What is a Process?

Major project objectives are achieved by processes. A **process** is a series of **tasks** that results in one or more critical project deliverables. The tasks in a process are usually highly dependent upon one another, and often span much of the project.

A process in PJM is a set of related management tasks. PJM processes have been organized to highlight key aspects of the project which must be continuously managed over its life-cycle. They also organize project management skills, tools, and techniques into logical groups which facilitate process improvement, training, and communication within a consulting organization.

PJM processes collectively form a complete set of all tasks required to perform an information technology project, when used with a product or service delivery method in Oracle Method. The following table summarizes the five PJM processes and the aspects of the project managed by each.

PJM Process	Management Focus
Control and Reporting	Stakeholders, scope, and approach
Work Management	Schedule and cost
Resource Management	Organization and infrastructure
Quality Management	Client satisfaction
Configuration Management	Products

PJM Process Organization

Each PJM process organizes its tasks into three groups, as shown in Figure I-1. These support the Project Execution tasks which are taken from other processes in Oracle Method involved with producing the deliverable products and services for the project. The task groups are

linked according to the results they produce to form a project delivery model.

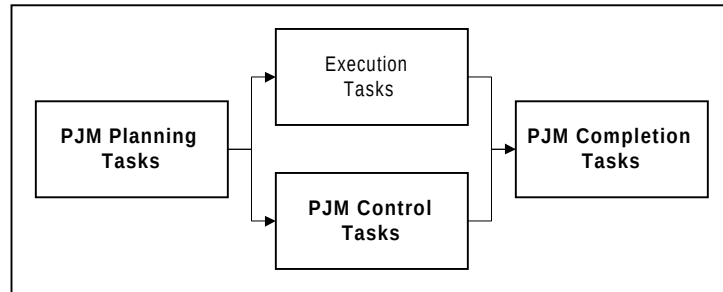


Figure I-1 Project Delivery Model

Planning tasks provides the definition of the project with respect to scope, quality, time, and cost. They also determine the appropriate level and organization of resources to execute the project.

Control tasks are performed concurrently with execution tasks. Control ensures that objectives are being met by measuring performance and taking corrective action, as needed. Its tasks are performed continuously, or discretely, as often as needed. Control tasks coordinate with each other by exchanging information and synchronizing their actions.

Completion tasks formalize acceptance of the project and bringing it to an orderly end. Completion results in the satisfactory conclusion of the project and settlement of all outstanding issues prior to handover of the project deliverables to the client.

Although the project delivery model in Figure I-1 is drawn with well-defined boundaries, in an actual project there will be overlaps. Planning in some form is performed at virtually all times during a project. It is first used in order for the client and consultant to reach an agreement to initiate the project. It is executed at the beginning of the project and also prior to beginning each phase of a project, to provide details of the tasks necessary to bring that phase to a successful completion. Planning must also provide a preliminary description of work to be done in later phases, and it must assess and define the impact of changes to plans due to a change in project scope. Completion tasks should occur at each major project milestone, including the conclusion of each project phase.

Process Guidelines

Each chapter of the *PJM Process and Task Reference* is devoted to a single process. The first part of each chapter gives guidelines on the process as a whole. It shows the relationships of tasks within the process, lists the critical deliverables of the process, and gives guidance on the skills needed to execute the process.

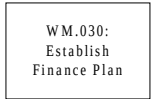
The process guidelines do not indicate exactly where each task falls in the project task plan, since this may vary by the development approach chosen. For more information on choosing and structuring a development approach using PJM processes, see the *PJM Method Handbook*.

Process Flow Diagrams

Each process is represented by a process flow diagram. This diagram shows the tasks that are part of the process, and the dependencies between those tasks. It also shows the key deliverables of the process, and the roles that are responsible for each task. The “Approach” section that follows explains the reason behind the organization of the tasks in the process flow diagram.

One thing to keep in mind is that a process flow diagram may indicate that two tasks are strictly dependent upon one another (task “B” may not begin until task “A” has completed) when, in fact, the two tasks will most likely overlap somewhat in real life. An example is in the Control and Reporting process. The task Issue/Risk Management is a predecessor task to Change Control. Ideally, we would like to analyze all problems before making any changes to the application so that we can make changes to modules efficiently. But this is rarely the case in the course of a project, where schedule demands usually require us to start making application changes as soon as possible.

The following describes the symbols that are used in the process flow diagrams.



Task. This shows a task that is contained in PJM. The text gives the PJM ID and the task name.



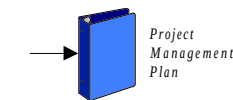
External Task. This symbol depicts a task that should be performed, but is not contained in PJM.



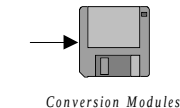
Decision. A decision indicates that there are two or more possible branches to the process flow, depending upon the outcome of the stated question. A decision symbol does not indicate another task -- all work that is done in order to indicate a particular outcome is included in previous tasks.



Process Flow. An arrow between two tasks signifies that the task at the end of the arrow generally should not start until the previous task has been completed. Example: you should not start the task *Define Training Plan* until you have finished the task, *Define Training Requirements*. In some cases, it may be desirable to overlap such tasks in an actual project plan.



Key Document Deliverable. This represents an important textual output of the process. It includes the name of the deliverable.



Key Software Deliverable. An important software product of the process.

Other symbols are used for key deliverables, as appropriate.



Major Prerequisite. A key input deliverable for a task. The name of the deliverable and the ID of the task that produces the deliverable are given. Different symbols may be used to represent the medium of the deliverable.



Attention: Major prerequisites and key deliverables do not correspond to the agent channel in which they are drawn.

What is a Project?

The work performed in organizations can generally be viewed as being a result of either *operations* or *projects* [1]. Operations and projects share common characteristics. Both involve work performed by people and are constrained by limited resources. They can also be described using process modeling techniques. Lastly, they both have objectives and are planned, executed, and controlled.

Work performed in operations tends to be ongoing and repetitive. Projects, on the other hand, involve work that is *finite* and *unique*. Every project has a finite ending point, when either the project's objectives have been achieved, or the project is canceled. Although operations may be discontinued for various reasons, they do not cease to exist when their objectives have been attained. Instead, they adopt a new set of objectives and continue to function.

Projects are unique because they involve doing something which has not been done before. The product or service may belong to a large class, such as customer information systems, but each individual system is different from all others in that class. Because the product of each project is unique, the characteristics that distinguish it must be progressively elaborated. These characteristics will be broadly defined early in the project and made more specific as the project team develops a more complete understanding of the product.

Types of Projects

PJM is designed to support any type of work that fits the preceding characteristics of a project in the information technology industry. The scale of such projects can range from a small team effort over a period of weeks to hundreds of person years to complete. Accordingly, PJM can be applied to project work known by various names such as focus teams, work packages, sub-projects, pilots, ventures, and programs.

A program is a group of projects managed in a coordinated way to obtain benefits not available from managing them individually [2]. Programs can also include elements of ongoing operations, such as support for products which have been developed by completed

projects. Although the management of a program is usually more complex than that of a single project, the essential project characteristics of finiteness and uniqueness still apply, and the management processes defined in PJM are therefore relevant to programs as well.

Organizations will usually divide a project into several phases to provide better management visibility and control. These phases should be viewed as projects in their own right, since they also satisfy the criteria of being finite and unique. The *PJM Method Handbook* provides guidance on tailoring PJM processes for project phase management.

Project Management Team

Information technology projects are typically performed within an organization larger than the project itself. The people who have influence over the products and conduct of the project may be drawn from within the organization, outsourced to another organization, or a combination of both. A contract may or may not be involved. In PJM, the *consultant* and the *client* represent the two parties which together form the project management team responsible for the project's success.

The client represents the customer, or primary beneficiary of the project's products, as well as the acquirer, or funding source for the project. The client is also assumed to be capable of providing both physical and human resources to the project.

The consultant represents an information services organization with its own management structure and systems, which performs projects as a core competency. The consultant may be acting either as a profit center, performing the project for a profit, or as a cost center, sharing project costs with the client. The consultant is also assumed to have an organization made up of service practices, or business units, which supply consulting staff resources and sub-contractors to the project.

You should note that the tasks performed in reaching and maintaining a contractual agreement between the client and consultant are not covered in PJM. PJM assumes that a contract may be established prior to the start of the project, and identifies where contractual impacts can occur during the project. However, a contract is not a prerequisite for the use of PJM. PJM also assumes that both the client and the consultant may have internal management policies governing project

conduct. You should tailor these aspects of the client- consultant relationship to your project's specific situation whenever you use PJM.

Project Management Roles

The key management roles performed by the client in PJM are the Project Sponsor and Client Project Manager. The Project Sponsor is the client role that holds the budget for the project, and may be a single individual or a committee. The Project Sponsor ensures organizational commitment to the project and validates project objectives. The Client Project Manager is expected to be assigned to the project where client commitments or business interests require a daily client management presence. This role is responsible for providing client resources, resolving problems, and monitoring the consultant's progress.

The key management roles performed by the consultant in PJM are the Consulting Business Manager and the Project Manager. The Consulting Business Manager role represents the consulting manager whose practice is responsible for the successful execution of the project. The Consulting Business Manager will also represent the consultant if a contractual agreement exists with the client. The Project Manager is the consultant role which is held ultimately responsible for the project's success or failure. The Project Manager must manage the various aspects of time, cost, scope, and quality to satisfy client expectations and meet the business objectives of the consulting practice.

PJM Relationship to Industry Standards

PJM addresses the need to provide a well-defined management methodology for information technology projects. The organization and terminology used in PJM are complimentary with other management, quality, and software process standards which represent best practices in their respective disciplines. Using PJM should not preclude your use of these standards on your project. Each model provides a unique perspective and contribution which can elaborate on or supplement PJM. Conversely, where these models impose requirements on the project, PJM will support many of their requirements.

The following sections provide a high-level comparison of PJM with four such standards. Use this information as a guide for further understanding and tailoring of PJM processes and tasks. The models discussed are:

- ISO 9000 Series - Quality Systems
- PMI - Project Management Body of Knowledge
- ISO / IEC Standard 12207 - Software Life-Cycle Processes
- SEI - Capability Maturity Model

ISO 9000 Series - Quality Systems

ISO 9000 is a series of quality systems standards and associated guidance material. The definition of a *quality system* used by ISO 9000 is “the organizational structure, procedures, processes, and resources needed to implement quality management”. The term *quality system* is generally used to refer to a process which ensures and demonstrates the quality of the products and services it produces. The term *quality management system* is sometimes used to emphasize the need for active management of the quality process.

For the information technology industry, the primary ISO 9000 documents of interest are ISO 9001 [3] and ISO 9000-3 [4]. ISO 9001 is a *standards model* which applies to quality systems in organizations which design, develop, produce, install, or service their products. ISO 9000-3

is the *guideline* which interprets the requirements of ISO 9001 specifically for information technology.

An acceptable structure for a quality system which satisfies ISO 9001 certification requirements is:

- Level I - Quality Manual: describes how the organization’s overall quality system works.
- Level II - Standards and Procedures: provide a sufficient level of control over organizational processes. Standards apply to deliverables and procedures to tasks. These are tailored by the organization to support the quality manual.
- Level III - Reference Materials: consist of methods defining processes, tasks, guidelines, checklists, templates, and forms. Methods must address the development of products, as well as the management of development.

PJM is designed to satisfy the requirements of Level III for management and supporting activities, such as quality assurance and configuration management, as interpreted in ISO 9000-3. The following table provides a cross-reference of the ISO 9000-3 quality system framework, life-cycle activities, and supporting activities applicable to PJM processes.

Quality System Framework	PJM Process
Management Responsibility	Control and Reporting, Quality Management
Quality System	Quality Management
Internal Quality System Audits	Quality Management
Corrective Action	Quality Management

Quality System Life-Cycle Activity	PJM Process
Development Planning	Control and Reporting

Quality System Life-Cycle Activity	PJM Process
Design and Implementation	Quality Management
Testing and Validation	Control and Reporting
Acceptance	Control and Reporting
Maintenance	Control and Reporting, Configuration Management

Quality System Supporting Activity	PJM Process
Configuration Management	Configuration Management
Document Control	Configuration Management
Quality Records	Quality Management
Measurements	Quality Management
Tools and Techniques	All
Purchasing	Resource Management
Training	Resource Management

PMI - Project Management Body of Knowledge

The Project Management Institute is a nonprofit professional association which provides project management standards, education programs, and professional certification. It seeks to define and organize a Project Management Body of Knowledge (PMBOK), which describes the sum of knowledge within the profession of project management. The PMBOK Guide [1] organizes generally accepted project management practices into nine knowledge areas, each of which is described in terms of its component processes. In the PMBOK, a process is “a series of actions bringing about a result”.

PMBOK knowledge areas and PJM processes collectively cover very much the same aspects of project management. However, the PMBOK must be general enough to apply to projects across the entire spectrum of industry, government, and academia, while PJM focuses on information technology projects. PJM processes corresponding to PMBOK Guide knowledge areas are indicated in the table on the following page:

PMBOK Guide Knowledge Area	PJM Process
Project Integration Management	Control and Reporting Work Management
Project Scope Management	Control and Reporting
Project Time Management	Work Management
Project Cost Management	Work Management
Project Quality Management	Quality Management
Project Human Resource Management	Resource Management
Project Communications Management	Control and Reporting
Project Risk Management	Control and Reporting
Project Procurement Management	Resource Management

PMI also identifies five basic management process groups - Initiating, Planning, Executing, Controlling, and Closing. The Initiating process is included in PJM Planning. The four remaining groups correspond to the components of the PJM project delivery model.

ISO/IEC Standard 12207- Software Life-Cycle Processes

ISO / IEC 12207 [5] is intended to establish a common framework for viewing the software life-cycle, with well-defined terminology that can be referenced by the software industry. It covers the entire software

life-cycle from conceptualization through supply, development, operation, maintenance, and retirement.

ISO / IEC 12207 defines a process as “a set of interrelated activities, which transform inputs into outputs”. Each activity is further divided into a set of tasks. Tasks are listed, without further detail. ISO / IEC 12207 identifies three groups of software life-cycle processes: Primary, Supporting, and Organizational. Primary processes serve the primary parties of acquirer, supplier, developer, operator, and maintainer. A party utilizes its primary process and any supporting or organizational processes necessary for the specific circumstances.

Project management related activities and tasks are found mainly in the Supporting and Organizational processes. Although the standard does not have a model corresponding to the PJM project delivery model, the Organizational Management process identifies activities for planning, execution and control, review and evaluation, and closure.

The following table lists all of the processes in ISO / IEC 12207, and the PJM processes which most closely apply.

Primary Life-Cycle Process	PJM Process
Acquisition	Resource Management (for subcontracting)
Supply	Control and Reporting, Work Management
Development	Other OM processes
Operation	Other OM processes
Maintenance	Other OM processes

Supporting Life-Cycle Process	PJM Process
Documentation	Configuration Management
Configuration Management	Configuration Management

Supporting Life-Cycle Process	PJM Process
Quality Assurance	Quality Management
Verification	Quality Management
Validation	Quality Management
Joint Review	Quality Management
Audit	Quality Management
Problem Resolution	Control and Reporting

Organizational Life-Cycle Process	PJM Process
Management	All
Infrastructure	Resource Management
Improvement	Quality Management
Training	Resource Management



SEI - Capability Maturity Model

The Software Engineering Institute’s Capability Maturity Model (CMM) [6] provides a framework that describes the key elements of an effective software process. The CMM characterizes five levels of increasing software process maturity, from Level 1 (Initial) to Level 5 (Optimizing). Each maturity level describes the extent to which an organization’s processes comply with specified key practices. The CMM provides a path which can be used by organizations to guide continuous process improvement, rather than a method composed of processes, such as PJM, which is employed on individual projects.

CMM Level 2, “Repeatable”, defines how an organization establishes basic project management processes to track cost, schedule, and

functionality. PJM provides tasks and deliverables which directly support Level 2 activities on individual projects. Level 3, “Defined”, focuses on the ability to document, integrate, and standardize management and engineering processes across an organization. PJM supports Level 3 key practices by providing standard management plans, standards, and procedures which can be adopted organization-wide.

Level 4, “Managed”, focuses on the organization’s use of quantitative measures to improve software quality. The PJM Quality Management process supports this level by defining how quality measurements are collected on projects. At Level 5, “Optimizing”, an organization continuously improves software processes through qualitative analysis and innovation. Within a project, PJM supports the Level 5 Defect Prevention key process area through quality reviews and audits.

The following is a general comparison of the CMM key process areas supported by PJM processes. The table indicates which PJM processes provide contributing tasks and deliverables to each process area.

Level	CMM Key Process Area	PJM Process
2	Requirements Management	Control and Reporting
2	Software Project Planning	Control and Reporting, Work Management
2	Software Project Tracking and Oversight	Control and Reporting, Work Management, Quality Management
2	Software Subcontract Management	Resource Management
2	Software Quality Assurance	Quality Management
2	Software Configuration Management	Configuration Management
3	Training Program	Resource Management

Level	CMM Key Process Area	PJM Process
3	Integrated Software Management	Control and Reporting, Work Management
3	Intergroup Coordination	Work Management
3	Peer Reviews	Quality Management
4	Quantitative Process Management	Quality Management
4	Software Quality Management	Quality Management
5	Defect Prevention	Quality Management

References and Publications



Reference: [1] Project Management Institute. 1996. *A Guide to the Project Management Body of Knowledge*. Sylva, N.C.: PMI Communications.



Reference: [2] Turner, J. Rodney. 1992. *The Handbook of Project-Based Management*. New York, N.Y.: McGraw-Hill.



Reference: [3] International Organization for Standardization. 1994. *International Standard ISO 9001:1994(E): Quality systems - Model for quality assurance in design, development, production, installation, and servicing*. Geneva, Switzerland: ISO.



Reference: [4] International Organization for Standardization. 1991. *International Standard ISO 9000-3:1991(E): Quality management and quality assurance standards - Part 3: Guidelines for the application of ISO 9001 to the development, supply and maintenance of software*. Geneva, Switzerland: ISO.



Reference: [5] International Organization for Standardization and International Electrotechnical Commission. 1995. *International Standard ISO/IEC 12207:1995(E): Information Technology - software life-cycle processes*. Geneva, Switzerland: ISO / IEC.



Reference: [6] Carnegie Mellon University / Software Engineering Institute. 1993. *CMU/SEI-93-TR-25: Key Practices of the Capability Maturity Model, Version 1.1*. Pittsburgh, PA: Research Access.

Control and Reporting (CR)

This chapter describes the Control and Reporting process.

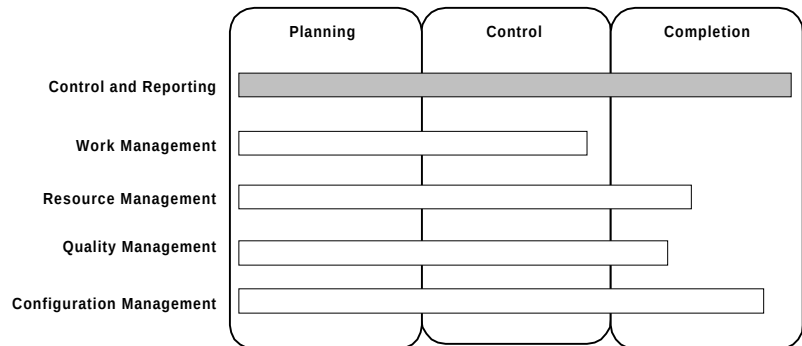


Figure 1-1 Control and Reporting Context

Process Flow

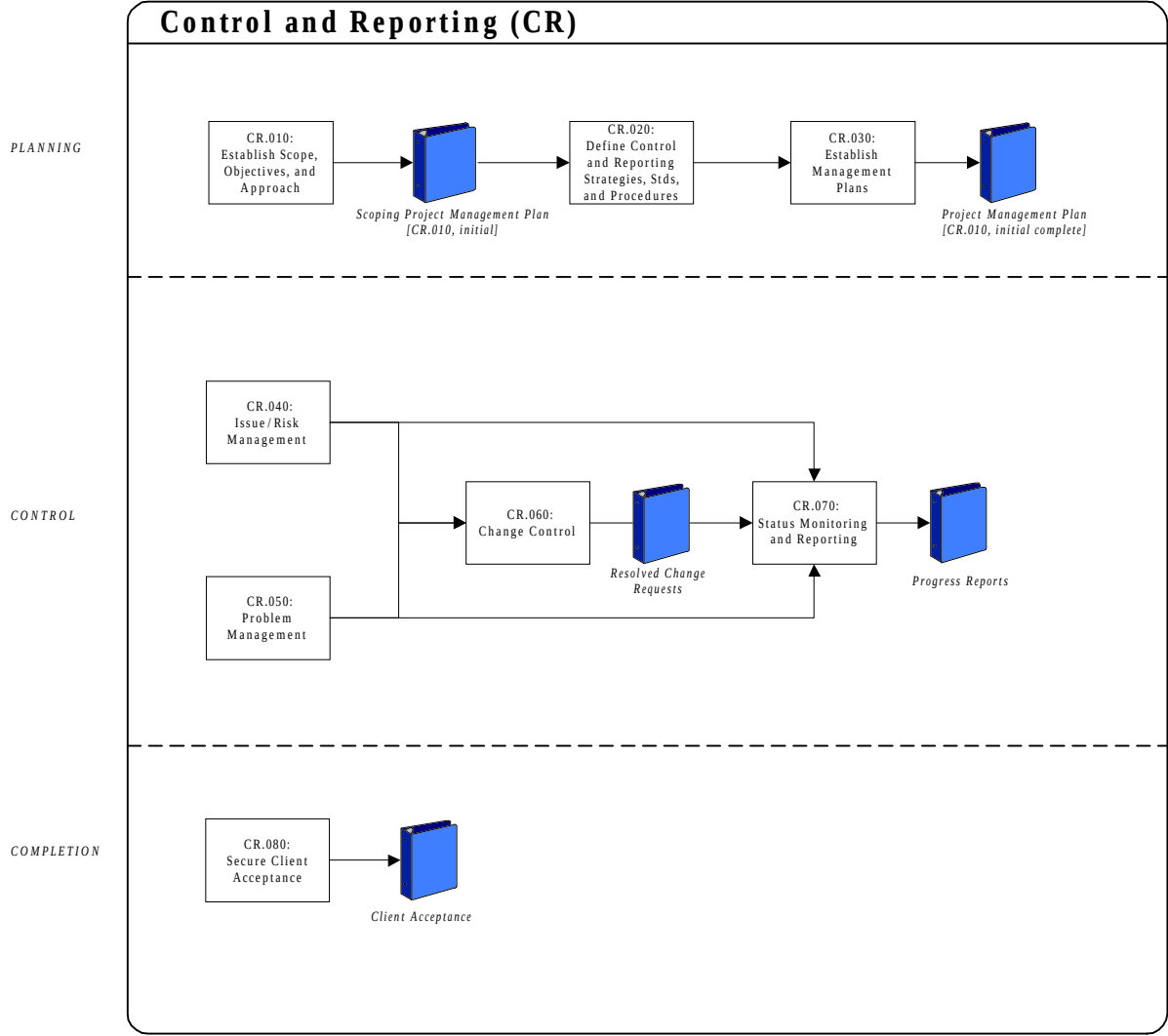


Figure 1-2 Control and Reporting Process Flow Diagram

Approach

The Control and Reporting process manages the overall scope of the project and client relationship through change control, issue and risk management, and progress reporting. A clear conclusion to the project is controlled through client acceptance.

In the planning phase, a clear definition of the scope, objectives, and approach, documented in the Scoping Project Management Plan, sets the scene for the project and identifies project constraints. Standards and procedures ensure consistency among team members and define quality assessment criteria. The complete Project Management Plan defines how all of the management processes work together to accomplish the project's scope, objectives and approach.

In the control phase, issue and risk management ensures that risks are cataloged at the outset of the project, understood in terms of likelihood and impact, and monitored throughout the project. Status monitoring and reporting ensures that progress is visible to team members, client management, and consulting management, by keeping everyone who is responsible for the success of the project informed.

During completion, phase or project deliverables are reviewed, approved and accepted by the client. Effort is focused on obtaining written sign-off by the client for work completed to that point.

Tasks and Deliverables

The tasks and deliverables for this process are shown in this table:

ID	Task Name	Deliverable or Output	Type*
Planning			
CR.010	Establish Scope, Objectives, and Approach	Scoping Project Management Plan [CR.010]	IT
CR.020	Define Control and Reporting Strategies, Standards, and Procedures	Control and Reporting Strategies, Standards, and Procedures	IT
CR.030	Establish Management Plans	Project Management Plan [CR.010, complete]	IT
Control			
CR.040	Issue / Risk Management	Resolved Issues, Risk Containment Measures, Change Requests	O
CR.050	Problem Management	Resolved Problems, Issues / Risks, Change Requests	O
CR.060	Change Control	Scope Changes, Approved Change Requests	O
CR.070	Status Monitoring and Reporting	Client Progress Reports, Consulting Progress Reports, Action Items, Issues / Risks	O
Completion			
CR.080	Secure Client Acceptance	Client Acceptance	MO

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary in the *PJM Method Handbook*.

Objectives

The objectives of the Control and Reporting process include:

- Initiate the project.
- Define and control the scope.
- Manage client expectations.
- Assess and manage risks, issues, changes, and problems.
- Report progress.
- Establish completion of the project.
- Ensure that future projects can benefit from the lessons learned in this project.
- Ensure that project intellectual capital is made available for reuse within the consulting organization.

Key Deliverables

The key deliverables of this process are:

Deliverable	Description
Scoping Project Management Plan [CR.010, initial]	The Scoping Project Management Plan contains the background, objectives, scope, constraints, and assumptions for the project, in terms of the client's organization and the involvement of consulting.
Project Management Plan [CR.010, complete]	The Project Management Plan contains the project's management approach, refined with additional detail and supported by necessary standards and procedures.

Deliverable	Description
Resolved Change Requests	Documented and approved or rejected changes to project scope and baselines.
Progress Reports	Periodic status of the project for communication inside and outside the project, they are used in meetings and describe the progress of the project in terms of task completion, milestone achievement, and summary of risk, issues, and problems.
Client Acceptance	Approval of satisfactory project work and deliverables.

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Project Sponsor	Approve Project Management Plan, scope changes, and verify milestones.
Consulting Business Manager	Agree on the Project Management Plan, Workplan, and Finance Plan.
Client Project Manager	Agree on all issues, changes, risks, and problems.
Project Support Specialist	Maintain originals and distribute all project standards and plans. Maintain logs of risks, change requests, issues, and problems, and ensure that they progress according to due dates. Draft progress reports.

Role	Responsibility
Project Manager	Write Project Management Plan, control scope and risks, report on progress of project, and secure acceptance of deliverables.

Critical Success Factors

The critical success factors of the CR process are:

- The scope, objectives, and progress of the project are understood and agreed upon by all parties.
- The project culture and climate are conducive to a mutually beneficial philosophy.
- Risks are identified and containment measures are put in place.
- Changes to scope are agreed on and approved.
- Risks, issues, and problems are escalated in a timely manner to avoid detriment to the project.
- Project records are accurate, complete, and auditable (internally and externally).

CR.010 - Establish Scope, Objectives, and Approach

In this task you define and agree on the project scope, objectives, and approach. This task confirms that both consulting and the client have a clear understanding of the scope, objectives, and risks of the work to be undertaken before beginning more in-depth planning. Use risk assessment to identify risks and determine how to contain them.

Deliverable

The deliverable for this task is the **Scoping Project Management Plan** [CR.010, *initial*]. It is the initial version of the Project Management Plan, which is completed in CR.030.

Prerequisites

Required

You need the following input for this task:

☐ **Client Requirements Documentation**

This documentation (including invitation to tender, request for proposal, correspondence, and requirement specifications) is produced by the client as part of the bid and gives their definition of the scope of work in their own terminology.

☐ **Consulting Proposal**

This documentation (including: Bid Approval Forms, Risk Assessment, Proposal Presentations, Profitability Analysis, High-Level Project Workplan, and correspondence) is produced by consulting as part of the bid, and gives the definition of the scope of work in consulting terminology.

☐ **Contractual Agreement Between Consulting and the Client**

The contractual agreement (including any subcontracts) is the joint statement of the scope of work to be undertaken by consulting in response to the client's requirements, and it has legal standing. This agreement is the culmination of the bid process and represents a

consensus between the client and consulting of the requirements and proposal.

☐ **Applicable Consulting Documentation**

Review examples of deliverable documentation from successful projects. These can be obtained from the consulting Knowledge Repository. They will illustrate the type and depth of content required for successful completion of an engagement.

Optional

The following input may be relevant to your project for this task:

☐ **Approved Change Requests (CR.060)**

This documentation includes any scope changes which affect the agreed contractual baseline (approved since most recent planning) and have been produced during previous execution of the project.

☐ **Scope, Objectives, and Approach**

Scope, objectives, and approach documentation may exist from previous execution of the project. This material provides a baseline for refining management plans.

☐ **Risk Assessment**

Risk Assessment should already exist as part of the bid and may have been updated through execution of the project.

Task Steps

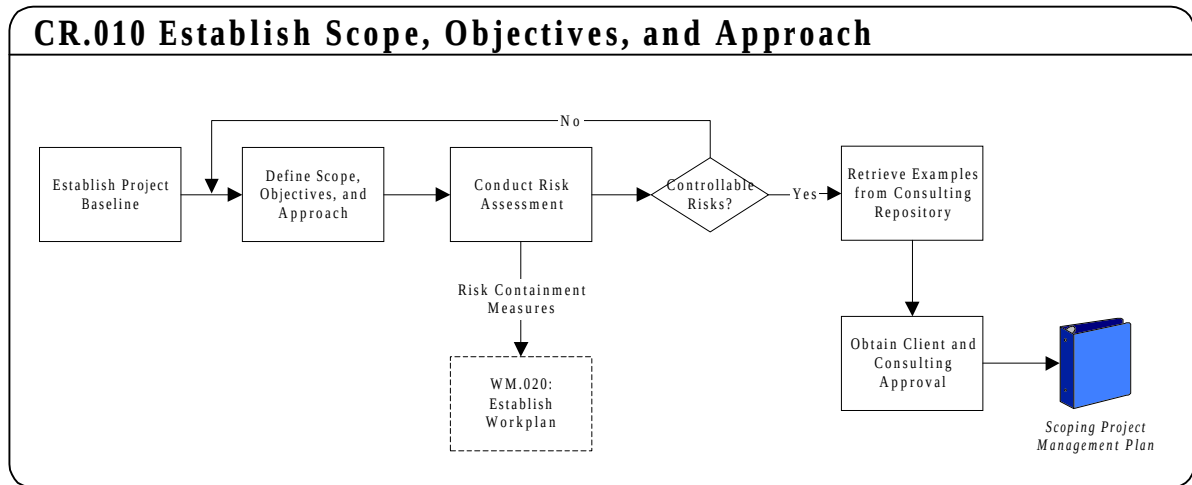


Figure 1-3 Task Flow for Establish Scope, Objectives, and Approach

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Establish a project baseline.	Project Baseline
2.	Define the scope, objectives, and approach.	Preliminary Scope, Objectives, and Approach
3.	Conduct a risk assessment.	Risk Assessment, Risk Containment Measures
4.	Retrieve examples from the consulting repository.	Completed Sample Deliverables
5.	Obtain client and consulting approval.	Scoping Project Management Plan

Approach and Techniques

Establish Project Baseline

This task step follows the bid.

- Collect documentation from the bid manager or practice library. Collect all the source material from the bid manager. Where there has been a gap between proposal submission and contract negotiation, you may need to search for subsequent material, such as letters, scope changes, and legal schedules, in the archives or practice library.
- Review documentation with the bid manager.
- Conduct stakeholder analysis. Discuss the commitments made by the bid manager and explore the client's political climate.
- Use the Project Start Up Check List to facilitate the successful completion of all of the required tasks and task steps.
- File baseline documentation in a temporary project filing system. The filing system established is temporary until it can be incorporated into the Project Library established by the Configuration Management (CM) process.

Define Scope, Objectives, and Approach

In this task step you create the Scoping Project Management Plan.

- Conduct scope meetings with the client. The proposal may be used as the basis for the definition of the scope, objectives, and approach. Update the proposal with any changes that have occurred since it was commissioned. It is essential to meet with the client staff to agree on the project definition and what constitutes completion of the project. Several meetings may be required.
- Define the scope and objectives of the project. In some cases the proposal may already be outdated due to changes during contract negotiation. If these changes impact the costs or time scales, then consulting and the client must come to an agreement.

- Identify constraints and assumptions. It is essential to document and agree on any constraints and assumptions associated with the project. These will often be required later to clarify where things have changed and to identify changes in scope.
- Define the approach. Usually this will be defined as an Oracle Method route (such as Custom Development Method, CDM, Application Implementation Method, AIM, etc.). Further refinement of the route may be required as it applies to this project. For example, there is more than one CDM approach. See the appropriate Method Handbook for the selected route for details of the options. Obtain example deliverables from the Knowledge Repository to assist with the definition of the approach.
- Draft the Scoping Project Management Plan.

Conduct Risk Assessment

In this task step you identify the risks associated with the project (or update risk assessments developed during the bid) and formulate containment measures:

- Review previous risk assessment where available. New risks may have emerged and old risks may no longer be appropriate. Risks can also have changed in their severity and impact.
- Conduct risk assessment. Use an appropriate risk assessment tool and document in the Risk and Issue Log. Where the risks have now become unacceptable, it may be necessary to go back to defining the scope, objectives, and approach.
- Formulate containment measures for each risk. Containment measures all have an associated cost. These must be planned in to the work, and into the financial plans of the work to be undertaken. For example, a typical risk may be where the client is providing and maintaining the development environment. You may decide to expend some effort assisting the client in setting up their procedures and you may wish to audit these periodically. These are risk containment measures that carry a cost.

Retrieve Examples from the Knowledge Repository

In this step, review information from the Knowledge Repository for possible reuse in your current project to leverage successfully completed project deliverables:

- Review available completed documents from similar engagements.
- Incorporate reuse requirements into the development of the approach used to create the deliverables.
- Use sample deliverables to assist the project team in determining the depth and content of each of the deliverables in the context of a successfully completed engagement.
- Incorporate the information whenever possible to reduce costs, reduce risks, and facilitate the timely and effective completion of the deliverables.

Obtain Client and Consulting Approval

The steps required to obtain client and consulting approval are:

- Review the scope, objectives, and approach internally. If containment measures are not acceptable, consider *provisioning* or *contingency*. Risk provision is money set aside to cover recognized quantified risks. Contingency is money set aside to cover risks that are not quantifiable. Both risk provision and contingency are insurance against factors which should not occur but over which you may have limited control.
- If risks are still unacceptable or the containment measures are too costly, then return to redefining the scope, objectives, and approach.
- Reflect review decisions in the Scoping Project Management Plan.
- Schedule and conduct a project kick off meeting to ensure mutual understanding of the scope, objectives, and approach, establish the roles and responsibilities of each of the parties on the project, and to gain acceptance of the scope, objectives and approach.
- Publish the Scoping Project Management Plan.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Establish project baseline, define the scope, objectives, and approach, conduct risk assessment, and secure approvals. Review and approve reusable deliverables obtained from the Knowledge Repository	70
Project Support Specialist	Set up tools and a library of baseline materials. Obtain reusable deliverables from the Knowledge Repository.	30
Bid Manager	Make all relevant bid materials available, and pass on knowledge of the client and stakeholders.	*
Client Project Manager	Approve scope, objectives, and approach.	*
Project Sponsor	Agree on the scope, objectives, and approach.	*
Consulting Business Manager	Agree on the scope, objectives, approach, and risk assessment.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Scoping Project Management Plan are:

☐ **Introduction**

☐ **Scope**

- Scope of Project
- Client Organization
- Locations and Network
- Milestones
- Deliverables
- Constraints and Assumptions
- Risks
- Scope Control
- Relationship to Other Systems and Projects

☐ **Objectives**

- Mission Statement
- Critical Success Factors
- Project Objectives

☐ **Approach**

- Project Methods
- Strategy
- Requirements Tracing
- Acceptance
- Project Administration

Component Descriptions

This section describes each component.

Introduction

This component documents the purpose, background, scope and application and any related documents. In order to draw the reader's attention, any changes that have occurred since the first consulting proposal was prepared should be reflected in the purpose and background.

Scope

Scope of Project: The scope of the project includes the functionality of the solution for your client, together with the boundary conditions that apply, such as the phase(s) of the project. Try to express the scope in specific terms such as to *upgrade Oracle General Ledger from 9.4.1 to 10.4.2*. The remainder of this section then augments specific aspects of this scope.

Client Organization: Put the project in the perspective of the client organization and key contacts, including users.



Attention: The project organization will be defined later, in the Project Management Plan.

Locations and Network: Specify the site where development will take place and the live environment, if this is different. Include hardware and operating system details and network connections.

Milestones: List the milestone tasks and deliverables or checkpoints, together with proposed dates. Indicate any critical dates for the client.

Deliverables: List the deliverables specific to this contract. Indicate those that require client acceptance.

Constraints and Assumptions: List any constraints the project has, such as site working hours limited, client providing the development environment, and limited access to users.

Also differentiate from assumptions, which would include such items as: client approval of deliverables within 2 weeks of delivery and client organization to keep daily backups of the system environment.

Risks: Define any risks that the project will have to overcome or protect against, such as an inflexible implementation date, lack of client experience, new technology or application area, and production dates.

Scope Control: Define how changes to this defined scope will be controlled, or specify that this will be included in the Scoping Project Management Plan.

Relationship to Other Systems and Projects: Put the contract in the perspective of any interfacing or related systems that the client has or is developing.

Objectives

Mission Statement: If the contract has a specific Mission Statement or forms part of a broader agreement, then put this here.

Critical Success Factors: Specify the critical success factors for the contract, including those relating to the client and those relating to the consulting company.

Project Objectives: List the key objectives of the contract, preferably in specific and measurable terms.

Approach

Project Methods: List any consulting methods or other client-specific methods that will be used, together with any tool support, such as *Project Workbench*.

Strategy: Define here the strategy that you will employ to ensure project success, such as teaming agreements, prototyping, utilization of users / client staff, etc.

Requirements Tracing: Cross-reference the Scope Project Management Plan for the contract. Include a contents list, if there is going to be a delay in production of the Scope Project Management Plan.

Acceptance: Define the method you will use to secure Client Acceptance, including who will develop the acceptance tests, what your involvement will be, and the point where acceptance criteria and tests need to be defined.

Project Administration: Specify any administration points, such as site working hours, photocopying limitations, phone calls, system administration arrangements, stationery, etc.

Audience, Distribution, and Usage

The Scoping Project Management Plan is used to:

- confirm a baseline for the project with client staff
- communicate the project baseline to project staff
- record changes that may have occurred between the time a consulting proposal was supplied to a client and when the contract was awarded, as well as any delays in starting the project

Distribute the Scoping Project Management Plan to:

- project manager for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable
- any key client users, identified by the client project manager or line managers, who are to accept the deliverable
- key members of the project team after client approval

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are the scope, objectives, and approach clearly defined?
- ☐ Have you taken into account the impact of dependent tasks from other processes, and information from other documents, such as the consulting proposal and contract?
- ☐ Is this a suitable definition for the project to proceed, or are there unknown items that require resolution?

CR.020 - Define Control and Reporting Strategies, Standards, and Procedures

In this task you create or update the plans which define how the CR process support and implement the Scope, Objectives, and Approach. These will include the detail procedures for such things as issue management, change control, problem management, and reporting of progress.

Deliverable

The deliverable for this task is the set of **Control and Reporting Strategies, Standards, and Procedures**.

Prerequisites

Required

You need the following input before you begin this task:

☐ **Project Management Plan (CR.010)**

The Scoping Project Management Plan is used to set the context for the Control and Reporting strategies, standards, and procedures.

☐ **Policies**

Applicable consulting policies regarding Control and Reporting should be further refined in how they apply to this project. The client may have their own policies regarding Control and Reporting. Take these into account when developing the strategies, standards, and procedures for this process.

Optional

You may need the following input for this task:

Project Management Plan

Prior phases of the project will have produced a Project Management Plan which includes or refers to the deliverable from a previous iteration of this task. You can use it as a baseline for this task.

Task Steps

CR.020 Define Control and Reporting Strategies, Standards, and Procedures

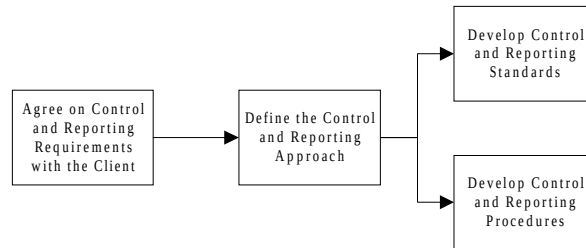


Figure 1-4 Task Flow for Define Control and Reporting Strategies, Standards, and Procedures

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Agree on Control and Reporting requirements with the client.	Control and Reporting Requirements
2.	Define the Control and Reporting approach.	Control and Reporting Strategy
3.	Develop Control and Reporting standards.	Control and Reporting Standards
4.	Develop Control and Reporting procedures.	Control and Reporting Procedures

Approach and Techniques

Agree on CR Requirements with the Client

To agree on CR requirements with the client, complete these steps:

- Review the Scoping Project Management Plan.
- Identify any consulting policies affecting Control and Reporting.
- Identify any client policies affecting Control and Reporting.

Define the CR Approach

To define the CR approach:

- Identify consulting policies, tools, and standards.
- Document a Control and Reporting strategy.

Develop CR Standards

To develop CR Standards:

- Identify the standards that are required.
- Customize the standards to satisfy the approach.
- Add additional standards to meet client requirements.

Develop CR Procedures

- Write detailed instructions for change control, including approval authorities and escalation procedures. Consider the time allocated for investigation and preparation of change control documentation, and whether this is billable to the client.
- Write detailed instructions for issue and risk management, including approval authorities and escalation procedures. Issues may have far reaching business implications for both the client and the project and often result in invoking the change control procedure.

- Write detailed instructions for problem management. Include definition of priorities, approval authorities, and escalation procedures. The definition of a problem *is a non-conformance of a deliverable*. Generally a problem can be fixed with no change to scope. Include an investigation stage to determine the scope will be affected.
- Write detailed instructions for status monitoring, reporting and communication. These instructions will specify the various levels of status reporting required, format, and frequency, as well as any additional communication required to obtain support from any of the stakeholders.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Support Specialist	Prepare strategies, standards, and procedures.	70
Project Manager	Review and approve all strategies, standards, and procedures.	30
Client Project Manager	Agree to Control and Reporting requirements, and identify client policies affecting the approach.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Control and Reporting Strategies, Standards, and Procedures are:

- ☐ Risk and Issue Management
- ☐ Problem Management

☐ **Change Control**

☐ **Status Monitoring and Reporting**

Component Descriptions

This section describes each component.

Risk and Issue Management

This component defines how issues and risks are identified and managed through to resolution or non-impact on the contract. Some risks may have been identified during proposal preparation. These should be provided to the project manager at the start of the project.

Problem Management

This component defines how problems will be managed, including identification, analysis, resolution, and escalation procedures. The problem management procedure is usually used for deliverable-related problems.

Change Control

This component defines how changes to project tasks and deliverables will be handled, from identification through analysis, review, and approval. Where a separate procedure defines this, such as a client system, this section should refer to the client procedure, or copy the client procedure.

Status Monitoring and Reporting

This component defines how project status will be monitored, corrective actions taken, and the project progress reports reviewed and distributed. Reporting includes reporting within the project, to the client, and to consulting management.

Audience, Distribution, and Usage

The Control and Reporting Strategies, Standards, and Procedures are used to:

- provide Control and Reporting inputs to the Project Management Plan
- detail the approach to the various topics for medium to large projects
- define common ways of working for the project team

Distribute the documents to:

- project manager and team leaders for review
- project library for control
- project distribution list

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are the scope and objectives clearly defined?
- ☐ Have you taken into account the impact of dependent tasks from other processes?

CR.030 - Establish Management Plans

In this task you create or update project management plans and document them in the Project Management Plan. The plans created consist of the approaches taken by each management process to address the project scope, objectives, and approach. The Project Management Plan also refers to detailed standards and procedures which serve to implement the processes during project execution, and is updated as necessary.

Deliverable

The deliverable for this task is the Project Management Plan [CR.010, *complete*], or simply the **Project Management Plan**. It contains the definition of project strategies, standards, and procedures aimed at assuring the success of the project. The Project Management Plan defines how the Oracle Method processes are to be applied to this particular project and is an essential management document.

Prerequisites

Required

You need the following input before you begin this task:

- ☐ **Management Strategies, Standards, and Procedures (CR.020, WM.010, RM.010, QM.010, CM.010)**

Management Strategies, Standards, and Procedures are produced by each of the five management processes.

- ☐ **Workplan (WM.020)**

The Workplan provides a time line and work estimate which the supports the project scope, objectives, and approach.

- ☐ **Scoping Project Management Plan (CR.010)**

The background, objectives, scope, constraints, and assumptions for the project, in terms of the client's organization and the involvement of consulting.

Optional

You may need the following input for this task:

Project Management Plan (CR.010)

Prior phases of the project will have produced a Project Management Plan. You can use that as a baseline for this task.

Task Steps

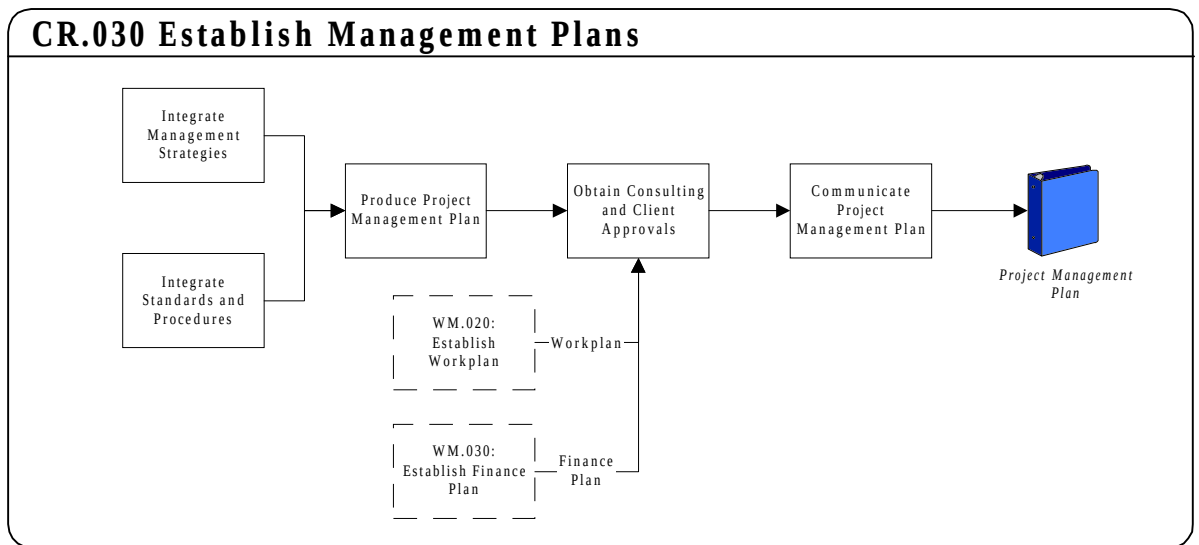


Figure 1-5 Task Flow for Establish Management Plans

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Integrate management strategies.	Reviewed and Coordinated Strategies
2.	Integrate standards and procedures.	Reviewed and Coordinated Standards and Procedures

No.	Task Step	Deliverable Component
3.	Produce the Project Management Plan.	Draft Project Management Plan
4.	Obtain consulting and client approvals.	Approved Plans
5.	Communicate the Project Management Plan.	Project Management Plan

Approach and Techniques

Integrate Management Strategies

To integrate management strategies:

- Review each set of Management Strategies for completeness and approach.
- Review all plans for consistency and coordination.
- Update plans as required.

Integrate Standards and Procedures

To integrate standards and procedures:

- Review each Standard and Procedure for consistency with Management Strategies.
- Review all Standards and Procedures for consistency and coordination. Ensure that approval authorities are consistent and appropriate.
- Update as required.

Produce Project Management Plan

To produce the Project Management Plan:

- Integrate Management Strategies into the Project Management Plan.

- Integrate Standards and Procedures into the Project Management Plan.
- Create the document according to standard format.

Obtain Consulting and Client Approvals

To obtain consulting and client approvals, complete these steps:

- Use the Quality Review procedure to obtain consulting approval for Project Management Plan (and any associated detail documents), Workplan, and Finance plan. Agree on estimates, resources, and work schedule, and update plans as needed.
- Use the Quality Review procedure to obtain client approval for the Project Management Plan and Workplan. Obtain agreement from the client on resource requirements, work schedule, and client commitments.
- Adjust the plans as required, and place under configuration control.

Communicate Project Management Plan

To communicate the Project Management Plan:

- Publish and distribute the Project Management Plan and any associated detail documents throughout the organization as defined for communication.
- Plan startup meeting(s).
- Conduct startup meeting(s).



Attention: Project startup is the culmination of the planning meetings that take place to agree on such things as the scope, objectives and approach. The purpose of these startup meetings is to gain commitment to the project by all the stakeholders, and therefore may take many forms and time.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Manager	Review and approve all management strategies, standards, and procedures. Review and approve the Project Management Plan. Prepare for and conduct reviews and startup meeting(s).	75
Project Support Specialist	Prepare plans and review for coordination and consistency.	25
Consulting Business Manager	Review and approve the Workplan, Finance Plan, and Project Management Plan.	*
Project Sponsor	Review and agree on the Project Management Plan.	*
Client Project Manager	Review and approve deliverable components.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Project Management Plan are:

☐ **Introduction**

☐ **Scope**

Scope of Project
Client Organization
Locations and Network
Milestones

- Deliverables
- Constraints and Assumptions
- Risks
- Scope Control
- Relationship to Other Systems and Projects

Objectives

- Mission Statement
- Critical Success Factors
- Project Objectives

Approach

- Project Methods
- Strategy
- Requirements Tracing
- Acceptance
- Project Administration

Control and Reporting

- Control and Reporting Standards and Procedures
- Issue/Risk Management
- Problem Management
- Change Control
- Status Monitoring and Reporting
- Meetings

Work Management

- Work Management Standards and Procedures
- Time Recording
- Workplan Control
- Financial Control

☐ **Resource Management**

- Resource Management Standards and Procedures
- Staff Roles and Organization Management
- Project Team
- Project Roles and Responsibilities
- Education and Training
- Project Software / Tools
- Hardware
- Project Working Environment
- Environmental Maintenance
- Software Backup Procedures

☐ **Quality Management**

- Quality Management Standards and Procedures
- Quality Reviewing
- Quality Auditing
- Quality Measurement
- Test Management

☐ **Configuration Management**

- Configuration Management Standards and Procedures
- Document Control
- Configuration Definition
- Configuration Control
- Release Management
- Configuration Status Accounting
- Configuration Audit

Component Descriptions

This section describes each component.

Introduction

This component documents the purpose, background, scope and application, and any related documents.

Scope

Scope of Project: The scope of the project includes the functionality of the solution for your client, together with the boundary conditions that apply, such as the phase(s) of the project. Try to express the scope in specific terms such as "to upgrade Oracle General Ledger from 9.4.1 to 10.4.2". The remainder of this section then augments specific aspects of this scope.

Client Organization: Put the project in the perspective of the client organization and key contacts, including users.

Locations and Network: Specify the site where development will take place and the live environment, if this is different. Include hardware and operating system details and network connections.

Milestones: List the milestone tasks and deliverables or checkpoints, together with proposed dates. Indicate any critical dates for the client.

Deliverables: List the deliverables specific to this contract. Indicate those that require client acceptance.

Constraints and Assumptions: List any constraints the project has, such as site working hours limited, client providing the development environment, and limited access to users.

Also differentiate from assumptions, which would include such items as: client approval of deliverables within 2 weeks of delivery and client organization to keep daily backups of the system environment.

Risks: Define any risks that the project will have to overcome or protect against, such as an inflexible implementation date, lack of client experience, new technology or application area, and production dates.

Scope Control: Define how changes to this defined scope will be controlled, or specify that this will be included in the Project Management Plan.

Relationship to Other Systems and Projects: Put the contract in the perspective of any interfacing or related systems that the client has or is developing.

Objectives

Mission Statement: If the contract has a specific Mission Statement or forms part of a broader agreement, then put this here.

Critical Success Factors: Specify the critical success factors for the contract, including those relating to the client and those relating to the consulting company.

Project Objectives: List the key objectives of the contract, preferably in specific and measurable terms.

Approach

Project Methods: List any consulting methods and / or other client-specific methods that will be used, together with any tool support, such as *Project Workbench*.

Strategy: Define here the strategy that you will employ to ensure project success, such as teaming agreements, prototyping, utilization of users or client staff, etc.

Requirements Tracing: Cross-reference the Project Management Plan for the contract. Include a contents list, if there is going to be a delay in production of the Project Management Plan.

Acceptance: Define the method you will use to secure Client Acceptance, including who will develop the acceptance tests, what your involvement will be, and the point where acceptance criteria and tests need to be defined.

Project Administration: Specify any administration points, such as site working hours, photocopying limitations, phone calls, system administration arrangements, stationery, etc.

Control and Reporting

This component defines overall strategies for control of the project, and reporting internally and externally to the project.

Control and Reporting Standards and Procedures: The procedures for Control and Reporting may be specified separately, or, for small projects, defined in this section. Where procedures are separately defined, this section can be either bulleted to show which procedures

exist, or may be written at a higher level, so the reader can get an appreciation of the approach.

Issue/Risk Management: This defines the way Issues/Risks are identified and managed through to resolution or non-impact on the project.

Problem Management: This defines the approach to problem management, including identification, analysis, resolution, and escalation mechanisms. Problem management may use the Problem Report (and Log), a client-specific procedure, or a software tool. The problem management procedure is generally used for deliverable-related concerns, during review or testing of project products.

Change Control: This procedure defines how changes to project scope will be handled. Options include using the Change Request Form (and Log), using a software tool, or using an existing client system. Where a separate procedure defines change control measures, such as in a client system, this section should refer out to the client procedure, or copy the client procedure as is.

Status Monitoring and Reporting: This defines the way in which project status will be monitored and reported on a regular basis. Reporting includes reporting within the team, to the client, and to consulting.

Meetings: Define meeting arrangements for the team, with the consulting practice, and with the client in terms of where the meetings will be held, how often, and who will attend.

Work Management

Work Management Standards and Procedures: The procedures for Work Management may be specified separately, or, for small projects, defined in the remainder of this section. Where procedures are separately defined, the remaining parts of this section can simply be bulleted to show which procedures exist, or may be defined at a higher level, so the reader can get an appreciation of the content.

Time Recording: Specify the arrangements for time recording, including the use of a timesheet to record details, including work performed, holidays, training days, and how often these are to be completed, together with billing arrangements. Define responsibility for

inputting time details, including how often the project workplan will be updated with actual time spent.

Workplan Control: Define any specific arrangements for controlling work, such as the responsibilities and tasks of the project coordinator, where one is appointed. Include estimating, adjustment of the project plan, scheduling, and limitations of authority. Include responsibility for raising issues and action items based on assessment of the project's progress against the workplan.

Financial Control: Define how expenses, labor, and non-labor costs are collected on a monthly basis. Specify how earned value will be used (where applicable). Define how tracking financial information will be used in invoicing, and how the Financial Progress Statement will be used.

Resource Management

Resource Management Standards and Procedures: The procedures for Resource Management may be specified separately, or, for small projects, defined in the remainder of this section. Where procedures are separately defined, the remaining parts of this section can simply be bulleted to show which procedures exist, or may be defined at a higher level, so the reader can get an appreciation of the content.

Staff Roles and Organization Management: Specify any procedural aspects of how the organization will be managed, including key roles and responsibilities, staff review schedules, and client relationship management.

Project Team: Define the project organization, including key client staff and any user representatives nominated for the project. Identify the specific project team members (management and key roles, such as configuration manager, will usually be appropriate, rather than every team member).

Project Roles and Responsibilities: List the key responsibilities of staff in this section, or use an appendix where a large organization exists. This section then acts as a communication to team members for their responsibilities.

Education and Training: Specify any training that has been identified for the project team, such as specific skills training, familiarization with

the project environment, or any assumptions about client education. Identify project training requirements here.

Project Software Tools: List the baseline project tools (including applications) and their versions.

Hardware: Define the hardware platform and operating system for both the development and production environments.

Project Working Environment: Define the requirements for physical resources to support project work, such as PCs and office automation software. List environment set-up details, such as directory structures, or refer out to a document defining this.

Environmental Maintenance: Define the responsibility for environment maintenance, together with constraints, such as backups outside working hours. Include specific arrangements for service requests or fault reporting.

Software Backup Procedures: Where the team is performing backups, specify the specific procedure for this activity here, including responsibility.

Quality Management

Quality Management Standards and Procedures: The procedures for Quality Management may be specified separately, or, for small projects, defined in the remainder of this section. Where procedures are separately defined, the remaining parts of this section can be bulleted to show which procedures exist, or may be defined at a higher level, so the reader can get an appreciation of the content.

Quality Reviewing: Reviews are carried out on all project deliverables using techniques which identify errors as early as possible. These include: peer reviews, inspections by an individual or team, and walkthroughs. Define here the techniques that are going to be used and who is involved. Modify them to suit the project and include client involvement.

Healthchecks: Complete a Healthcheck to review project progress to date and to ensure consistent application of policies and procedures early in the engagement. Healthchecks are conducted by independent consultants that are not part of the project team to bring an objective

focus on the engagement. Subsequent Healthchecks may be required but are discretionary. The results of the Healthcheck are to be reviewed by the project manager and the client.

Quality Auditing: Audits focus on whether the project processes are functioning according to plans and procedures. Usually a project needs at least one audit, and large projects may need many, for example one per phase, or one per major milestone. Audits use resources independent of the project team.

List here the audits that will be conducted during the project.

Quality Measurement: In order to improve the accuracy of the project estimates, both as the project progresses and for future projects, keep some basic metrics. Specific metrics include time required to design and build modules of all types, number of modules under development, number of problems during system test, number of problems during acceptance test, and also other types of metrics, such as setup time for application. Define the metrics which are to be kept here.

Test Management: Define the overall strategy of testing for the project, including the scaling of the test program to suit the size, complexity, and criticality of the system under development. Explain the use of any testing tools. Specifically, define:

- Test Strategy
- Test Stages
- Test Procedure

Configuration Management

Configuration Management Standards and Procedures: The procedures for Configuration Management may be specified separately, or for small projects, defined in the remainder of this section. Where procedures are separately defined, the remaining parts of this section can be bulleted to show which procedures exist, or may be defined at a higher level, so the reader can get an appreciation of the content.

Configuration Definition: Describe any project naming conventions for each type of item to be produced and the documentation naming conventions. Describe directory names and storage arrangements, including versioning, if different from item naming. Define:

- Configuration item types of significance to a project.
- Interrelationships between item types.
- A life-cycle for each item type. Identify a hierarchy of possible states for the item type. For each state transition (promotion), document a quality criteria and authorization requirement to be enforced.

Document Control: This defines how to create, maintain, update, and distribute documents. The project library, in physical terms, is usually part paper and part electronic, so the key aspect of this component is delimiting the paper from the electronic items, while defining control mechanisms for both.

Configuration Control: Describe arrangements for the control of configuration items, such as placing in a controlled area when tested, and the mechanisms to be employed for managing technical changes, including approval for the change and correction due to problems being found. Describe when baselines will be issued and who will be responsible for it.

Release Management: Describe the procedure for preparing a release, including responsibilities and deliverables. Describe the arrangements for baselining and releasing software to the client, for example for acceptance or for final delivery. Include how to maintain a copy in the project environment to support acceptance testing and problem solving.

Configuration Status Accounting: Define any item completion reporting designed to record progress and track problems.

Audience, Distribution, and Usage

The Project Management Plan is used to:

- define ways of working for the project team
- explain the client involvement in the project
- define management and control mechanisms

Distribute the Project Management Plan to:

- project manager and team leaders for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable
- any key client users, identified by the client project manager or line managers, who are to accept the deliverable

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Are the scope and objectives clearly defined?
- ☐ Have you taken into account the impact of dependent tasks from other processes and deliverables, such as the contract and the Scoping Project Management Plan?
- ☐ Are review types, testing, and audits specified?

CR.040 - Issue/Risk Management

In this ongoing task you identify, assess, and take actions to control issues and risks. Issues often relate to the risks originally identified for the project and require discussion, tracking, and resolution. New risks may emerge during project execution that require discussion, containment measures, and tracking.

Risks and issues have similar characteristics. Risks are things that *could* happen and *might* have an impact on the project if not contained. Issues are usually more specific, and will have an impact on the project unless action is taken to resolve them. Do not allocate too much time defining the difference between an issue and a risk. The important point is to track and manage anything that may threaten the success of the project.

Issues may trigger new risks and changes for which you will need to apply the appropriate tasks. Risk containment measures may have schedule and cost implications, giving rise to changes to the Workplan and Finance Plan.

Prerequisites

Required

You need the following input before you begin this task:

☐ **Issue/Risk Management Procedure (CR.020)**

The Issue / Risk Management procedure is created during planning for CR.

Optional

You may need the following input for this task:

☐ **Risk and Issue Log (CR.040)**

Risks and issues identified during the previous project control activity are tracked in this log.

Task Steps

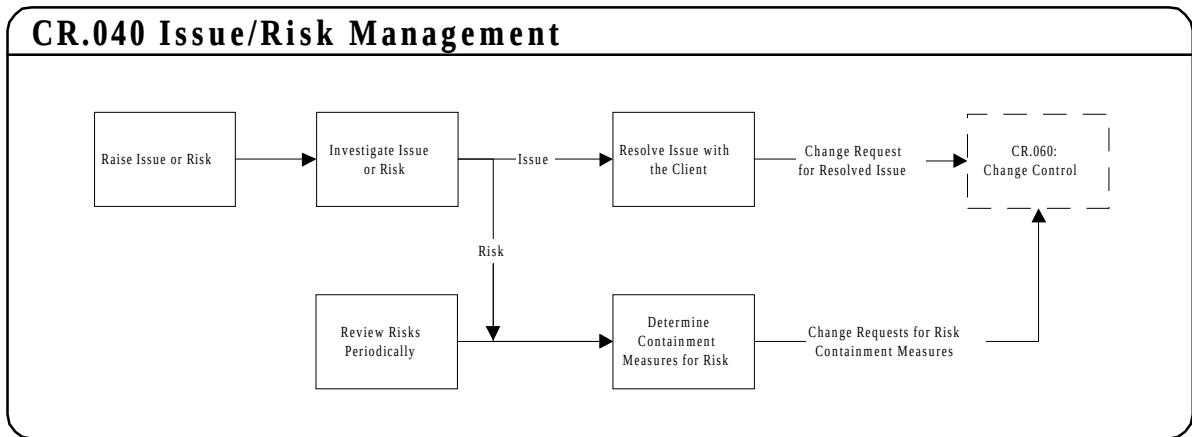


Figure 1-6 Task Flow for Issue/Risk Management

The steps of this task are:

No.	Task Step	Result or Output
1.	Raise issue or risk.	New Issue or Risk
2.	Investigate issue or risk.	Resolution or Containment Options
3.	Resolve issue with the client.	Resolved Issue, Change Request
4.	Determine containment measures for risk.	Agreed Risk Containment Measures, Change Requests
5.	Review risks periodically.	Revised Risk Containment Measures

Approach and Techniques

Raise Issue or Risk

Issues and risks may be raised by the client or by consulting staff. Log and assign responsibility for issues and risks.



Suggestion: It may be appropriate to use two Issue/Risk Logs where there are issues/risks that only relate to consulting and are resolvable internally. Consulting issues/risks are reflected on the client log but only in the way that they affect the project. The client would thus be protected from consulting issues.

Investigate Issue or Risk

To investigate Issue or Risk:

- Investigate each issue/risk for the potential impact on the project in terms of effort, timescales, and finance. Explore alternative scenarios, and offer options for resolution.
- Include the option of doing nothing.
- Where possible, this investigation should be carried out with the client in a joint problem-solving exercise. There may be alternative resolutions involving client business policies and practices where only the client would be able to assess the impact.
- An issue resolution may result in a change in scope. In this case, invoke the Change Control procedure to determine the estimated cost of implementing the resolution.

Resolve Issue with the Client

To resolve the issue with the client:

- Review the options for resolution for each issue with the client, and agree on the most appropriate resolution. If a resolution cannot be agreed upon until a later date, then a deferment date should be agreed upon and interim actions taken to advance the issue.

- The review can take the form of a regular meeting where all new issues and issues that have reached their deferment date are discussed.
- Create a Change Request, if necessary, to implement resolution.
- If necessary, escalate issues that prove intractable to the appropriate level according to the escalation procedure in the Issue/Risk Management procedure.

Determine Containment Measures for Risk

To determine containment measures for risk:

- Review each risk with the client so that everyone is aware of the possible impact on the project. If the client is aware of the risks at an early stage, then they can manage their community’s expectations accordingly.
- Review the possible containment measures for each risk, and agree on the appropriate containment measures. Use a Change Request to implement containment measures as necessary.
- Consider the cost of the containment measure for each risk versus the possible impact on the project.

Review Risks Periodically

To review risks periodically:

- Keep risks open until there is no possibility of them becoming a reality.
- Review all risks and containment measures periodically to ensure that the strategies for each are still appropriate.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Resolve issues and risks.	60

Role	Responsibility	%
Project Support Specialist	Ensure that all issue and risk documentation is filed in the Project Library. Distribute this documentation according to procedure.	40
Consulting Business Manager	Agree on issue resolutions and risk containment measures.	*
Client Project Manager	Raise issues and risks; agree on issue resolutions and risk containment measures.	*

* Time not estimated.

Outputs

The outputs for Issue/Risk Management are:

- ☐ **Resolved Issues**
- ☐ **Risk Containment Measures**
- ☐ **Change Requests**

Output Descriptions

This section describes each output for this ongoing task.

Resolved Issues

Resolved issues are issues for which a documented course of action has been approved by project management as a remedy. Issues normally are investigated one or more times to determine various courses of action and their impact to the project. Once a course of action has been selected, both the consultant and the client record and approve it. Approved issues require that action be taken. Rejected issues are decisions to take no action.

The Risk / Issue Form is an example of a form which can be used to document an issue through its life-cycle. Such a form is useful on small to medium sized projects, where issue volume and coordination demands are low. A tracking Issue Log can be maintained using a simple spreadsheet. On larger projects, with a number of teams and a larger volume of issues to manage, an automated tool which can produce issue details or summaries on demand, and which facilitates coordination across teams, is recommended in order to reduce administrative demands.

Issue management is an important communication tool for projects. The most critical aspect of tracking issues is maintaining visibility for both project leaders and staff. Ensure that all project members can raise and track the status of project issues without administrative impediments.

Risk Containment Measures

Risk containment measures are the documented decisions reached by management to establish contingencies for, or reduce the likelihood of, an undesirable situation occurring on the project, which might affect its success.

The Risk / Issue Form can also be used to track risks. Since the number of risks tracked is usually relatively small, this form and the corresponding log are usually sufficient to maintain risk visibility. One effective technique is to use a composite Risk / Issue Log and extract a list of the most urgent issues and risks for regular use in management reviews.

Change Requests

When a course of action is agreed upon by management to resolve an issue or contain a risk, one or more actions are assigned. If an action affects schedule, resources, or cost, then a change request is created to track implementation.

RISK/ISSUE FORM	
Ref No: ____/____/ RIF/ ____	
Originator's name: Functional Area Type: Risk/Issue*	Priority: Critical / High / Medium / Low* Date Raised: Target Resolution Date:
Status: Open / Assigned / Investigated / Resolved / Approved / Deferred / No Action*	
Description:	
Possible action:	
Estimated Impact:	
Possible action:	
Estimated Impact:	
Possible action:	
Estimated Impact:	
Recommendation:	
Accepted (Consultant):	Accepted (Client):
Date:	Date:
Associated Problem Report:	Problem Report Date:
Associated Change Request Form:	Change Accepted on:
* Circle as appropriate	

Figure 1-7 Risk and Issue Form

Audience, Distribution, and Usage

Issue/Risk Management outputs are used to:

- document risks and issues as they occur, for subsequent analysis, resolution, and discussion
- act as a checklist for the project manager and team leaders

Distribute the outputs to:

- project member who is assigned to investigate the risk or issue
- project managers in summary form on a regular basis, or made available on demand
- project library for control and project team visibility
- project leaders prior to issue and risk review meetings

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Is the risk or issue clear?
- ☐ Has sufficient attention been paid to understanding the problem and detailing courses of action and their impacts?
- ☐ Is the option to do nothing included?
- ☐ Does the investigation include analysis of the root cause of the issue and risk to preclude it from recurring?
- ☐ Are risks and issues resolved quickly?

CR.050 - Problem Management

In this ongoing task you identify and manage problems that occur throughout the project. Manage these problems through to resolution.

Problems are raised when deliverables do not meet an expectation. Frequently this will occur during software testing. Investigate problems to verify that they are not actually changes. If the problems are changes, then invoke the Change Control task. If issues need to be raised for client resolution, then invoke the Issue /Risk Management task.

During the course of the project, record and track problems resolution. The Problem Report and Log is one way of doing this, although most projects adopt an automated tools as the problem counts rise.

The use of Problem Reports helps to quantify, schedule, and prioritize problems during a project. Progress reporting also serves as a mechanism to ensure that problems are resolved.

Prerequisites

Required

You need the following input for this task:

☐ **Problem Management Procedure (CR.020)**

The Problem Management procedure is defined in planning for Control and Reporting.

Optional

You may need the following input for this task:

☐ **Problem Report Log (CR.050)**

Previously identified problems and their progress are tracked in the Problem Report Log. This is the key management document for tracking problems.

Task Steps

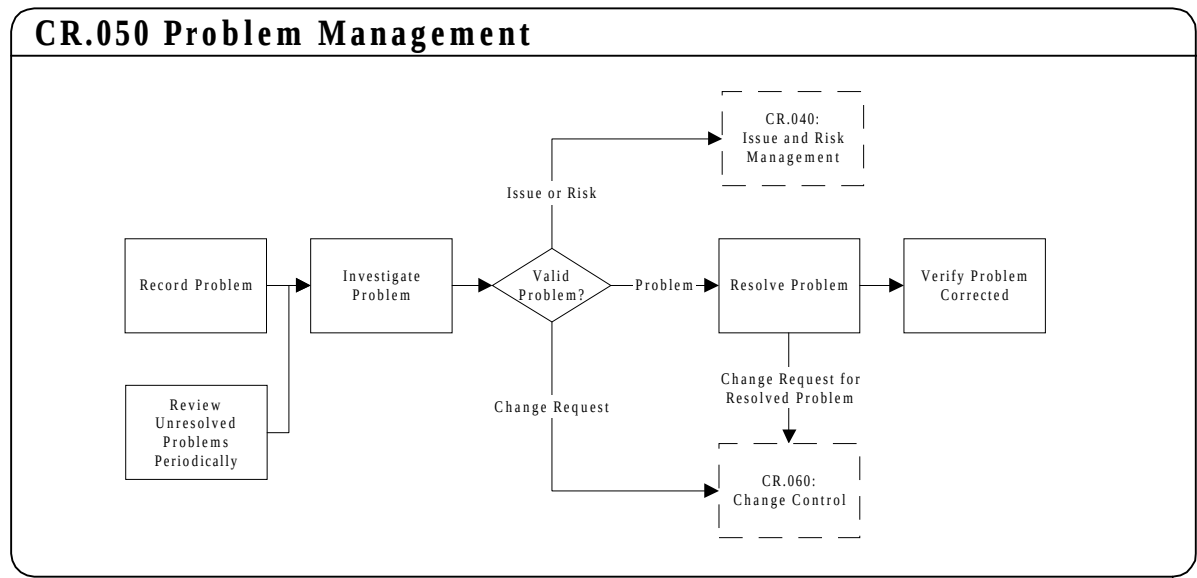


Figure 1-8 Task Flow for Problem Management Process

The steps involved in this task are:

No.	Task Step	Result or Output
1.	Record problem details.	New Problem
2.	Investigate the problem.	Investigated Problem, New Issue/Risk or Change Request (as appropriate)
3.	Resolve the problem.	Resolved Problem
4.	Verify that the problem has been corrected.	Closed Problem
5.	Review unresolved problems periodically.	None

Approach and Techniques

Record Problem

Anyone associated with the project may raise problems. It is helpful if there is a central point where problems are investigated for mistaken understandings and duplications. When a wide number of people are exposed to a deliverable (such as during acceptance testing), it is preferable that the client undertake this initial investigation, so that unnecessary time is not lost to trivial problems and duplications. If the client does not agree to undertake this task, then include the additional effort in effort and cost estimates for the project. Also, project staff may use emotive terminology when raising a problem; it is important to establish and record specific, factual data about the problem.

- Record problem details using the Problem Report and Problem Report Log.
- Project staff may use emotive terminology when raising a problem; it is important to establish and record specific, factual data about the problem.

Investigate Problem

- Determine if the issue is really a problem. A problem is defined as a non-conformance against the latest baseline. If it is an issue that needs client resolution, or a change in scope, then log an Issue or Change Request and pass to the appropriate task. In this case, reference the problem on the Issue or Change Request, and then close it.
- Record the priority of the problem. Detail guidelines for setting priorities and escalation in the Problem Management procedure.
- Establish the possible resolutions of the problem and the impact of these against the Workplan, cost, and quality.
- Remember that one option is to do nothing about the issue.
- If the resolution to a problem requires a change to the latest agreed baseline, then raise and apply the appropriate change requests and ongoing task, and then close the problem, irrespective of whether the client is to be charged for the change.



Attention: When the Issue / Risk Management or Change Control ongoing task has been applied to a problem, close the problem. Subsequent tracking will take place through the other ongoing task.

Resolve Problem

To resolve a problem:

- Take the appropriate action to fix the problem.
- Where the problem will take significant effort to fix, it may be necessary to pass the problem to Work Management for scheduling. There should already be an allowance within the Workplan for fixing routine problems.

Verify Problem Corrected

To verify that the problem is corrected:

- Gain confirmation from the person who raised the problem that it has been corrected.
- Update the Problem Report and the Problem Report Log.

Review Unresolved Problems Periodically

Review all open problems periodically to ensure progress on resolution. A large number of problems may indicate a quality problem that may need to be addressed.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Analyze and resolve problems. Log problems and provide status reports.	50
Project Manager	Arrange for analysis and resolution of problems, and escalate resolution where necessary.	50
Client Project Manager	Approve problem resolutions.	*

* Time not estimated.

Outputs

The outputs for Problem Management are:

- ☐ **Resolved Problems**
- ☐ **Issues/Risks**
- ☐ **Change Requests**

Output Descriptions

This section describes each output for this ongoing task.

Resolved Problems

Resolved problems are those problems which have been corrected or for which no action is deemed necessary. Problems are investigated by appropriate project members after being raised. The problem is

validated and its priority, cause, and possible corrective actions are proposed.

Problems found during testing are often further classified into categories for management, such as defect, bug, documentation error, or data error. For problems that could require considerable investigation, a management review after initial validation is conducted to assign a priority and resource limits. Once a course of action has been agreed upon, the problem is assigned to a project member for execution, and tracked until action is complete and verified.

The Problem Report is an example of a way to track a problem through its life-cycle. Such a form is useful on small to medium projects where problem volume and coordination demands are low. A tracking Problem Report Log can be maintained using a simple spreadsheet. On larger projects, with a number of teams and a larger volume of problems to manage, an automated tool which can produce problem details and summaries on demand, and which facilitates coordination across teams, is recommended in order to reduce administrative demands.

Issues and Risks

New issues and risks are identified during investigation of problems, or by reclassifying a problem. The underlying cause of a problem may uncover an issue or risk, or the problem itself may not apply to a deliverable but to a broader scope of concern. In the latter case, close the original problem and reference the new issue or risk.

Change Requests

If the investigation of a problem determines that it is actually a request to change project scope or a project baseline, then raise a change request in its place. Make this decision at the time that the problem resolution is initially investigated. Ensure that you establish clear rules for initiating a change request in the Problem Management procedure. If the problem is passed to a change request, then close the problem and reference the change request which was raised.

PROBLEM REPORT Priority: Immediate / Urgent / Routine # Ref No: ____ / ____ / PRT / ____	
Originator's name:	Investigator's name:
Date:	
Problem details including item affected if applicable: 	
Investigation: 	
Suggested / Actual action: 	
Estimate of required work: 	
Authorization to Proceed (Consultant): Date:	Authorization to Proceed (Client)*: Date*:
Person who tested the change:	Date change tested:
Problem Closed by:	Associated Change Request Form*:
On:	Raised On*:
Associated Risk/Issue Form:	
* Complete if Applicable # Delete as applicable	

Figure 1-9 Problem Report

Audience, Distribution, and Usage

Problem Management outputs are used:

- to document problems as they occur for subsequent analysis, approval, and resolution
- as control information for the project manager and team leaders
- for discussion at project meetings both internally and with the client

Distribute outputs to:

- project member who is assigned to investigate a problem
- project managers, in summary form on a regular basis, or made available on demand
- project library for control and project team visibility
- to project leaders prior to problem review meetings

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Is the problem clearly stated?
- ☐ Does the analysis of the problem impact cover all likely aspects of the change, including documentation, interfaces, and tools?
- ☐ Has the option to do nothing been considered?
- ☐ Has the problem been validated against a project deliverable? Have any issues, risks, or change requests been raised, if needed?
- ☐ Are problem trends evaluated regularly to help identify underlying issues or risks to the project?

CR.060 - Change Control

In this ongoing task you manage requests to change a project baseline, including changes to project scope.

Requests for change can arise either from within the project or from the client. The resolution of an issue or problem, or the recognition of a potential improvement by a project member, can give rise to a Change Request. The client may introduce a change due to a change in business circumstances or a potential improvement identified during the execution of the project. The key in this task is to identify scope changes and ensure that they are validated, approved, and contained in the Workplan.

Reach agreement at the beginning of the project, and document it in the Project Management Plan as to how to account for the time spent on change impact investigation.



Attention: The estimates for changes may require consulting business management authorization before review with the client. Refer to consulting bid review procedures.

Prerequisites

Required

You need the following inputs for this task:

☐ **Change Control Procedure (CR.020)**

The Change Control procedure is agreed upon during the planning of Control and Reporting.

Optional

You may need the following inputs for this task:

☐ **Change Request Log (CR.060)**

The Change Request Log has records of all previous requests to change the baseline. Use this to track the progress of changes.

Task Steps

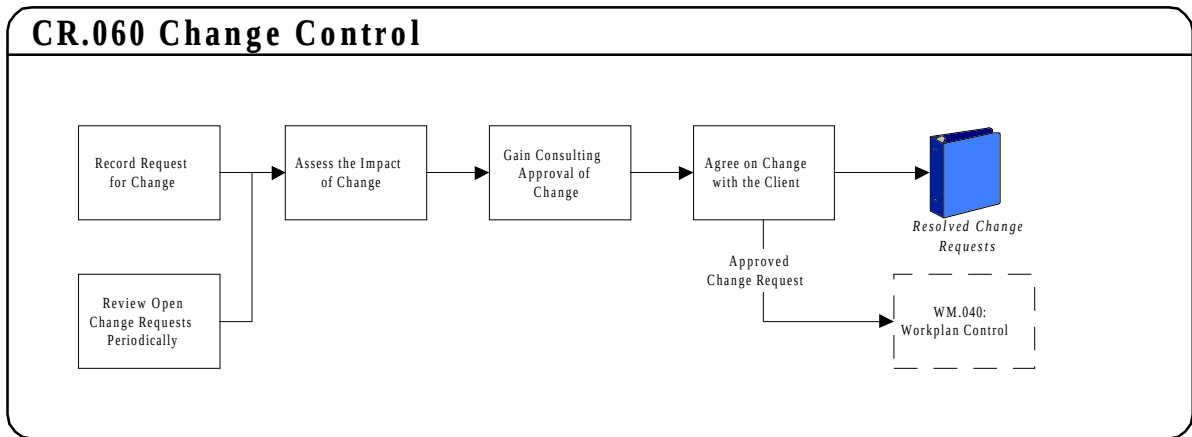


Figure 1-10 Task Flow for Change Control

The steps of this ongoing task are:

No.	Task Step	Result or Output
1.	Record new request for change.	New Change Request
2.	Assess the impact of change (cost, schedule, resources).	Change Impact Assessments
3.	Gain consulting approval of change.	Change Impact Estimate
4.	Agree on change with the client.	Resolved Change Request
5.	Review open Change Requests periodically.	None

Approach and Techniques

Record Request for Change

Changes may be raised by the client or by consulting. The client may have their own procedure to challenge changes before any further action is taken.

- Challenge changes at their inception, as they may be the result of a misunderstanding or may require changes in business practices which render them unnecessary.
- Complete the Change Request Form and the Change Request Log.

Assess the Impact of Change

To assess the impact of the change:

- Identify the configuration items affected and the tasks necessary to perform the change.
- Determine how the change would affect the project in terms of schedule, deliverables, and cost.
- Ask the client to undertake an investigation of what the impact of the change is to them in terms of project schedule, business practices, and costs.



Attention: Where the investigation is lengthy, it may be appropriate to request a Workplan change to cover the investigation. If the investigation is considered a proposal effort, then record the time spent as such and not as an impact on project profitability.

Gain Consulting Approval for Change

To gain consulting approval for the change:

- Prepare an estimate of the impact to the project using an estimating technique employed in Work Management. Document the estimate in the Change Request.

- Ensure that the estimates and impact of the change are approved by consulting by following the consulting practice bid review procedure.



Attention: It is essential to use the same amount of rigor in reviewing estimates for changes as for the original bid estimates.

Agree on Change with the Client

Alert the client of changes even where there are no cost implications, and the change control procedure does not require client approval. Note that the client may have their own approval cycle dependent on the impact of the change.

- Present the benefits and impact of the change to the client. This may take place at a regular review meeting in which the progress of all changes is reviewed.
- Reach agreement on whether or not to proceed with the change, and update the Change Request Log accordingly. Document any change to project scope in the Change Request.
- If the client has a separate approval cycle for change, then document as approval authorities in the Change Control procedure.
- Pass the change when the client approves it to Work Management for inclusion in the Workplan.

Review Open Change Requests Periodically

Periodically review all open Change to ensure that they are being processed.

- Close changes that have been completed.
- Collect statistics on changes, and report them in progress reports.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Identify and ensure full assessment of Change Requests. Agree with client management on the need for changes, and address the impact on schedules and cost, if implemented.	40
Project Support Specialist	Investigate the impact of Change Requests. Ensure that Change Requests are properly recorded in the Change Request Log and managed according to the Change Control procedure.	60
Client Project Manager	Validate the requirement for the change and agree on Change Requests.	*
Project Sponsor	Approve scope changes and their cost and schedule impact.	*
Consulting Business Manager	Approve scope changes which warrant invocation of the bid review procedures.	*

* Time not estimated.

Outputs

The output for Change Control is:

☐ **Resolved Change Requests**

Output Descriptions

This section describes the output for this ongoing task.

Resolved Change Requests

The change request is the mechanism by which a change is requested, investigated, estimated, and approved or rejected. An approved change request conveys project management authorization to implement the change and adjust the Workplan as specified. A rejected change request indicates that no change is to be made and that the change request will not be considered further.

Treat any change request that potentially affects the Workplan as a scope change. Estimate its impact in cost, schedule, and resource terms. Agree with the client on how to handle scope changes using the Change Control procedure contained in the Project Management Plan. The client and consulting authorities who approve scope changes are identified in the Change Control procedure.

The Change Request Form shown in Figure 1-11 is an example of how to track a change through its life-cycle. Such a form is useful on small to medium sized projects where change volume and coordination demands are low. A tracking Change Request Log can be maintained using a simple spreadsheet. On larger projects, with a number of teams and a larger volume of changes to manage, an automated tool which can produce change request details and summaries on demand, and which facilitates coordination across teams, is recommended in the interest of reducing administrative demands.

CHANGE REQUEST FORM		Ref No: ____/____/CRF/____
Originator's name:	Priority:	
Signature:	Date Raised:	Date Expires:
Change Details (description of proposed change, the reason for the proposed change, the impact of the proposed change and the implications of not performing the proposed change):		
Resolution (The history of this note):		
Possible action:		
Accepted (Consultant): Date:	Accepted (Client): Date:	
Completion Verified By:	Completion Date:	

Figure 1-11 Change Request Form

Audience, Distribution, and Usage

Change Control outputs are used:

- to document requested changes as they occur, for subsequent analysis, approval, and implementation
- as control information for the project manager and team leaders
- for discussion at project meetings, both internally and with the client

Distribute outputs to:

- project member who is assigned to investigate a change request
- project leaders in summary form on a regular basis, or made available on demand
- project library for control and project team visibility
- project leaders prior to change request review meetings

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Is the change clearly stated? Are the benefits of the change to the client well understood and quantifiable?
- ☐ Has the option to do nothing been considered?
- ☐ Does the analysis of the change impact cover all likely aspects of the change, including documentation, interfaces, tools?
- ☐ Are time scales and costs realistic?

CR.070 - Status Monitoring and Reporting

In this ongoing task you monitor and report on progress. Status reports are given by each process. For example, in the Work Management process, you will update the project plans with actual effort and progress against individual tasks. In this task you consolidate these reports and produce the overall progress report(s) for the project. The period, level (e.g., daily, weekly, etc.), format, and recipient of the reports should be agreed upon with the client and specified in the Project Management Plan.

The Project Management Plan may dictate different levels of progress reporting. This will include the formal reports and communication meetings. Each level requires different levels of detail and emphasis. These levels may include: consulting project team, wider project team, client project manager, project sponsor, consulting business manager, and steering committee.

Prerequisites

Required

You need the following input before you begin this task:

☐ **Status Monitoring and Reporting Procedure (CR.020)**

The Status Monitoring and Reporting Procedure details how to collect, review, and communicate progress on the project.

☐ **Project Management Plan (CR.010)**

The Project Management Plan contains the strategies for monitoring and reporting.

☐ **Progress Statements (WM.040, WM.050, QM.030, CM.050)**

Other processes will produce progress statements as detailed in the Project Management Plan.

Task Steps

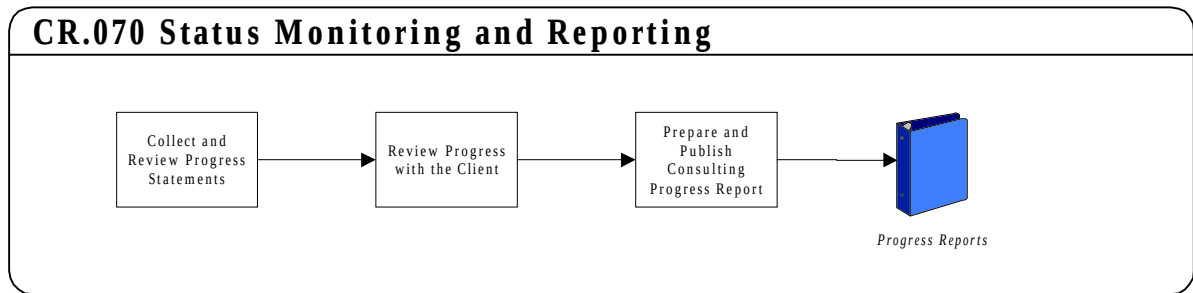


Figure 1-12 Task Flow for Status Monitoring and Reporting

The steps involved in this ongoing task are:

No.	Task Step	Result or Output
1.	Collect and review progress statements from other processes.	Accumulated Progress Statements, Action Items
2.	Review progress with the client.	Client Progress Report, Meeting Minutes, Action Items, Issues/ Risks
3.	Prepare and publish the consulting progress report.	Consulting Progress Report

Approach and Techniques

Collect and Review Progress Statements

Progress Statements are prepared for specific processes. These detail progress, problems, priorities, and plans. Report progress verbally or in writing, depending on the procedures delineated in the Project Management Plan.

Review progress and assign corrective actions. Qualitative information can be even more valuable than the quantitative information. The qualitative information answers questions such as:

- Is the organization functioning as a unit? Is the process functioning satisfactorily?
- Is there an individual with any particular problems? (and is the solution in the manager's control?)
- Are there any problems of resource or environment that can be solved?
- Are there any new management issues or risks?

Review Progress with the Client

The levels and the frequency of reporting progress to the client are defined in the Project Management Plan. Some reporting may be done jointly by the client project manager to the project sponsor.

- Consolidate the progress statements from each process, add information of progress from Control and Reporting, and create a draft of the Client Progress Report.
- Review the draft Progress Report with the client prior to publication. The format of this review is defined in the Project Management Plan and should always include a Progress Meeting with recorded minutes.

Prepare and Publish Consulting Progress Report

The level and format of reporting within consulting will be determined by practice policies. Determine the application at the start of the project. Where a variety of consulting stakeholders are influential in the project, and the levels of risks and issues merit it, consider a board comprising all these stakeholders.



Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Manage status monitoring and reporting.	50
Project Support Specialist	Draft the report(s) and record meeting minutes.	50
Consulting Business Manager	Read and review reports and attend meetings.	*
Client Project Manager	Read and review reports, and attend meetings.	*
Project Sponsor	Read and review reports, and attend meetings.	*

* Time not estimated.



Outputs

The outputs for Status Monitoring and Reporting are:

- ☐ **Client Progress Report**
- ☐ **Consulting Progress Report**
- ☐ **Action Items**
- ☐ **Issues/Risks**

Output Descriptions

This section describes each output for this ongoing task.

Client Progress Report

Progress Summary: Describe the progress made this month on the project, answering the question "is the project running according to plan?" and describing any major achievements this period.

Deliverables and Milestones: List (cumulatively across progress reports) the deliverables, milestones, and dates versus planned dates. Deliverables and milestones are those listed in the Project Management Plan.

Problems and Issues Outstanding: List summary statistics for problems raised or cleared at the time of writing the report. Summarize any significant problems or issues that you want to bring to the client's attention. Internal issues are listed later in the report in the consulting section.

Change Requests for Period: List any change requests raised this period.

List of Changes: List approved change requests (cumulatively) together with value and status.

Major Activities for Next Period: Describe the major tasks to be undertaken in the next period. Discuss these during any progress meeting with the client.

External Audits, Reviews, and Healthchecks: List any external audits, reviews, or Healthchecks carried out during the period. List the date of previous and next reviews.

Next Client Progress Meeting: List the next client progress meeting and venue.

Consulting Progress Report

This report includes all of the Client Progress Report, as well as the following sections.

Status Summary: Use the following checklist items to record status:

- ☐ Is the Complexity Matrix for Bespoke Modules updated?
- ☐ Are the Risk Analysis and Containment Measures updated?
- ☐ Is the latest Baseline Project Plan agreed to by client?
- ☐ Is the current Project Plan updated?
- ☐ Is the current Project Budget agreed to by business unit and client?
- ☐ Is the client Progress Report delivered?
- ☐ Have you held the client Progress Meeting?

Project Tracking Controls: List the following project details for plan and financial progress tracking:

- Approved Plan
 - Original project cost consulting (labor only)
 - Original project days (excluding reserves)
 - Approved change days
 - Approved total days in plan (excluding reserves)
- Current Plan
 - Number of consulting staff planned or required
 - Number of consulting staff currently on project
- Earned Value Cost (where this is used for tracking)
 - Unapproved change days (not in plan)
 - Actual days spent this month (chargeable to client)
 - Actual days spent this month (not chargeable to client)
 - Actual days spent to date
 - New estimated days to complete (excluding reserves)
 - New total days in plan (excluding reserves)
- Risk and Contingency Reserves
 - Unused risk provision days
 - Unused contingency days
 - Total reserves days

- Variance (approved total days less new total days in plan)
- Explanations of Variance
- Additional meetings

Use of Reserves: Describe any use of reserves (contingency) during the period (cumulatively across the project).

Quality Measures: List project (quality) measures as defined in the Quality Plan, for example:

Design defects found after the Design Phase
Code defects found after Unit Test

Internal Issues: List any internal issues that require resolution.

Action Items

Action items are assignments to the project staff to take corrective actions identified in progress reviews. Track completion of these at succeeding progress reviews.

Issues and Risks

Issues and risks are raised during the review by project management or the client. Record them using the Issue / Risk Management ongoing task.

Audience, Distribution, and Usage

Status Monitoring and Reporting outputs are used to document:

- project progress for consulting management assessment
- progress for discussion with the client

Distribute the outputs to:

- project manager for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Does the report include key information from the project workplan?
- ☐ Is the report complete?
- ☐ Has the report been reviewed by the project manager (usually compiled by the project manager or project coordinator)?

CR.080 - Secure Client Acceptance

In this task you obtain approval of deliverables, to the mutual satisfaction of consulting and the client. A well-managed acceptance demonstrates that you have met your client's needs, and, when all deliverables have been accepted, allows you to negotiate a successful conclusion to the project. A satisfied client is more likely to act as a future reference, as well as a future customer.

Deliverable

The deliverable for this task is the **Client Acceptance** of the phase or project.

Prerequisites

Required

You need the following input for this task:

☐ **Workplan and Progress Reports (WM.040, CR.070)**

The Workplan and Progress Reports tell you what deliverables are due and what deliverables have been completed. Progress Reports and associated meeting minutes detail agreed upon progress.

☐ **Project Management Plan (CR.010)**

The Project Management Plan directs you to the acceptance criteria for deliverables.

☐ **Risk and Issue Log, Problem Report Log, and Change Request Log (CR.040, CR.050, CR.060)**

Use these logs to assess the completeness of the work. Changes in scope are recorded in the Change Request Log.

☐ Quality Reports, Healthcheck Reviews, and Audit Reports (QM.050, QM.045, QM.030)

These reports identify any shortfalls in meeting quality standards.

Task Steps

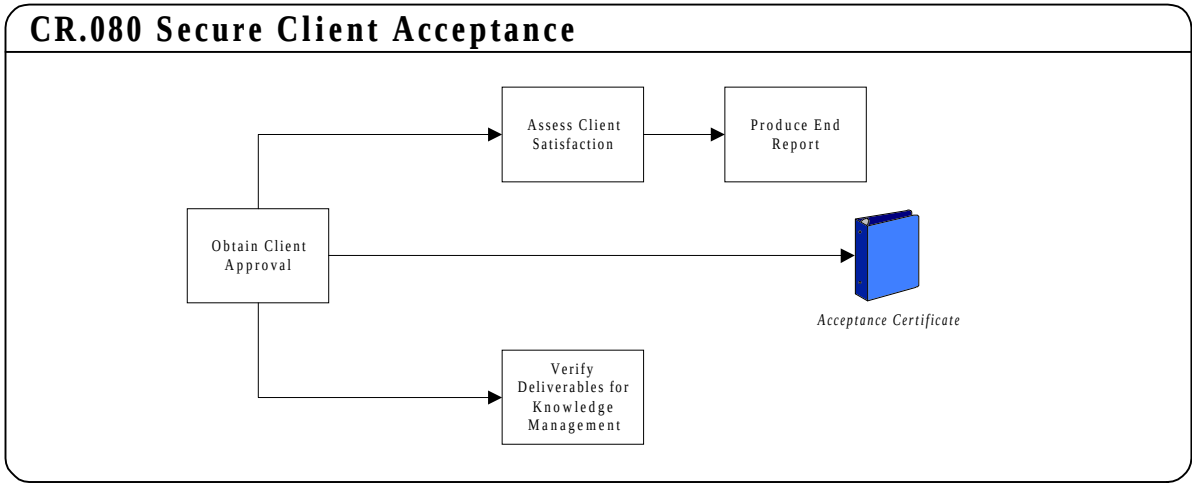


Figure 1-13 Task Flow for Secure Client Acceptance

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Obtain client approval.	Acceptance Certificate
2.	Verify deliverables for Knowledge Management.	Candidate Deliverables for Reuse
3.	Assess client satisfaction.	Client Satisfaction Report
4.	Produce an end report.	End Report

Approach and Techniques

Obtain Client Approval

In this task step you reach agreement with the client that the effort is complete. The client includes key stakeholders in the client organization, such as the client project manager, project sponsor, and steering committee.

- Prepare for the acceptance review meeting.
- Hold one or more client meetings to agree on acceptance.
- Complete the acceptance certificate with the client.



Attention: It is essential that you agree on the acceptance criteria with the client at the beginning of the project. Acceptance of deliverables should be ongoing throughout the project such that the final acceptance is just a formality.

Verify Deliverables for Knowledge Management

To verify deliverables for knowledge management:

- At each major milestone, the end of each phase, and at the end of the project, identify the deliverables which are to be candidates for assessment by the Knowledge Management task (CM.035). Consider scheduling a knowledge review with the responsible consulting practice.



Attention: Scheduling and conducting a knowledge review helps the project manager facilitate gathering reusable deliverables and helps ensure that they are properly cataloged. Knowledge reviews also allow the project team to become aware of similar projects and the potential use of intellectual capital from those projects.

- Review the Project Management Plan with the Client Project Manager to confirm intellectual property rights to the project deliverables.

- Review the Knowledge Management Procedure with the client to ensure that it is in compliance with the Project Management Plan and client expectations regarding deliverable cleansing required for intellectual capital release.

Assess Client Satisfaction

In this task step, gather valuable quality and marketing information by surveying client satisfaction. This step can be performed at any time but typically occurs at the end of a project. If it is formally conducted, then it may be more appropriate to use an external party to conduct it in order to ensure an objective analysis.

- Tailor the questions to reflect the objectives of the project.
- Reach agreement with the client as to who the respondents of the survey will be.
- Send out a questionnaire or interview the respondents.
- Analyze the responses.
- Prepare the draft satisfaction report.
- Discuss the results internally.
- Discuss the results with the client.

Produce End Report

In this task step, document the lessons learned during the project. The End Report includes these sections:

- Metrics (time and cost for modules and, if possible, for each technology area, such as Forms 4.5)
- Strengths (areas which proved useful and repeatable)
- Weaknesses (areas to be done differently and why they should be performed differently)
- Problems (with specifications, system test, acceptance test, and residual problems)
- Other information (staffing issues, platform and tools issues, technology issues, and location issues)
- Marketing (could it become a reference site, is the business problem a commonly occurring one, etc.)

- Knowledge Management (components that were reused from the consulting Knowledge Repository and the value and time saving benefit derived from reusing these components)

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Conduct client satisfaction survey, prepare the end report, and secure client approval. Ensure that all candidate deliverables for reuse are identified. Advise consulting management when deliverables should be considered for a knowledge review.	75
Project Support Specialist	Compile logs, records, and statistics. Prepare client satisfaction report, and end report documents. Schedule knowledge reviews. Prepare the acceptance certificate.	25
Client Project Manager	Coordinate client response to satisfaction assessment and approval. Discuss and contribute material to the end report. Confirm intellectual property rights and deliverable cleansing procedures.	*
Project Sponsor	Sign the acceptance certificate.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Client Acceptance are:

- ☐ **Client Satisfaction Report**
- ☐ **End Report**
- ☐ **Acceptance Certificate**

Component Descriptions

This section describes each component.

Client Satisfaction Report

The Client Satisfaction Report promotes the survey and formal review of client satisfaction with the project to date. The Client Satisfaction Report is an effective tool in uncovering client dissatisfaction concerns which could hinder acceptance if not addressed early. It is normally undertaken at the end of a project or completion of a key project milestone, but can be used at any time to provide the consulting business manager with information about the progress of the project.

The survey and report can be conducted informally on small and medium sized projects. On larger projects, it may be more appropriate to combine these with a quality audit from an external organization. In this way, it can be used as part of an ongoing practice quality system.

End Report

The End Report is the means by which the project conveys key project lessons learned to the consulting practice. The End Report should have the following sections:

- Metrics (time and cost for modules and, if possible, for each technology area, such as Forms 4.5, etc.)
- Strengths (areas which proved useful and repeatable)
- Weaknesses (areas to be approached differently and why they should be performed differently)

- Problems (with specifications, system test, acceptance test, and residual problems)
- Other information (e.g., staffing issues, platform and tools issues, technology issues, and location issues)
- Marketing (could it become a reference site?, is the business problem a commonly occurring one?, etc.)

Acceptance Certificate

The Acceptance Certificate is used to close a part of the project, on the basis that the client has accepted the completion of the work.

Describe the Deliverable (can be multiple lines). Include the following statement:

“The above deliverable has been reviewed by <Customer Long Name> and fully meets the objectives expressed by <Customer Short Name> and <Consultant Name> and passes the acceptance criteria specified by <Customer Short Name>.”

Conditional acceptance, where there may be some problems outstanding that will be fixed in a defined timescale, is a common variant of acceptance catered to in the design of the Acceptance Certificate. You may want to use either of the following paragraphs if partial acceptance is to be recorded:

“Acceptance will be complete when the Problem Reports attached have been cleared.”

ACCEPTANCE CERTIFICATE	
Ref No: ____ / ____ / ACC / ____	
Client: <Customer Long Name>	
Project: <Project Name>	
Initiated by: <Requester> Date: <Date>	
Deliverable Reference: <Deliverable Reference>	Type: ☐ Proposal ☐ Plan ☐ Specification ☐ Form ☐ Manual ☐ Other _____
Deliverable Description: _____ <Deliverable description - can be multiple lines>	
The above deliverable has been reviewed by <Customer Long Name> and fully meets the objectives expressed by <Customer Short Name> and<Consultant Name> and passes the acceptance criteria specified by <Customer Short Name>.	
Acceptance will be complete when the Problem Reports attached have been cleared*.	
Problems during testing have shown that acceptance is not complete. I undertake to inform <Consultant Name> of these problems in writing within two working days, or else I understand that acceptance will be deemed complete*.	
Signed for <Customer Short Name>	Date
_____	_____
_____	_____
Signed for <Consultant Name>	Date
_____	_____
_____	_____

Figure 1-14 Acceptance Certificate

Audience, Distribution, and Usage

Client Acceptance components are used in the following ways:

- to gain acceptance of the completion of any deliverable or milestone activity, or at the end of the project
- to communicate the client's overall assessment of the quality of the project products and services
- to communicate lessons learned from the project which would improve the consulting practice's ability to deliver future projects

Distribute the components as follows:

- distribute the Client Satisfaction Report to the consulting business manager and then to the client
- distribute the End Report to the consulting business manager for review and forwarding to appropriate consulting practice improvement organizations
- distribute the Acceptance Certificate to the client project manager to accept the components and agree on the scope, detail, and quality of the deliverable

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Check the accuracy of the Acceptance Certificate.
- ☐ Use a conditional variant of the Acceptance Certificate when full acceptance is not immediately possible.
- ☐ Ensure that assessments of client satisfaction are performed objectively, if necessary, by a third party.

The End Report should be prepared while project resources are still available to provide accurate qualitative content.

CHAPTER

2

Work Management (WM)

This chapter describes the Work Management process.

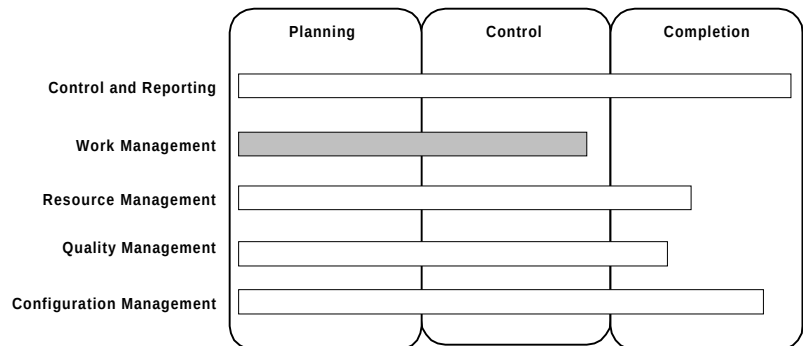


Figure 2-1 Work Management Context

Process Flow

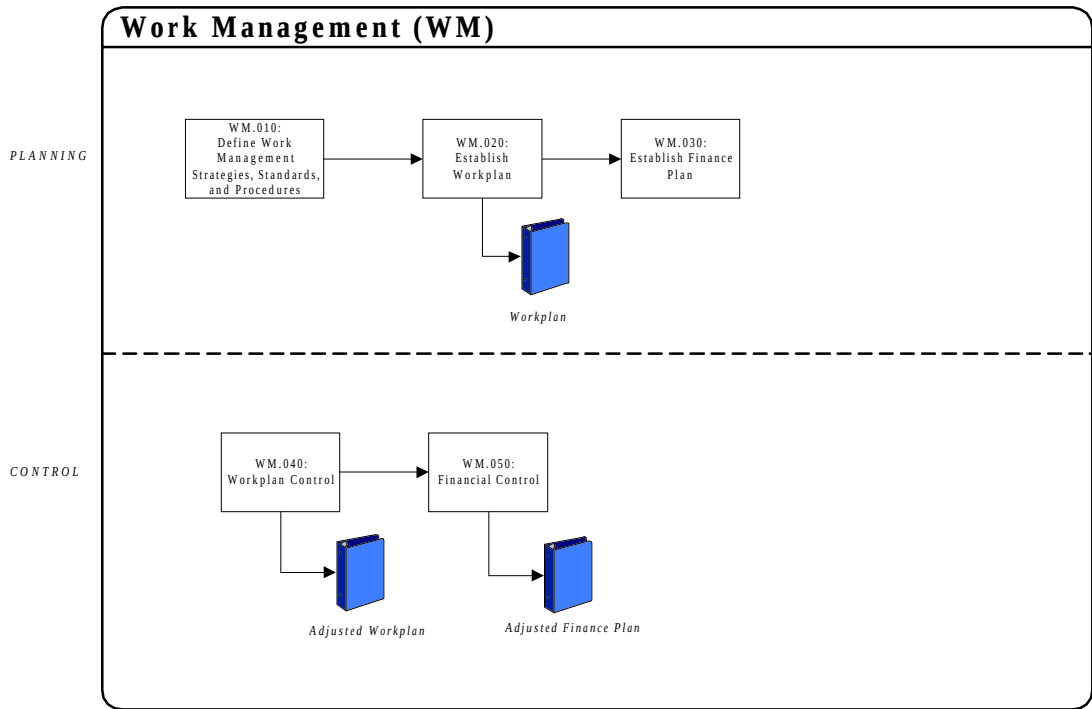


Figure 2-2 Work Management Process Flow Diagram

Approach

The Work Management process emphasizes planning. Planning is a continuous process during the entire life-cycle of the project.

Work Management planning tasks establish a baseline Workplan in accordance with the accepted Scope, Objectives, and Approach. Obtaining understanding and approval from consulting and the client for your WM plans are critical. Your Workplan will demonstrate to the client the thoroughness of your planning and preparation and provide tangible verification of the Scope, Objectives, and Approach of the project. Your client will likely have responsibility for resources or tasks on which your Workplan depends. The approval of the Workplan ensures you have commitments from your client for this support.

During project execution, the Workplan you have prepared will be used regularly by you and your project members to monitor, control, report, and adjust the project. Coordinate a regular control cycle for the Workplan based on the requirements of the Control and Reporting (CR) process. Continuous, careful monitoring and re-planning of the Workplan enables you to spot and respond to trouble early. Additionally, it allows you to promote progress information for your Work Progress Statement.

The consulting practice responsible for the project and the client will require that you maintain a clear picture of your project's financial condition. The Finance Plan established during planning will verify that the financial objectives set forth for the project during the bid are still achievable. This plan is simply an expanded view of the Workplan that accounts for items such as non-labor costs and payment schedules. During project execution, translate Workplan progress into labor hours and expenditure statistics. Report these statistics to the consulting practice and client according to Work Management procedures.

Tasks and Deliverables

The tasks and deliverables for this process follow:

ID	Task Name	Deliverable Name	Type*
Planning			
WM.010	Define Work Management Strategies, Standards, and Procedures	Work Management Strategies, Standards, and Procedures	IT
WM.020	Establish Workplan	Workplan	IT
WM.030	Establish Finance Plan	Finance Plan	IT
Control			
WM.040	Workplan Control	Adjusted Workplan, Work Progress Statements	O
WM.050	Financial Control	Adjusted Finance Plan, Financial Progress Statements	O

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary in the *PJM Method Handbook*.

Objectives

The objectives of the Work Management process are:

- Define the organization of, and dependencies between, tasks to be performed.
- Identify key milestones of the work and ensure that they are consistent with the scope, objectives, and approach.
- Establish and maintain a realistic work schedule, taking into account expected staff availability and assignments.
- Verify that the expected financial return to consulting is within practice guidelines, when risk is considered.

Key Deliverables

The key deliverables of this process are:

Deliverable	Description
Workplan	A network of interdependent tasks representing all work, with staff work assignments and schedules consistent with the scope, objectives, and approach.
Adjusted Workplan	Ongoing updates to the Workplan which contain the current work status in detail and the current plan to complete the project.
Adjusted Finance Plan	Ongoing updates to the finance plan which contains the current financial status in detail and the current financial estimate to complete the project.

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Consulting Business Manager	Review and approve the Workplan and Finance Plan with respect to profitability, risk, and resource requirements.

Role	Responsibility
Project Support Specialist	Assist in preparing review materials; file signed off Workplan after review; file updated Finance Plan. Prepare and maintain all process deliverables.
Project Manager	Review and approve all work management deliverables and present them for approval to client and consulting management. Delegate work planning and control tasks to subordinate leaders and assign them corrective actions to maintain control of the Workplan.
Client Project Manger	Review and approve the Workplan and commit client resources to the plan.
Project Sponsor	Review and approve the Workplan in the context of scope, objectives, and approach, and approve client resource requirements.

Critical Success Factors

The critical success factors of the WM process are:

- The Workplan is organized according to Oracle Method routes, with adequate and appropriate risk contingencies and mitigation strategies.
- The client accepts the Workplan and Finance Plan in the context of the project's scope and risk assessment, and consistently provides client staff resources according to the Workplan.
- The project manager and client project manager execute a disciplined, thorough review of the Workplan and Finance Plan on a weekly basis, and take aggressive actions to correct problems as soon as they are identified.
- Consulting management actively participates in the management of risk, and commits adequate resources to execute the Workplan.

WM.010 - Define Work Management Strategies, Standards, and Procedures

In this task you create or update the plans which define how the Work Management process will support and implement the Project Management Plan.

Deliverable

The deliverable for this task is the set of **Work Management Strategies, Standards, and Procedures**. These include time and expense reporting, work and financial planning, controlling, monitoring, and status reporting.

Prerequisites

Required

You need the following input for this task:

☐ **Scoping Project Management Plan (CR.010)**

This provides the requirements context for defining Work Management Standards and Procedures.

☐ **Policies**

Examine applicable client and consulting policies regarding WM and document how they apply to the project.

Optional

You may need the following input for this task:

☐ **Project Management Plan (CR.010)**

Use the Project Management Plan from previous planning as a baseline for this task, if one exists.

Task Steps

WM.010 Define Work Management Strategies, Standards, and Procedures

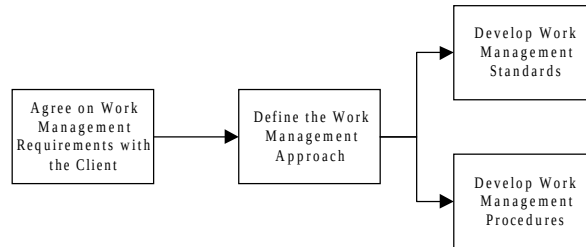


Figure 2-3 Task Flow for Define Work Management Strategies, Standards, and Procedures

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Agree on work management requirements with the client.	Work Management Requirements
2.	Define the work management approach.	Work Management Strategy
3.	Develop work management standards.	Work Management Standards
4.	Develop work management procedures.	Work Management Procedures

Approach and Techniques

Agree on Work Management Requirements with the Client

- Review the Scope, Objectives, and Approach section of the Project Management Plan.
- Review any consulting policies affecting Work Management.
- Review any client policies affecting Work Management.

Define the Work Management Approach

- Identify consulting best practices and standards.
- Document Work Management strategy.

Develop Work Management Standards

Document the standards to be followed by the project in progress monitoring, re-planning, and reporting of work and finances.

- Identify any existing consulting standards for work management.
- Customize standards to satisfy approach.
- Add additional standards to meet client requirements.
- Integrate where necessary with other standards.

Develop Work Management Procedures

Document the procedures to be followed in planning and monitoring work and for financial status reporting. Obtain examples of time collection procedures from the corporate library and modify to accommodate this engagement. The time collection process will impact client billing and expense estimates since it determines when time is reported.

- Write detailed instructions for time recording, work progress status reporting, monitoring, and re-planning of project work.
- Develop the time reporting deliverable. Include the necessary level of detail required to support the expense estimation process.
- Write detailed instructions for financial tracking.

- Review the time collection procedures with the project team.
- Implement the time collection procedures.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Support Specialist	Prepare strategies, standards, and procedures.	90
Project Manager	Review and approve all strategies, standards, and procedures.	10
Client Project Manager	Agree to work management requirements and identify client policies affecting approach.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Work Management Strategies, Standards, and Procedures are:

- ☐ **Time Recording**
- ☐ **Workplan Control**
- ☐ **Financial Control**

Component Descriptions

This section describes each component.

Time Recording

This component specifies the arrangements for time recording, including the use of a timesheet to record details about work performed, holidays, training days, etc., and how often these are to be completed. It also defines responsibilities for providing time details, including how often the Workplan and Finance Plan will be updated with actual time spent.

Workplan Control

This component defines how work is planned through the Workplan and updated using time reporting procedures to record actual time spent on tasks and estimates to complete. The procedures for work progress reporting are defined here. Specific organizational responsibilities for work planning are also defined here.

Financial Control

This component defines how the Finance Plan will be used to track and report financial information for management control during the life of the project.

Audience, Distribution, and Usage

The Work Management Strategies, Standards, and Procedures are used in the following ways:

- to provide Work Management input to the Quality Plan
- to detail the approach to the various topics for medium-to-large projects
- to define common ways of working for the project team

Distribute the Work Management Strategies, Standards, and Procedures as follows:

- to the project manager and team leaders for review
- to the Project Library for control

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are the procedures clearly defined?
- ☐ Have you taken into account the impact of dependent tasks from other processes?
- ☐ Have you changed tasks and associated deliverables which do not pertain to your project? Have you consistently changed them in the other template deliverables?

WM.020 - Establish Workplan

In this task you determine, organize, estimate, link, and schedule all of the tasks representing the work to be performed on the project. This task can be performed with varying degrees of scope, rigor, and detail, depending on the circumstances in which it is used. It can be used for forecasting, estimating, planning and impact analysis.

Deliverable

The deliverable for this task is the **Workplan**.

Prerequisites

Required

You need the following input for this task:

☐ **Scoping Project Management Plan (CR.010)**

This is the primary prerequisite of this task. It is used to determine the appropriate WBS (phases, tasks, deliverables, milestones), Task Dependency Network, Project Timeline, and Resource Requirements.

☐ **Project Management Plan (CR.010)**

The Project Management Plan identifies the quality audits and reviews, and the Healthchecks that are required for the project. This is also used to determine the appropriate WBS.

☐ **Risk Assessment (CR.010)**

The Risk Assessment determines the project complexity and estimates the effort required for contingency and how to apply the contingency against the Workplan.

Optional

You may need the following input for this task:

Workplan (WM.020)

The existing Workplan from previous planning is the baseline for subsequent re-planning.

Task Steps

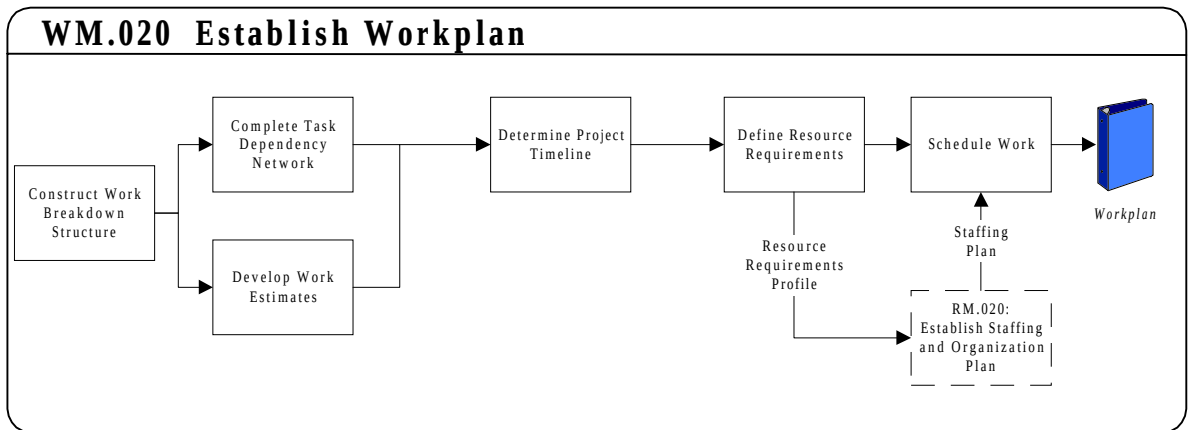


Figure 2-4 Task Flow for Establish Workplan

The steps of this task follow:

No.	Task Step	Deliverable Component
1.	Construct the work breakdown structure.	WBS
2.	Complete the task dependency network.	Task Dependency Network
3.	Develop work estimates.	Work Estimates
4.	Determine the project timeline.	Project Timeline

No.	Task Step	Deliverable Component
5.	Define resource requirements.	Resource Requirements Profile
6.	Schedule work based on expected resources.	Work Schedule

Approach and Techniques

These instructions assume that the person who performs this task will be using a project planning and management tool such as ABT Project Bridge Modeler and ABT Project Workbench.

Construct Work Breakdown Structure

Develop an organization of all work to be performed, and how that work supports the initial Project Management Plan and milestones.

- Create a WBS by either selecting the appropriate Oracle Method route (or routes) for the project approach (or approaches) being employed on the project or by defining the appropriate phases, activities, tasks, deliverables, and roles with the percentage of effort to perform the work being planned.
- Add any additional project levels necessary to suit the project Scope, Objectives, and Approach, and Project Management Plan.
- Assign the appropriate roles with the percentage of effort to complete the additional project tasks.
- Tailor the WBS at each level to reflect project specific requirements.



Suggestion: Remove tasks from your WBS only when they are clearly not within the scope or required to be managed by the Workplan.

Complete Task Dependency Network

Complete the dependency network of tasks and identify key milestones which represent strategic points at which project objectives will be certified.

- Verify, create, or update task to task dependencies within the WBS.
- Verify the deliverables for each task are consistent with the project Scope, Objectives, and Approach.
- Create milestones which are consistent with the Scope, Objectives, and Approach.



Suggestion: Avoid adding dependencies merely to reflect a desired sequencing of tasks if there is no true deliverable dependency between them.

Develop Work Estimates

Estimate the work effort required for each task and role based on guidelines within the Method Handbook for the method being employed.

- Determine the project complexity.
- Verify or create estimating formulae for each task in the WBS.
- Assign values to all estimating factors.
- Calculate estimates for each task and role within the WBS.
- Check estimates by performing sensitivity analysis or using an alternate estimating technique if necessary.
- Apply contingencies for risks that are not quantifiable.
- Repeat above until satisfied to establish confidence intervals for the estimate.

Determine Project Timeline

Determine a project timeline that will achieve the desired overall project schedule. Calculate a project timeline which achieves the target duration of the project as defined by the Scope, Objectives, and Approach by determining (and assigning) the number resources required for each key role.

Define Resource Requirements

Compile a profile of staff and client resources required by role which will achieve the Scope, Objectives, and Approach based on the Project Timeline.

- Assign role placeholders, for example Analyst 1 - Principal, Analyst 2 - Senior, Designer 1 - Principal, Designer 2 - Senior to each resource name. Allocate to each task based on the number of resources allocated to each role as documented in the Project Timeline.
- Analyze the resource requirements and adjust (level) the Project Timeline and Work Estimate to determine a realistic utilization profile.
- Prepare a profile of resource requirements by listing each resource, start date, end date, effort, role (or roles) and by documenting the responsibilities and skills required.

Schedule Work

The Staffing Plan from Resource Management is needed to begin this task step. Apply the Staffing Plan to the Project Timeline and determine one or more achievable workplans which are candidates for acceptance by consulting and client management.

- Review the Staffing Plan and determine expected or actual staff availability.
- Validate and adjust the Staffing Plan based on the Project Timeline, Resource Requirements Profile, and staff availability.
- Assign resource names to replace the role placeholders; update the resource categories (consulting grade levels) if necessary, based on the staffing plan.
- Generate and adjust the work schedule until a best schedule solution is achieved which conforms to available staff and client resources.



Suggestion: If you plan on tracking financial progress through the workplan, then enter the appropriate fee rates for the resources based on their resource categories (consulting grade levels).

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Perform all task steps.	90
Project Manager	Review and approve deliverable components.	10
Client Project Manager	Review and approve deliverable.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Workplan are:

- ☐ **Work Breakdown Structure**
- ☐ **Task Dependency Network**
- ☐ **Work Estimate**
- ☐ **Project Timeline**
- ☐ **Resource Requirements Profile**
- ☐ **Work Schedule**

Component Descriptions

The following components represent the products of the steps used in the development of the Workplan, and will normally be produced using a work management tool.

Work Breakdown Structure (WBS)

This is an organization of project tasks into a hierarchy for scheduling and reporting progress. Tasks are based on those contained in appropriate Oracle Method route(s) for the method approaches being employed on the project.

Task Dependency Network

The task dependency network identifies the dependencies between tasks in the WBS.

Work Estimate

The work estimate is the estimate of the work to be undertaken, by task and by resource, using established estimating factors and algorithms.

Project Timeline

The project timeline is the number of resources required, by key roles, to achieve the desired overall project schedule. Milestones are incorporated into the Project Timeline.

Resource Requirements Profile

This component documents the responsibilities and skills required for each resource on the project. It lists the resource name or a role placeholder (e.g., Analyst 1, Analyst 2, Designer 1, Designer 2, etc.), start date, end date, effort, staff grade level/job title, role(s), responsibilities, and the skills required. This component is used by the Resource Management process to acquire the appropriate resources when they are needed.

Work Schedule

This applies the timeline to allocated resources to create a timetable of task execution.

Audience, Distribution, and Usage

The Workplan is used in the following ways:

- to plan the tasks and deliverables, phases, and milestones
- to identify the appropriate resources needed on the project
- to schedule resources
- to monitor progress by entering time information or estimates to complete

Distribute the Workplan as follows:

- to the project manager for assessment
- to the client project manager to accept the plan's milestones and client resource requirements
- to the project members and team leaders for their assigned tasks

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Is the plan consistent with an Oracle Method project approach?
- ☐ Have all appropriate tasks and deliverables been identified?
- ☐ Are milestones and phase end points clear?
- ☐ Are the task estimates based on appropriate estimating formulas and guidelines?
- ☐ Have quality reviews, quality audits, and training tasks been included?
- ☐ Does the project timeline and work schedule satisfy the project Scope, Objectives, and Approach?
- ☐ Have adequate contingencies been included based on the risk assessment?
- ☐ Are the appropriate resources allocated and smoothed to fit a sensible plan, with peaks and troughs avoided?

WM.030 - Establish Finance Plan

In this task you determine the expected margin of profit and financial schedules that reflect the anticipated revenue, expenses, costs, and payments to the project over its lifetime. The financial schedules you create in this task will serve as the basis for reporting project financial health to management on a regular basis. In addition, you will establish how the expense information is collected during the project.

Construct the Finance Plan according to the relationship between consulting and the client, and the type of contractual agreement that exists for the project. Use the task steps and components listed here as a guide.

Deliverable

The deliverable for this task is the **Finance Plan**.

Prerequisites

Required

You need the following input for this task:

☐ **Workplan (WM.020)**

The Resource Requirements Profile component contains the effort by resource or role placeholder that is used to create the Work Effort Distribution and to perform Profitability Analysis. The Project Timeline contains the milestone dates used in developing the Payment and Cost Schedules.

☐ **Project Orientation Guide (RM.025)**

The Project Orientation Guide establishes the travel and living expense policies for the project team members for this engagement. This document will be used as a reference for quantifying these expenses during the project.

☐ **Cost of Staff, Physical Resources, and Non-labor Expenses**

Used to price the resources and to estimate project expenses and perform profitability analysis. Staff costs are obtained from the consulting practice. The Physical Resource Plan from Resource Management provides the cost of physical resources.

☐ **Consulting Business Practices Policy**

This contains contract and agreement policies and administrative policies, procedures, and guidelines that govern the content and organization of the Finance Plan deliverable.

Optional

You may need the following input for this task:

☐ **Finance Plan (WM.030)**

An existing Finance Plan from previous planning is the basis for any subsequent re-planning.

Task Steps

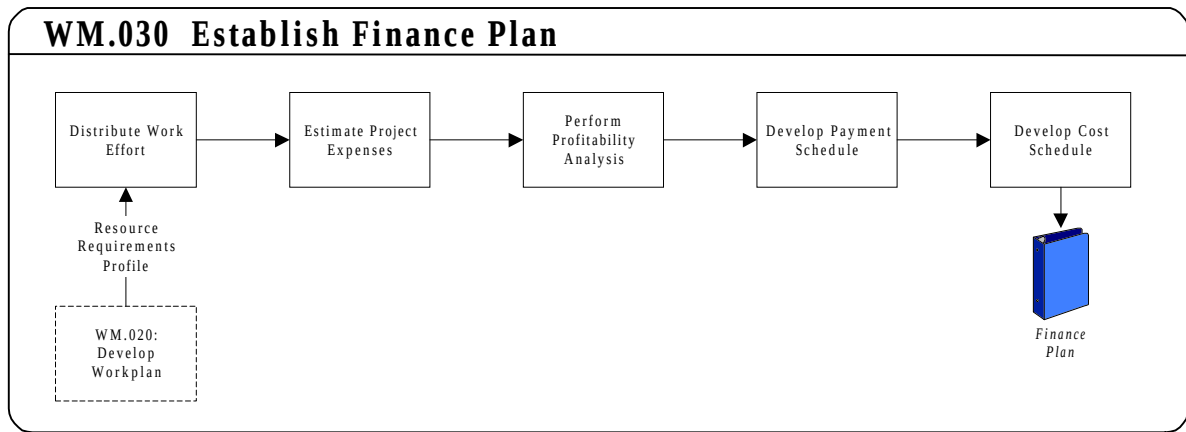


Figure 2-5 Task Flow for Establish Finance Plan

The steps of this task follow:

No.	Task Step	Deliverable Component
1.	Distribute work effort to consultant grade levels, subcontractors, and the client.	Work Effort Distribution
2.	Estimate project expenses.	Estimated Expenses
3.	Perform profitability analysis.	Profitability Analysis
4.	Develop payment schedule.	Payment Schedule
5.	Develop cost schedule.	Cost Schedule

Approach and Techniques

Distribute Work Effort

Distribute the work by role or resource to the appropriate consultant grade level, subcontractors, and client based on Scope, Objectives, and Approach and the amount of effort within the Workplan for each role or resource.

- Distribute the work effort by role or resource from the Resource Requirements Profile to the appropriate consultant grade levels, subcontractors and the client.
- Calculate the total effort for each consultant grade level, subcontractor, and client.

Estimate Project Expenses

Estimate the project expenses over the duration of the project with the expected dates on which they will occur.

- Itemize the billable and non-billable project expenses to be incurred monthly, by listing each expense item and the date that the expense will be incurred over the life of the project. The Physical Resource Plan from Resource Management provides the cost of physical resources.
- Calculate the total of the itemized billable and non-billable expenses.

Perform Profitability Analysis

Determine the fees (revenue), costs, and expected margin of profit for the effort based on the information within the Work Effort Distribution and Estimated Expenses.

- Transfer the work effort by consulting grade level from the Work Effort Distribution or determine the plan workdays (effort) for each consultant grade level and subcontractor.
- For fixed price projects, determine and apply the appropriate contingency factor.

- Determine the actual fee rates and labor fees to be charged for each grade level, subcontractor on the project, by determining the appropriate rate structures and discount percent to be applied to the standard fee rates or optionally, determine the total contract value for a fixed price project.
- Determine the labor costs for each grade level and subcontractor by applying an appropriate cost rate.
- Transfer the itemized monthly billable and non-billable expenses for non-labor costs from the Estimated Expenses or determine the expenses.
- Estimate the amount of billable expenses to be reimbursed by the client, including any markup.
- Calculate the total plan days, plan fees, plan costs, net expenses, and direct project costs and fees.
- Calculate the average daily fee and cost rates , multipliers, and overall effective daily fee and cost rates.
- Determine the appropriate base salary multiplier.
- Determine the target practice utilization factor.
- Calculate the overhead (practice non-utilization, division, corporate) factors and costs using the practice utilization factor.
- Document any license revenue and expenses associated with the project.
- Calculate the total project cost, practice margin, division margin, net contribution margin value and percentage.

Develop Payment Schedule

Create a schedule of expected payments to the project based on the information contained within the Estimated Project Expenses and the Profitability Analysis.

- For a time and material (T&M) engagement:
 - List each consultant grade level, rate, and estimated effort (in days), calculate the estimated labor fees, along with a total estimated effort and labor fees.

- Transfer the estimated billable expenses from the Estimated Expenses or determine the expenses.
- Optionally, list the resources, and document the payment terms.
- For a Fixed Price engagement:
 - List each project phase, payment number, the payment milestones / description of the payment, date payments are due, and payment amounts.
 - Transfer the estimated billable project expenses from the Estimated Project Expenses or determine the expenses.
 - Calculate the total of the project payments and estimated billable expenses and document the payment terms.



Suggestion: Ideally, you should schedule fixed price payments representing progress that are due monthly, to limit any cash flow issues. If payment dates are as occurs for the payment milestone, then you must document how to determine or calculate when the payment milestone is reached in the payment terms.

Develop Cost Schedule

Create a monthly schedule of fees, costs, and margin over the duration of the project to set the financial baseline (budget) of the project. The cost schedule will be constructed from the information contained within the Profitability Analysis and Payment Schedule and will be used by the Workplan Control ongoing task to track the financial progress of the project.

- List the budgeted effort, labor fees (revenue), labor costs, billable and non-billable project expenses, project expense reimbursement, overhead costs (practice non-utilization, division, corporate), margin value, margin percentage, number of deliverables, and project milestones for each phase, and month over the duration of the project.
- Calculate the total budgeted effort, labor fees, labor costs, billable and non billable project expenses, project expense reimbursement, margin value, margin percentage, and the number of deliverables for the project.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Perform all task steps.	90
Project Manager	Review and approve deliverable.	10
Client Project Manager	Review and approve applicable components.	*
Consulting Business Manager	Provide staff costs.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Finance Plan are:

- ☐ **Work Effort Distribution**
- ☐ **Estimated Expenses**
- ☐ **Profitability Analysis**
- ☐ **Payment Schedule**
- ☐ **Cost Schedule**

Component Descriptions

This section describes each component.

Work Effort Distribution

This component provides a distribution of work (effort) by role or resource across the consulting grade levels, subcontractors, and the client.

The component will be used to convert estimated days by role or resource from the project plan to estimated days by consulting grade level and subcontractor. The result is then used to determine the Profitability Analysis component. Alternatively, estimated days by grade level/subcontractor may be input directly into the Profitability Analysis component.

Estimated Expenses

This component itemizes the estimated non-labor project expenses over the duration of the project with the expected dates on which they will occur. Estimated expenses are divided into three categories:

- Incidental travel and living (T&L) expenses: airfare, hotel, meals, taxi fares, mileage, etc.
- Physical resources: hardware, software, facilities, office supplies, etc.
- Other: training, etc.

Estimated expenses include:

Date: the month and year in which the expense item is to be incurred.

Expense item: a short description of the expense item.

Number of items: the number of expense items to be incurred.

Rate: the rate charged for each expense item.

Unit: the unit of measure for each expense item, e.g., “monthly” for T&L expenses, “copies” for software, “each” for hardware, etc.

Extended cost: # of items x rate.

Non-billable project expenses: the project expense amount for each item to be incurred that are *not* billable to a client.

Billable project expenses incurred: the project expense amount for each item to be incurred that *are* billable to a client.

Net project expenses: the sum of the non-billable and billable project expenses for each item to be incurred.

Total non-billable, billable, and net project expenses: the sum of all billable, non-billable, and net project expense items.

Profitability Analysis

This component utilizes the information within the Work Effort Distribution and Estimated Expenses to determine the revenue and labor costs, expenses and overhead, and margin the project is expected to generate. The contents of the Profitability Analysis include:

Revenue and Labor Costs

Plan workdays: the total number of billable workdays in the project plan for each specific consulting grade level for all resources.

Contingency factor: a contingency factor may be used, primarily in the case of fixed price bids, to evaluate the effect on margin of a specific level of increase in project labor (work) effort.

Labor cost rate: this is a weighted daily labor cost rate by consulting grade level. It includes base salary cost plus burden, plus overheads associated with the local practice. It does not include divisional or corporate overheads.

Plan labor cost: labor cost for each consulting grade (number of plan workdays x labor cost rate).

Average daily cost rate: the average daily cost rate = total plan cost / total plan days.

Standard fee rate: consulting organizations standard daily fee rates, what the project should be striving to achieve.

Discount factor: discount factor to be applied to standard fee rates. This provides a quick way of computing actual fee rates and planned revenue.

Actual fee rate: proposed daily fee rate to be charged on this project for this consultant grade level.

Plan labor fees (revenue): projected fees (revenue) generated by each consultant grade level (plan workdays x actual fee rate).

Average daily fee rate: the average daily fee rate = total plan fees (revenue) / total plan days

Multiplier: in the case of individual consulting grade levels, this is the plan labor fees (revenue) divided by the plan labor cost. For labor as a whole, this is the total plan fees (revenue) divided by the total plan cost.

Base salary multiplier: a rough estimate of the average base salary multiplier for all consulting grade levels on the project.

Subcontractors: if subcontractors are included in the project, list the number of workdays estimated for each subcontractor grade level and labor cost rate. The labor costs, labor fees and related items are calculated the same way as for consulting labor.

Total plan days: this represents the sum of the consulting and subcontractor plan days for each grade level.

Total plan costs: this represents the sum of the consulting and subcontractor plan costs for each consulting grade level and subcontractor.

Total plan fees (revenue): this represents the sum of the consulting and subcontractor plan fees (revenue) for each consulting grade level and subcontractor.

Overall effective daily cost rate: an overall effective daily cost rate for the project is calculated by dividing the direct project costs by the total plan days for both the consulting grade levels and the subcontractors.

Overall effective daily fee rate: an overall effective daily fee rate for the project is calculated by dividing the direct projects fees (revenue) by

the total plan days for both the consulting grade levels and the subcontractors.

Contract value: for fixed price projects the contract value may be specified instead of a actual fee rate and planned labor fees (revenue) for each consulting grade level and subcontractor.

Direct project fees: this represents either the total plan fees for both the consulting grade levels and subcontractors, or the contract value depending if the value of the contract is specified.

Expenses and Overhead

Non-billable expenses: includes all expenses to be incurred on the project that cannot be billed to the client. Non-billable expenses only contribute to the cost of conducting the project.

Billable expenses: includes all categories of expenses to be incurred on the project that are billable to the client.

Billable expense reimbursement: the billable expenses to be reimbursed by the client. This may exceed the billable expenses if there is a markup.

Direct project costs: this represents the total direct labor and net out-of-pocket expense costs associated with the project.

Allocated overhead costs: besides the direct labor costs associated with a project, each project must also bear an equitable portion of the overall consulting organizations non-utilization, division and corporate allocated overhead costs. This overhead is calculated as a percentage of the total consulting extended labor costs.

Practice non-utilization overhead: derived from a practice utilization factor to be applied to the consulting extended labor cost to account for the consulting organizations non-utilization (practice management, business development, office space, supplies, training, vacation, & sick time, etc.) overhead.

Division overhead: derived from a division overhead factor to be applied to the consulting extended labor cost to account for the

consulting organizations divisional (division management, operations, finance, marketing, etc.) overhead.

Corporate overhead: derived from a division overhead factor to be applied to the consulting extended labor cost to account for the consulting organizations corporate (legal, MIS, human resources, etc.) overhead.

Practice utilization factor: the practice labor cost is based upon a potential number of billable workdays per year. In reality, utilization levels can be significantly lower and the true labor cost must be adjusted to include non-utilized workdays.

Total project cost: the total cost of the project including the practice non-utilization overhead and allocated division and corporate overheads.

Margin

Practice margin: margin amount and margin percentage at the practice level, excluding divisional and corporate overhead.

Division margin: margin amount and margin percentage at the division level, excluding corporate overhead.

Net contribution amount: the excess of total revenue over total costs including ALL allocated overhead.

Net contribution margin percentage: net contribution amount expressed as a percentage of total project revenue.

Payment Schedule

The payment schedule presents the rate tables and /or payment schedules (the client financial budget) over the duration of the project to the client. The Payment Schedule includes:

For Time and Material Engagements:

Rate table: consulting grade level, daily fee rate, estimated effort and labor fees for each consulting grade level. The resource names may also (optionally) be listed.

Payment schedule: budgeted labor fees and estimated billable project expenses by phase and month over the duration of the project. The project milestones / deliverables may also be listed.

Payment terms: specific conditions and assumptions on which the Payment Schedule is based.

For Fixed Price Engagements:

Payment schedule: phases, payment number, payment milestones / deliverables, date payments are due with payment amounts (labor fees and fixed project expenses), and estimated billable project expenses at each payment point over the duration of the project.

Payment Terms: specific conditions and assumptions on which the Payment Schedule is based.

Cost Schedule

The Cost Schedule presents the fees, costs and margin (the consulting financial budget) monthly over the duration of the project to the consulting practice. It utilizes selected information from the Profitability Analysis and Payment Schedule. The cost schedule contains:

- Budgeted effort, labor fees (revenue), labor costs, billable and non-billable project expenses, project expense reimbursement, overhead costs (practice non-utilization, division, corporate), margin amount, margin percentage, number of deliverables, and project milestones / deliverables by phase, month over the duration of the project.
- Total budgeted effort, labor fees, labor costs, billable and non billable project expenses, project expense reimbursement, margin amount, margin percentage, and number of deliverables.

Audience, Distribution, and Usage

The Finance Plan is used in the following ways:

- to set the financial baseline on the project
- to track financial information for inclusion in the Financial Progress Statement

Distribute the Finance Plan as follows:

- to the project manager for review
- to the Project Library for control
- to the consulting business manager responsible for the project
- the Work Effort Distribution and Payment Schedule components to the client project manager
- the Cost Schedule component to the client project manager, if cost responsibility is shared

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Is the plan traceable to the consulting proposal?
- ☐ Is the plan consistent with the WBS, project schedule, and staffing requirements?
- ☐ Is the project going to deliver required profitability?

WM.040 - Workplan Control

In this ongoing task you regularly measure the progress achieved during the project against the Workplan, take corrective actions, and prepare progress information for reporting to client and consulting management.

Prerequisites

Required

You need the following input for this task:

☐ **Workplan (WM.020, WM.040)**

Use the current Workplan to measure and report progress, and adjust the plan accordingly.

☐ **Work Management Strategies, Standards, and Procedures (WM.010)**

Work Management Strategies, Standards, and Procedures define how the Work Plan Control task will be conducted.

Task Steps

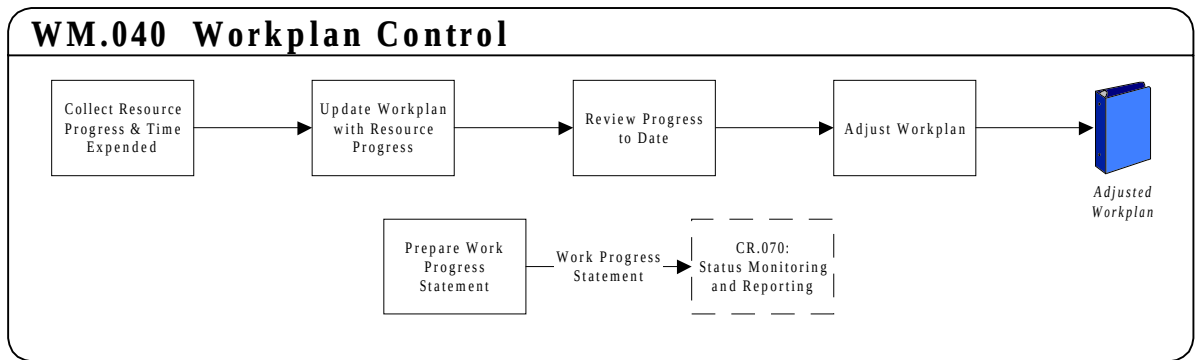


Figure 2-6 Task Flow for Workplan Control

The steps involved in this ongoing task follow:

No.	Task Step	Result or Output
1.	Collect resource progress and time expended.	Project Time Record or Timesheet, Task Progress
2.	Update the Workplan with actual resource time expended and /or task completion status information.	Progress to Date
3.	Review progress measured to date against plan, raise new management issues and risks, and assign corrective actions.	Action Items, Issues / Risks
4.	Adjust the Workplan using the Work Management procedures.	Adjusted Workplan
5.	Prepare work progress statement per procedure.	Work Progress Statement

Approach and Techniques

Collect Resource Progress and Time Expended

- Collect quantitative information on resource progress including:
 - time spent on project tasks by resource
 - a revised estimate to complete each task in progress
 - any changes to task start and end dates
 - availability of project members
- Project members submit this information according to the Time Reporting Procedure.

Update Workplan with Resource Progress

- Copy the Workplan to track progress at each control interval throughout the life of the project.
- Input the resource progress and/or time expended on each task into the project planning and management tool used on the project if not entered directly by the project team members.

Review Progress to Date

- Use the tracking Workplan as the basis for the ongoing Project Workplan status review meeting with your project leaders.
- Use the review to raise new management issues and risks and assign corrective actions, and determine any adjustments needed to the current Workplan.

Adjust Workplan

- Using the quantitative and qualitative information and the input from the project leaders during the review team meeting, adjust the Workplan.



Attention: Use the Establish Workplan task to adjust Workplan.

Prepare Work Progress Statement

Prepare Work Progress Statements based on the Workplan Control Procedure.

Roles and Responsibilities

The roles and their associated responsibilities for this ongoing task are:

Role	Responsibility	%
Project Support Specialist	Execute or manage all revisions to the Workplan and prepare progress statement.	50
Project Manager	Review progress, manage corrective actions, and approve progress statement. Submit progress reports for assigned tasks, participate in progress reviews, and execute corrective actions, as assigned.	50
Client Project Manager	Monitor progress against plan, resolve problems with client obligations.	*

* Time not estimated.

Outputs

The outputs for Workplan Control are:

- ☐ **Adjusted Workplan**
 - Time Reports
 - Task Progress
 - Updated Work Schedule
- ☐ **Work Progress Statement**
- ☐ **Action Items**

Output Descriptions

The section describes each output for this ongoing task:

Adjusted Workplan

The Adjusted Workplan is derived from the Workplan, for tracking work progress since the last baseline date. It represents an ongoing update to the Workplan, and contains the current work status, in detail, and the current plan to complete the project.

Time Reports: Project members submit time spent on project tasks, by person. This can be accomplished using a project timesheet, or can be entered directly into the work management tool being employed on the project, in accordance with the Workplan Control procedure.

Task Progress: This contains the following quantitative information collected on each task which is in progress or was completed in the period:

- time expended on each task, by project member
- a revised estimate to complete each task-in-progress
- actual task start and end dates

Updated Work Schedule: The Updated Work Schedule is the most frequently updated portion of the Workplan, although any of its components may be adjusted, if necessary. The Updated Work Schedule contains task progress and adjustments to the scheduled start and end dates of unfinished tasks, compared with the last Workplan baseline. The confidence level of task estimates to complete will normally vary inversely with the length of their planning horizon in the schedule.

Work Progress Statement

The Work Progress Statement is derived from the Adjusted Workplan, and may be either issued independently, or included in the Progress Report. It provides a summarized view of the work progress, based on the information entered into the Workplan. The Work Progress

Statement is normally produced using reports from the work management tool being employed on the project. The following information constitutes a minimum requirement for the Work Progress Statement:

- Approved days in plan
- Total days in plan
- Actual days spent
- Estimated days to complete
- Project Manager's view of percent complete
- Forecast earliest end date
- Forecast latest end date.

Action Items

During preparation and review of the Work Progress Statement any new issues, risks, or actions raised should be recorded.

Issues/Risks

These are raised during management review of work progress.

Audience, Distribution, and Usage

The Adjusted Workplan is used in the following ways:

- as the baseline of the current plan to complete the project
- to assist in the preparation of the Progress Report

Distribute the Adjusted Workplan as follows:

- to the project manager and client project manager for review
- to the Project Library as a baseline

The Work Progress Statement is used in the following ways:

- to assist in the preparation of the Progress Report
- to accompany the progress report, where specified, in the Quality Plan or Work Management procedures

Distribute the Work Progress Statement as follows:

- to the project manager for review, and inclusion in any appropriate reports
- to the Project Library for control
- to the client project manager to accept the components and agree on the scope, detail, and quality of the deliverable (where agreed that this information is useful to the client)

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Is the workplan kept up-to-date?
- ☐ Is the level of detail sufficient?
- ☐ Are there any issues or actions that need raising, such as missed milestones?
- ☐ Does the project timeline and detailed schedule still meet the project's Scope, Objectives, and Approach?

WM.050 - Financial Control

In this ongoing task you monitor financial performance during the project and prepare financial progress information for reporting to client and consulting management.

Prerequisites

Required

You need the following input for this task:

☐ **Work Management Strategies, Standards, and Procedures (WM.010)**

These contain the standards and procedures that define how the Financial Control task will be conducted.

☐ **Adjusted Workplan (WM.040)**

The Adjusted Workplan contains the labor fees (actuals, planned, and estimates to complete) for the current period.

☐ **Finance Plan (WM.030)**

Use the current Finance Plan to measure and report progress, and adjust the plan as necessary.

☐ **Financial Reports from Project Accounting System**

The current period labor fees and billable expenses and budget remaining are reconciled against the work and financial plans.

Optional

You may need the following input for this task:

☐ **Invoices and Payments**

The invoices sent and payments received from the project accounting system since the last reporting cycle are used to track outstanding invoiced revenue and revenue received. Payment information is needed only where a payment schedule is monitored.

Task Steps

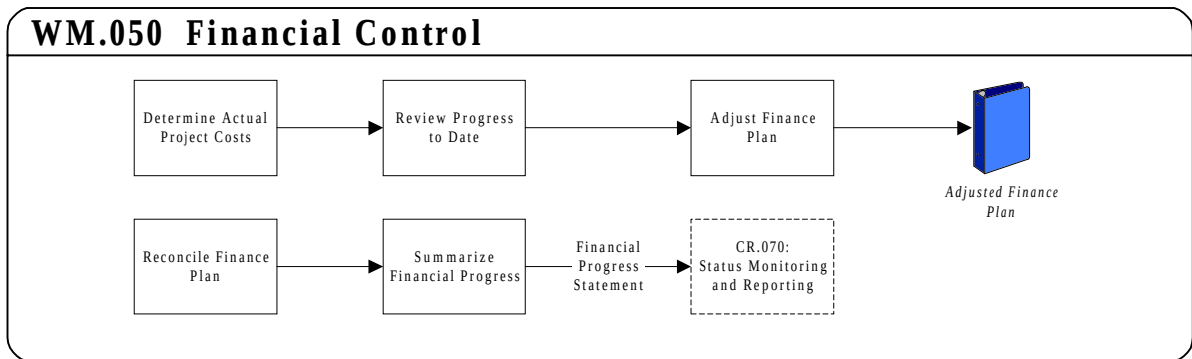


Figure 2-7 Task Flow for Financial Control

The steps involved in this ongoing task are:

No.	Task Step	Result or Output
1.	Determine actual labor fees and project costs for the current tracking period.	Financial Progress Detail
2.	Review progress measured to date against budget, raise new management issues and risks, and assign corrective actions.	Action Items, Issues / Risks
3.	Adjust Finance Plan, as needed, based on current period actuals and estimates to complete.	Adjusted Finance Plan
4.	Reconcile the Finance Plan to the internal billing statements.	Reconciled Finance Plan.
5.	Summarize financial progress statement per procedure.	Financial Progress Statement

Approach and Techniques

You can only track a portion of the project's financial progress through the workplan. The cost of labor can be recorded against each resource; therefore you can track the financial status for labor fees from the project workplan. However, you cannot track items such as non-labor expenses, invoice amounts, and payments received from the workplan. This additional financial information will need to come from a project accounting system, such as Oracle Project Accounting.

Since labor effort and fees are invoiced from the project accounting system, and monitored and reported from the workplan, reconcile the two sources of information as part of this task, prior to updating the Finance Plan and preparing Financial Progress Statements.

Determine Actual Project Costs

- You will need to collect and record the following quantitative information from the Workplan and/or financial reports to determine the actual labor fees and project costs for the current period for input into the summarized financial schedules of the project:
 - billable time (spent) by grade level and subcontractor for the current period
 - non-billable (net out-of-pocket) expenses for the current period
- Calculate the labor fees and project costs for each grade level for the current period based on the associated fee and cost rates, and overhead factors.
- Calculate the total project fees and costs for the current period.

Review Progress to Date

- The project manager should use the Financial Progress Statement as the basis for reviewing financial progress with the appropriate management team. The review will raise new management issues and risks, assign corrective actions, and determine any adjustments needed to the Finance Plan.

Adjust Finance Plan

- Adjust the Estimated Expenses if the actual non-billable (out-of-pocket) surpasses the budget, or the estimated project expenses to complete the project varies significantly from the budgeted project expenses. When making the update, be sure to update the estimated project expenses in the Profitability Analysis and Cost Schedule.
- Adjust the Profitability Analysis if the budgeted amount of work, distribution of work, fee rates, cost rates, or the value of the contract changes, or if the actuals or estimates to complete the project vary significantly from the budget.
- Adjust the Payment Schedule if the budget has changed at the client's request.
- If the Estimated Expenses, Profitability Analysis, or Payment Schedule is updated, then the Cost Schedule will need to reflect those changes.

Reconcile the Finance Plan to the Billing Statements

- To adequately review the Finance Plan with the client, you must obtain copies of the periodic billing statements. Reconcile these or match them against the engagement's Finance Plan.
- Resolve any reconciling items. Most timing differences will be resolved by the passage of time. However, any other differences must be identified, and appropriate actions taken to rectify them.

Prepare Financial Progress Statement

- Prepare the Financial Progress Statement according to procedure.

Roles and Responsibilities

The roles and associated responsibilities for this ongoing task follow:

Role	Responsibility	%
Project Support Specialist	Perform plan updates and prepare Financial Progress Statements.	70

Project Manager	Review and approve financial progress, raise issues / risks, and execute corrective actions.	30
Client Project Manager	Monitor progress against plan, and resolve problems with client obligations.	*

* Time not estimated.

Outputs

The outputs of Financial Control are:

☐ **Financial Progress Detail**

☐ **Adjusted Finance Plan**

Adjusted Estimated Expenses
Adjusted Cost Schedule
Adjusted Profitability Analysis
Adjusted Payment Schedule

☐ **Financial Progress Statement**

Labor Fee Summary
Cost and Margin Summary
Labor Fee Analysis
Cost Analysis
Margin Analysis

☐ **Action Items**

☐ **Issues/Risks**

Outputs Descriptions

The section describes each output for this ongoing task:

Financial Progress Detail

This contains all effort, labor fees, and costs (labor, net out-of-pocket expenses, overhead) incurred during the current and prior reporting periods for input into the financial schedules or financial views of the project (document below). The method of capturing the financial progress is detailed in the Financial Control Procedure, and may be based on time records, a work planning / management tool, or a project accounting system.

Adjusted Finance Plan

The Adjusted Finance Plan is derived from the Finance Plan based on the Financial Progress Statement. It represents an ongoing update to the Finance Plan, and contains the current financial status (budget), in detail, and the current financial estimate (plan) to complete the project.

Adjusted Estimated Expenses: This is an update to the estimated expenses for the project, based on current actuals and forecasts to complete.

Adjusted Profitability Analysis: The Profitability Analysis in profit-based projects is generally only adjusted when there are significant changes to the work effort distribution or estimated expenses, or as required by the current financial performance of the project. If it is necessary to revise this component, use the current estimated expenses, and work effort distribution as a basis for the updates.

Adjusted Payment Schedule: This reflects any changes, agreed to between consulting and the client, to dates, milestones / deliverables or amounts of payments for profit-based consulting agreements.

Adjusted Cost Schedule: This reflects any changes to the cost schedule for the project, based on revised profitability analysis, payment schedule, and current financial performance of the project.

Financial Progress Statement

The Financial Progress Statement is used to track financial activity since the last baseline date. It provides schedules or views of financial performance on a periodic basis, and is used to update the Finance Plan and to report financial performance. The method of reporting financial progress is detailed in the Financial Control Procedure.

Labor Fee and Deliverable Summary: This is a summary of the labor fees for the project measured monthly against the project phases and payment milestones / deliverables. It should include budgeted labor fees, actual labor fees incurred, planned labor fees, budget vs. plan, and cumulative labor fees, variances, and percent of completion. Indexes showing cumulative actuals divided by cumulative budget can be presented to quantify progress.

Cost and Margin Summary: This is a monthly summary of the project costs (labor costs, net out-of-pocket project, overhead) and margin. It should include budgeted labor costs and margin, actual labor costs incurred and margin generated, planned labor costs and margin, budget vs. plan, and cumulative costs, margin, variances, and percent of completion. Indexes showing cumulative actuals divided by cumulative budget can be presented to quantify progress.

Labor Fee Analysis: This graphically represents the cumulative budget, actuals to-date and planned monthly labor fees over the life of the project.

Cost Analysis: This graphically represents the cumulative budget, actuals to-date combined with the plan, for monthly project costs over the life of the project.

Margin Analysis: This graphically represents the budget, actuals to-date combined with the plan, for the monthly cumulative margin amount and margin percent over the life of the project.

Action Items

During preparation and review of the Financial Progress Statement any new action items raised should be recorded.

Issues/Risks

Issues and risks raised during management review of financial progress should be captured for follow-up by the Issue/ Risk Management sub-process.

Audience, Distribution, and Usage

Financial Control outputs are used in the following ways:

- to baseline, track, and report the financial performance of a project

Distribute the Adjusted Finance Plan as follows:

- to the project manager for review
- to the Project Library for baselining (as needed)
- to the Project Library for updating project progress
- to the consulting business manager responsible for the project
- the Adjusted Work Effort Distribution and Payment Schedule components to the client project manager
- the Adjusted Cost Schedule component to the client project manager, if financial responsibility is shared

Distribute the Financial Progress Statement as follows:

- to the project manager for review
- to the Project Library for baselining (as needed)
- to the consulting business manager responsible for the project
- to the client project manager in projects where the client shares financial responsibility

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Is the plan up-to-date?
- ☐ Is the level of detail sufficient?
- ☐ Is the information complete?
- ☐ Does the information raise any issues for the project, such as negative profitability or cashflow?

CHAPTER

3

Resource Management (RM)

This chapter describes the Resource Management process.

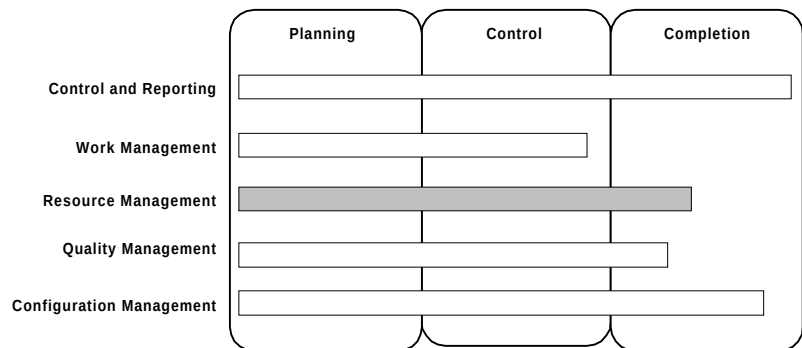


Figure 3-1 Resource Management Context

Process Flow

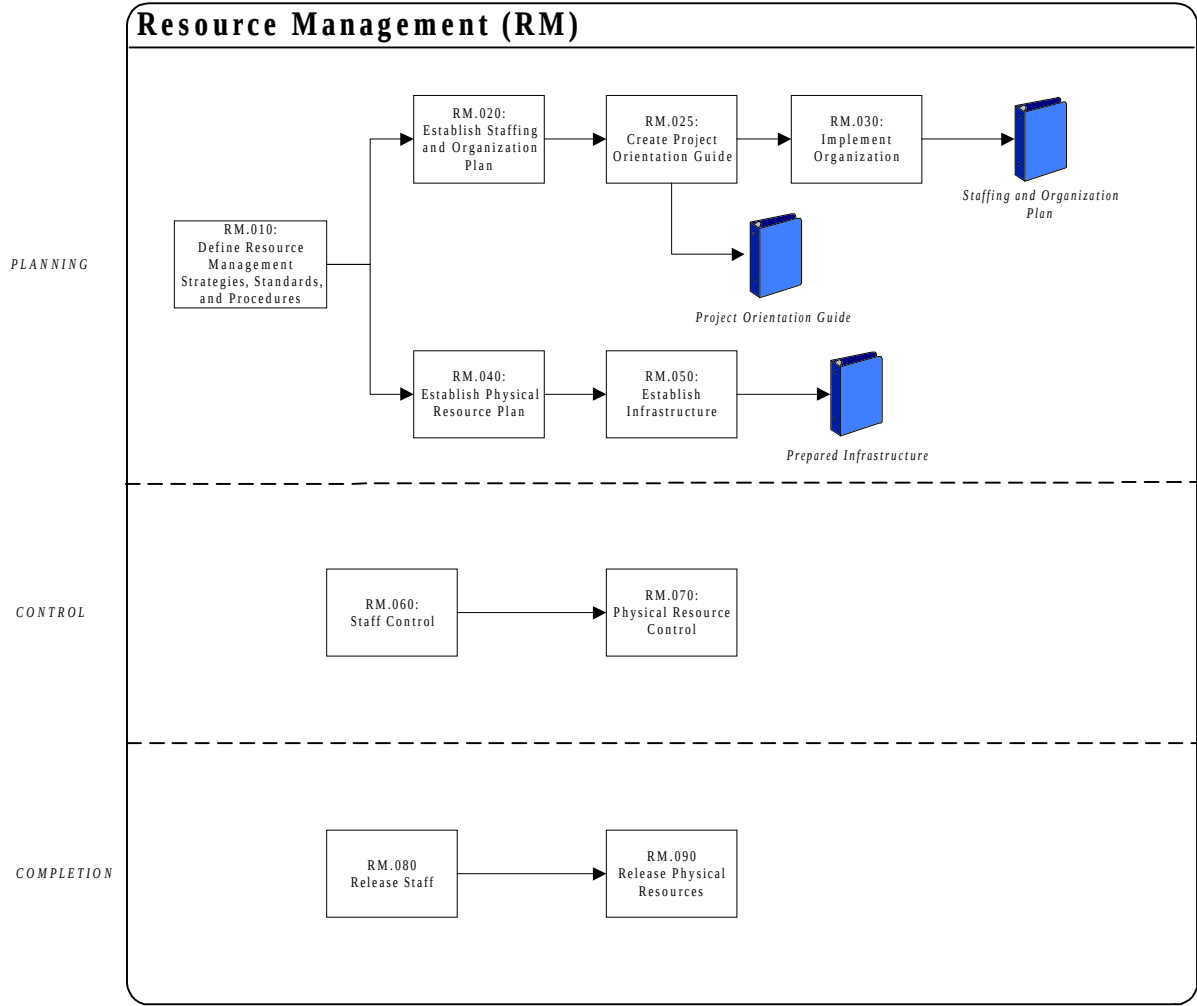


Figure 3-2 Resource Management Process Flow Diagram

Approach

The basis of staffing a project is the Staff Requirements Profile and Workplan produced by the Work Management process. Since the availability of resources will have an impact on scheduling the Workplan, there is a high level of planning interaction with the Work Management process.

Non-management processes and tasks from the chosen Oracle Method routes may also specify physical resource requirements procured through the Resource Management process. Local client and consulting policies and procedures may need to be taken into account.

Tasks and Deliverables

The tasks and deliverables for this process follow:

ID	Task Name	Deliverable or Output Name	Type *
Planning			
RM.010	Define Resource Management Strategies, Standards, and Procedures	Resource Management Strategies, Standards, and Procedures	IT
RM.020	Establish Staffing and Organization Plan	Staffing and Organization Plan	IT
RM.025	Create Project Orientation Guide	Project Orientation Guide	IT
RM.030	Implement Organization	Prepared Organization	IT
RM.040	Establish Physical Resource Plan	Physical Resource Plan	IT
RM.050	Establish Infrastructure	Prepared Infrastructure	IT
Control			
RM.060	Staff Control	Performance Appraisals, Adjusted Organization, Adjusted Assignment Terms of Reference	O
RM.070	Physical Resource Control	Installation Reports, Incoming Item Records, Equipment Release Records, Fault Reports	O

ID	Task Name	Deliverable or Output Name	Type *
Completion			
RM.080	Release Staff	Released Staff	MO
RM.090	Release Physical Resources	Released Physical Resources	MO

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary in the *PJM Method Handbook*.

Objectives

The objectives of the Resource Management process are to:

- Manage staff assignments in line with the staffing requirements and Workplan.
- Establish and maintain an appropriate infrastructure for the project, such that it can be delivered on time, within budget, and with the agreed deliverables.
- Maintain high team motivation and morale.

Key Deliverables

The key deliverables of this process are:

Deliverable	Description
Project Orientation Guide	Establishes and communicates the resource management policies and procedures for the engagement.
Prepared Organization	Confirmation from line managers of staff availability for assignments.
Prepared Infrastructure	All physical resources needed to support the project environments.

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Client Project Manager	Agree to the environment requirements and interview staff.
Consulting Business Manager	Supply staff availability.
Project Support Specialist	Draft the assignment terms of reference and research suitable resumes (CVs).
Project Manager	Manage the tasks within this process. Assess the staff performance and complete performance appraisals.

Critical Success Factors

The critical success factors of the Resource Management process follow:

- The project is adequately resourced in terms of staff and skills.
- Each team has adequate environment and equipment to deliver the project on time.
- Committed resources are not diverted by other projects.
- A constructive team atmosphere is established and maintained.

RM.010 - Define Resource Management Strategies, Standards, and Procedures

In this task you create or update the plans which define how the Resource Management process will support and implement the project scope, objectives, and approach.

Deliverable

The deliverable for this task is the set of **Resource Management Strategies, Standards, and Procedures**.

Prerequisites

Required

You need the following input for this task:

☐ **Scoping Project Management Plan (CR.010)**

The Scoping Project Management Plan is used to set the context for the Resource Management Strategies, Standards, and Procedures.

☐ **Policies**

Examine consulting policies regarding Resource Management to determine how they apply to your practice and your project. The client may have its own policies regarding Resource Management. Take these into account when developing the strategies, standards, and procedures for your project.

Optional

You may need the following input for this task:

☐ **Project Management Plan (CR.010)**

Previous planning, if any, will have produced a Project Management Plan. You can use that as a baseline for this task.

Task Steps

RM.010 Define Resource Management Strategies, Standards, and Procedures

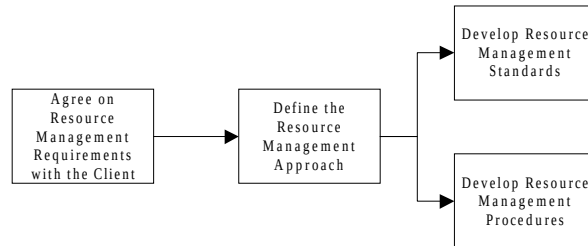


Figure 3-3 Task Flow for Define Resource Management Strategies, Standards, and Procedures

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Agree on Resource Management requirements with the client.	Resource Management Requirements
2.	Define the Resource Management approach.	Resource Management Strategy
3.	Develop Resource Management standards.	Resource Management Standards
4.	Develop Resource Management procedures.	Resource Management Procedures

Approach and Techniques

Agree on Resource Management Requirements with the Client

- Review the Project Management Plan.
- Identify any consulting policies affecting Resource Management.

There will be practice policies, such as staff terms and conditions, travel policy, frequency of appraisals, project roles, and project authorities.

- Identify any client policies affecting Resource Management

The client will have policies that affect the project, particularly where the work is to be carried out at the client site. These will include such things as working hours, access to secure areas, dress code, and car parking.

Define the Resource Management Approach

- Identify consulting best practices and standards.
- Document the Resource Management strategy.

Develop Resource Management Standards

- Identify any existing consulting standards.
- Customize standards to satisfy the approach.
- Add additional standards to meet client requirements.

Develop Resource Management Procedures

- Write detailed instructions for staff control.

These should include the induction process for staff joining the project, booking travel, allocating security passes, booking holiday time, and reporting absences.

- Write detailed instructions for physical resource control. These should include the allocation of workstations; how to obtain stationery; procedures for the receipt, maintenance, and release of equipment; and identification of responsibilities.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Prepare strategies, standards, and procedures.	70
Project Manager	Review and approve all strategies, standards, and procedures.	30
Client Project Manager	Agree to Resource Management requirements and identify client policies affecting approach. Agree on client responsibilities.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Resource Management Strategies, Standards, and Procedures are:

- ☐ Staff Roles and Organization Management
- ☐ Environment Maintenance

Component Descriptions

This section describes each component.

Staff, Roles, and Organization Management

This component defines the organization for the project and how it will be managed, including the roles and responsibilities of team members. It is an extension of the information in the Project Orientation Guide for large projects. It includes staff review scheduling and client relationship management.

Environment Maintenance

This component describes the procedure for environment maintenance, including account set up and administration, ordering of consumables, and back up arrangements, together with service request processing and fault reporting.

Audience, Distribution, and Usage

The Resource Management Strategies, Standards, and Procedures are used to:

- detail resource arrangements for the project, including physical resources
- communicate structure and responsibilities to the project team
- communicate working environment to the project team, including the client

Distribute the deliverable to:

- project manager for review
- Project Library for control
- defined distribution

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ The definition of roles extends to all project role types.
- ☐ The definition of environment includes hardware, software, and tools.

RM.020 - Establish Staffing and Organization Plan

In this task you plan how you will organize the staff to execute the project and verify that resources are available to satisfy the plan.

Deliverable

The deliverable for this task is the **Staffing and Organization Plan**.

Prerequisites

Required

You need the following input for this task:

☐ **Resource Management Strategy (RM.010)**

The Resource Management Strategy is produced in the Define Resource Management Strategies, Standards & Procedures task.

☐ **Resource Requirements Profile (WM.020)**

The Staff Requirements Profile is produced by the Work Management Process. It details the staff requirements to fulfill the Workplan in terms of grades, roles, skills, and dates.

Optional

You may need the following input for this task:

☐ **Staffing and Organization Plan (RM.020)**

A Staffing and Organization Plan may exist from previous planning.

Task Steps

RM.020 Establish Staffing and Organization Plan

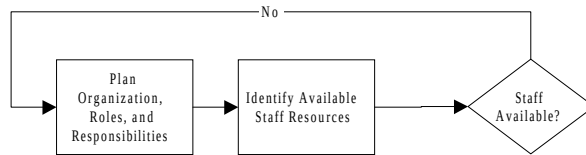


Figure 3-4 Task Flow for Establish Staffing and Organization Plan

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Plan organization, roles, and responsibilities.	Organization Plan, Assignment Terms of Reference
2.	Identify available staff resources.	Staffing Plan

Approach and Techniques

Plan Organization, Roles, and Responsibilities

- Review resource requirements profile.
- Review existing organization plan, if any.
- Design ideal going-forward project organization.
- Draft assignment terms of reference for each role.
- Draft the organization plan.

Identify Available Staff Resources

- Check with practice managers for availability and suitability of staff.
- Check with consulting staff concerning their fit to roles and training needs.
- Determine subcontractor requirements.
- Interview candidates and estimate training requirements.
- Send out assignment requests to practice managers and subcontractors.
- Receive assignment request commitments. If commitments cannot achieve organization plan requirements, perform task step 1 again. It may also be necessary to perform tasks in Work Management again to revise the staff requirements profile.
- If commitments achieve organization plan requirements, finalize the Staffing plan.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Manager	Determine staff availability and interview staff.	60
Project Support Specialist	Draft the assignment terms of reference and research suitable resumes (CVs).	40
Client Project Manager	Interview staff.	*
Consulting Business Manager	Supply staff availability.	*

* Time not estimated.

Deliverable Components

The components of the Staffing and Organization Plan are:

- ☐ **Organization Plan**
- ☐ **Assignment Terms of Reference**
- ☐ **Staffing Plan**

Component Descriptions

This section describes each component.

Organization Plan

The Organization Plan shows the project organizational structure, including consulting and client staff. This plan consists of an organization chart mapped to project roles. Include both consulting and client roles in the plan.

Assignment Terms of Reference

The Assignment Terms of Reference (TOR) deliverable can be used to define the tasks of any staff member, or, specifically, specialists joining the project team for a limited time, such as for a technical review.

Background: This is a short description of the purpose of the TOR and a general description of the assignment. Include the job title, job overview, and the location at which the consultant will work.

Objectives: What are the objectives of the assignment? For example, is the purpose to provide advice and guidance as requested by the client or to perform a specific task?

Scope: Define the scope of the assignment. Clearly state what is *not* in the scope where appropriate. Estimate the duration of the assignment.

Constraints: Define any constraints.

Assumptions: Define any assumptions. This includes the setup which the customer is supposed to be providing. If relevant, then include:

- platform
- operating system
- RDBMS version
- applications
- tools

Reporting Responsibility: Define reporting arrangements to both consulting and the client, including frequency of reporting , to whom, method (meetings, reports), and outputs (such as site visit reports, meeting minutes).

Tasks and Deliverables: Clearly state what the deliverables of the assignment are, including tasks and time scales involved.

Quality Issues: Define how quality control is to be applied to the project, for example if quality reviews are to be carried out and by whom. Include authorities (approvals).

Other Considerations: List anything else that affects the assignment and is not covered above.

Staffing Plan

This component defines how staffing requirements will be satisfied. The planning horizon may be the entire project or a project phase. Include the following information in the staffing plan:

- subcontractor requirements
- training requirements estimate
- assignment request commitments from practice managers and subcontractors

Audience, Distribution, and Usage

The Staffing Plan and Organization Plan are used to:

- communicate the organization of the project within the project team and to client staff
- define roles and responsibilities
- document staff commitments to the project organization

The Assignment TOR is used to:

- task project team members
- task specialists who join the team on a limited basis
- define individual terms of reference for any consulting assignment

Distribute the deliverable to:

- project manager for review
- project team for information
- to the Project Library for control

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ The plan is complete, with line and staff responsibilities defined.
- ☐ All roles have defined responsibilities.
- ☐ All TOR sections are completed and wording is clear and precise.

RM.025 - Create Project Orientation Guide

Within the context of this task, all of the information including standards, policies, and procedures for project team members will be collected and compiled. Include the policies for living expenses, indicate when and if per diem is applicable, and provide guidance on travel policies acceptable to both the consulting organization and the client. Also address items such as required identification for security and obtaining user accounts for the client's internal systems.

Ideally, complete this guide prior to the initial project kick off meeting. Distribute it to project team members to ensure consistency and prevent misunderstandings. This document allows each of the project team members to accomplish the first rule of project management, Start Right.

Deliverable

The deliverable for this task is the **Project Orientation Guide**.

Prerequisites

Required

You need the following input for this task:

☐ **Project Management Plan (CR.010)**

Contains the definition of the project strategies, standards and procedures developed to establish the project processes and the relevant application of those processes to the particular engagement. Guidance for the individual consultant is derived from the overall project objectives.

☐ **Staffing and Organization Plan (RM.020)**

Defines the resource requirements profile, reviews the project organization plan, constructs the planned project organization, and

develops the draft terms of reference for each role. Consulting and client project managers draft policies for living and travel expenses. Within this task, each of these components establishes guidelines for the consultants to be employed at this project.

Resource Management Strategies, Standards, and Procedures(RM.010)

Provides additional information on living and travel expense policies for the project.

Task Steps

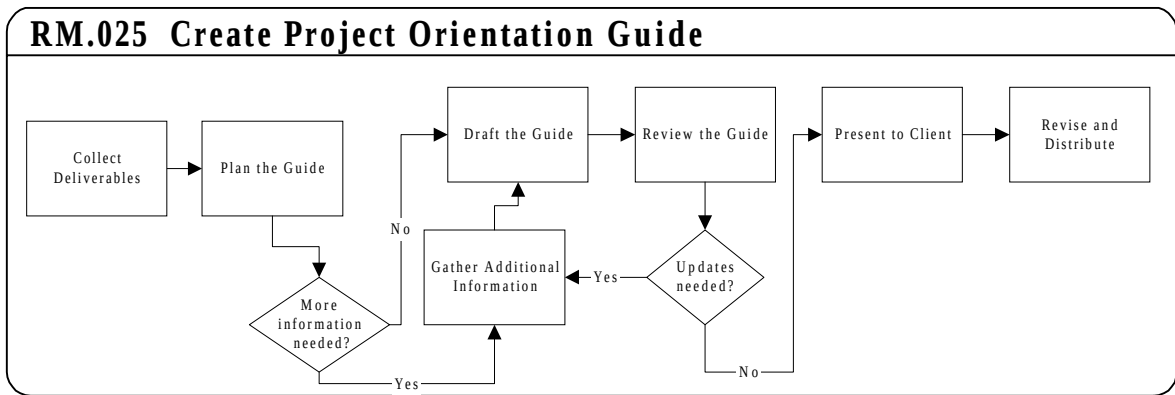


Figure 3-5 Task Flow for Create Project Orientation Guide

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Collect and compile the prerequisite deliverables.	Project Management Plan, Staffing and Organization Plan, Resource Management Strategies, Standards, and Procedures
2.	Plan the Project Orientation Guide.	Orientation Guide Template

No.	Task Step	Deliverable Component
3.	Gather additional information.	Client's Public reports or annual statements, maps, list of preferred hotels, list of restaurants
4.	Prepare draft Project Orientation Guide.	Project Orientation Guide draft
5.	Review Project Orientation Guide draft.	Updated Orientation Guide
6.	Present Project Orientation Guide to client.	Orientation Guide review notes
7.	Revise and distribute Project Orientation Guide.	Revised Orientation Guide

Approach and Techniques

Collect and Compile the Prerequisite Deliverables

- Obtain the Project Management Plan, Staffing and Organization Plan, and Resource Management Strategies, Standards, and Procedures from the project library.
- Review the deliverables for relevant information.
- Update the Project Orientation Guide template with the project information.

Plan the Project Orientation Guide

- Plan the Project Orientation Guide with input from the Project Management Plan, Staffing and Organization Plan, and the Resource Management Strategies, Standards, and Procedures.
- Identify the additional sources of information.
- Determine the most effective means of obtaining the information.

Gather Additional Information

- Obtain electronic versions of the information where possible.
- When possible, download electronic versions of the client's publicly available business reports or obtain copies of the client's annual statement, and view web sites for electronic copies of maps.
- Extract and compile all pertinent and relevant information.
- Update the Project Orientation Guide template with the information.
- Compile a consolidated deliverable.

Prepare Draft Project Orientation Guide

- Update the Project Orientation Guide template with the project information.
- Revise and update the information as needed to formulate a clear, concise guide.
- Incorporate electronic copies of company data, maps, and other information.

Review Project Orientation Guide Draft

- Review the updated Project Orientation Guide.
- Revise as needed to ensure clarity of policies.
- Note any revisions to policies, procedures or standards.
- Update the Project Orientation Guide in the Project Library.

Present the Project Orientation Guide to Client

- Schedule a review meeting with the client.
- Present and explain the Project Orientation Guide.
- Record comments, requested revisions from the client.

Revise and Distribute the Project Orientation Guide

- Revise the Project Orientation Guide with client review comments and requested revisions.
- Distribute the Project Orientation Guide to project team members and the project library.
- Schedule and present the Project Orientation Guide to project team members. This may be included in the Project Kick Off meeting.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Support Specialist	Compile all relevant documents, obtain additional information from external sources, and compose the draft deliverable.	60
Project Manager	Review the Project Orientation Guide for clear concise prose, clarity of purpose, and efficient dissemination of information. Schedule and conduct a walkthrough of the Project Orientation Guide with the appropriate client and any third party contractors.	40
Consulting Business Manager	Review the Project Orientation Guide.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Project Orientation Guide are:

- ☐ **Project Orientation Guide Definition and Purpose**
- ☐ **General Expectations**
- ☐ **Project Definition and Purpose**
- ☐ **Company Background**
- ☐ **Deliverables to Review List**
- ☐ **Checklist for new project members**

Component Descriptions

This section describes each components.

Project Orientation Guide Definition and Purpose

Provides an overview of the document, the intended audience, how the document is organized, and discusses how to use the document.

General Expectations

Describes any necessary project training, guidance on team dynamics, any cultural considerations expected of the project team, the logistics for the engagement, and the time reporting procedure.

Project Definition and Purpose

Provides a description of the scope, objectives and approach for the project in addition to an initial assessment of the key challenges identified for this engagement, the project goals and objectives, and overview of the project pilot definition, development and deployment where applicable. This section becomes more important to the new project team member when substantial changes in project scope, objectives or approach have occurred since commencement.

Company Background

Gives consulting staff general knowledge about the client's business. For public companies within the United States, the 10K report or other relevant securities filings are some of the best sources of the description, background, historical performance and current market position of the company. Other sources of information for public companies include Standard and Poor's and Moody's manuals as well as various Internet sites.

Deliverables to Review List

Lists the deliverable documents such as the Proposal, Project Management Plan, Staffing and Organization Plan, Project Workplan. Other deliverables include Issue and Change Management Procedures, Quality Management Procedures, and Configuration Management Procedures.

Checklist for New Project Members

Lists all the things a new project member needs to do. Include such things as obtaining site security pass, user accounts, collecting templates, and reading the Project Management Plan.

Audience, Distribution, and Usage

The Project Orientation Guide is used to:

- provide a single, concise source of information for project team members
- facilitate the initiation of new team members into the project
- define the project standards for project team members

Distribute the Project Orientation Guide to:

- project manager and team leaders for review
- client project manager for review and approval
- project library for control
- project distribution list
- to the Project Library for control

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Is the project organization well defined?
- ☐ Is the scope of the project clearly stated?
- ☐ Do project team members understand the time reporting procedures?
- ☐ Are the travel and living expense policies clear and understandable?
- ☐ Do consulting project team members know how to obtain office supplies, stationary, obtain security badges, and email accounts?
- ☐ Are standards of conduct clearly defined and stated?
- ☐ Are the project's goals and objectives clearly defined?
- ☐ Are the work days and working hours stated?

RM.030 - Implement Organization

In this task you plan how to acquire and train organization staff members for their designated roles.

Deliverable

The deliverable for this task is the **Prepared Organization**.

Prerequisites

Required

You need the following input for this task:

☐ **Staffing and Organization Plan (RM.020)**

The Staffing and Organization Plan includes the Resource Plan which includes levels, and timing as well as the number of resources required for the project.

☐ **Project Orientation Guide (RM.025)**

The Project Orientation Guide contain all of the policies and procedures for the engagement.

Task Steps

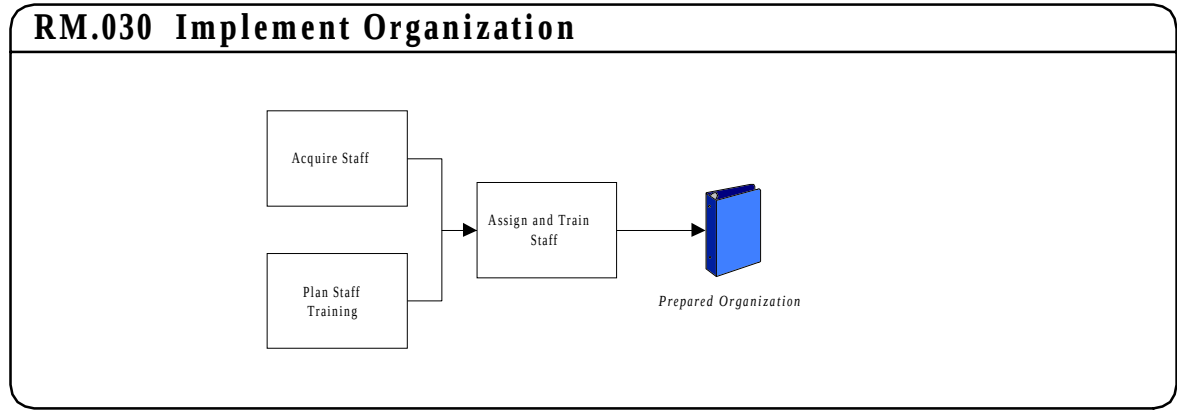


Figure 3-6 Task Flow for Implement Organization

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Acquire staff.	New Staff Members
2.	Plan staff training.	Staff Training Plan
3.	Assign and train staff.	Prepared Organization

Approach and Techniques

Acquire Staff

- Confirm start dates with practice managers and subcontractors.
- Interview substitutes when necessary.

Plan Staff Training

- Schedule training according to the requirements stated for each individual in the Staff Organization report.
- Prepare the Project Orientation Package.

Assign and Train Staff

- Invoke the orientation procedure for staff who are new to the project.
- Agree on individual assignment terms of reference.
- Verify that training has been undertaken, and update the training records.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Manager	Confirm start dates, agree to a training schedule, and welcome new staff.	60
Project Support Specialist	Send out confirmations, maintain training records, and ensure that the induction process is performed.	40
Client Project Manager	Participate in the induction process.	*
Consulting Business Manager	Provide staff.	*

* Time not estimated.

Deliverable Components

The deliverable components of the Prepared Organization are:

- ☐ **New Staff Members**
- ☐ **Staff Training Plan**
- ☐ **Project Orientation Package**
- ☐ **Prepared Organization**

Component Descriptions

This section describes each component.

New Staff Members

New staff members are assigned to the project based on commitments obtained from consulting practices or the client. Maintain an administrative log which records each staff member's start date, orientation date, and performance appraisal planned and actual dates. In addition, maintain home, email, and local project contact information for each member.

Staff Training Plan

This component lists the training which each staff member is expected to receive in order to be prepared to fulfill their assigned roles and responsibilities.

Project Orientation Package

This component provides administrative information that a new project member needs to know regarding project and site policies, standards, and procedures.

Prepared Organization

This component is represented by records indicating the role assignments and training of project members, as well as any other preparations made to ensure that staff members are qualified to perform their assigned roles.

RM.040 - Establish Physical Resource Plan

In this task you define the overall requirements and responsibility for all physical resources and supporting services needed to execute project tasks. You should consider the environment requirements from all processes being executed for the project.

Deliverable

The deliverable for this task is the **Physical Resource Plan**.

Prerequisites

Required

You need the following input for this task:

☐ **Resource Management Strategy (RM.010)**

The Resource Management Strategy outlines who is responsible for the procurement of equipment, and describes the type of equipment to be made available.

☐ **Staffing and Organization Plan (RM.020)**

The Staffing and Organization Plan identifies when staff assignments are going to start.

☐ **Environment Requirements**

All processes executed on the project identify environment requirements. These requirements are from management processes, such as Configuration Management, as well as execution processes, such as Testing.

Optional

You may need the following input for this task:

- ☐ **Physical Resource Plan**

A Physical Resource Plan may exist from previous planning.

Task Steps

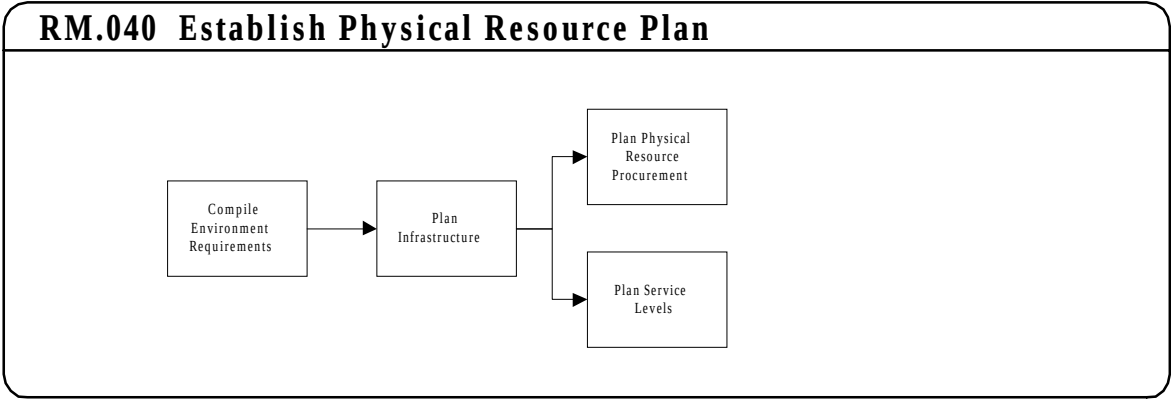


Figure 3-7 Task Flow for Establish Physical Resource Plan

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Compile environment requirements for all processes.	Environment Requirements
2.	Plan infrastructure to satisfy environment requirements.	Infrastructure Plan
3.	Plan physical resource procurement.	Procurement Plan
4.	Plan service levels.	Service Level Agreements

Approach and Techniques

Compile Environment Requirements

- Define working environment requirements for PCs, stationery, filing space, office software, phones, work areas, and meeting areas.
- Compile environment requirements for all project processes.
- Complete an infrastructure requirements list for all environments.

Plan Infrastructure

- Design facilities necessary to satisfy requirements.
- Design computer resources necessary to satisfy requirements.
- Define service levels necessary to satisfy requirements.

Plan Physical Resource Procurement

- Review consulting and client resource supply responsibilities with the client.
- List potential suppliers for any consulting responsibilities.
- Select suppliers for any consulting responsibilities (refer to practice policies).
- Draft a procurement plan for all physical resources.
- Review and approve the procurement plan within consulting.
- Review and agree on the procurement plan with the client.

Plan Service Levels

- Review consulting and client service supply responsibilities with the client.
- Identify project organizations responsible for services, from both consulting and client.
- Agree with the client on service levels to be provided externally to the project by the client.

- Draft service level agreements for each external service to be provided by the client.
- Identify suppliers of any services not provided by consulting or client, and update the procurement plan for these services.
- Review and approve service level agreements within consulting.
- Review and agree on service level agreements with the client.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Compile environment and infrastructure requirements, and draft the procurement plan and service level agreements.	80
Project Manager	Approve requirements, plans, and agreements with consulting and the client.	20
Client Project Manager	Agree to client responsibilities.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Physical Resource Plan are:

- ☐ **Environment Requirements**
- ☐ **Infrastructure Plan**
- ☐ **Procurement Plan**
- ☐ **Service Level Agreements**

Component Descriptions

This section describes each component.

Environment Requirements

Defines the environment requirements for the project, including software, tools, and hardware. Typical contents include:

- Project Environment Evaluation
- Support Tools Required
- Application and Development Environments
- High-level Database Topology
- Sizing Summary
- Hardware and Software Delivery Schedule
- Hardware Support Policy
- Instance Synchronization Approach
- Legacy Systems and Support

Infrastructure Plan

Describes the physical resources to implement the environment requirements, together with planned changes to the infrastructure during the life of the project, such as porting to other platforms and different environment needed for performance testing. It also defines network setup and system administration details (unless defined later, in the Service Level Agreement component).

Procurement Plan

This describes the arrangements for purchasing items for the project, including software, tools, and consumables, such as stationery and writing implements. Procurement may be performed by consulting or client's organization.

Service Level Agreements

Plans for and implements any Service Level Agreements needed to guarantee a maintained environment for the project. Includes

environment availability (times), access to project environment during weekends, machine startup, shutdown and maintenance arrangements, backup requirements (including off site media storage).

Audience, Distribution, and Usage

The Physical Resource Plan is used to:

- define arrangements for procuring and installing project hardware and software
- define controls for safeguarding and maintaining the project environments

Distribute the deliverable to:

- project manager for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable
- any key client users, identified by the client project manager or line managers, who are to accept the deliverable

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are the details sufficient to ensure that the environment will be available in the right time scale and with suitable available physical resources for the project team?
- ☐ Are maintenance arrangements and service level agreements sufficient to safeguard the environment?

RM.050 - Establish Infrastructure

In this task you establish, acquire, and install the physical resources required.

Deliverable

The deliverable for this task is a **Prepared Infrastructure** that is capable of sustaining all tasks being planned.

Prerequisites

Required

You need the following input for this task:

☐ **Physical Resource Plan (RM.040)**

The Physical Resource Plan defines the total set of requirements for procuring and installing the project infrastructure.

Optional

You may need the following input for this task:

☐ **Infrastructure**

An infrastructure may exist from previous project execution.

Task Steps

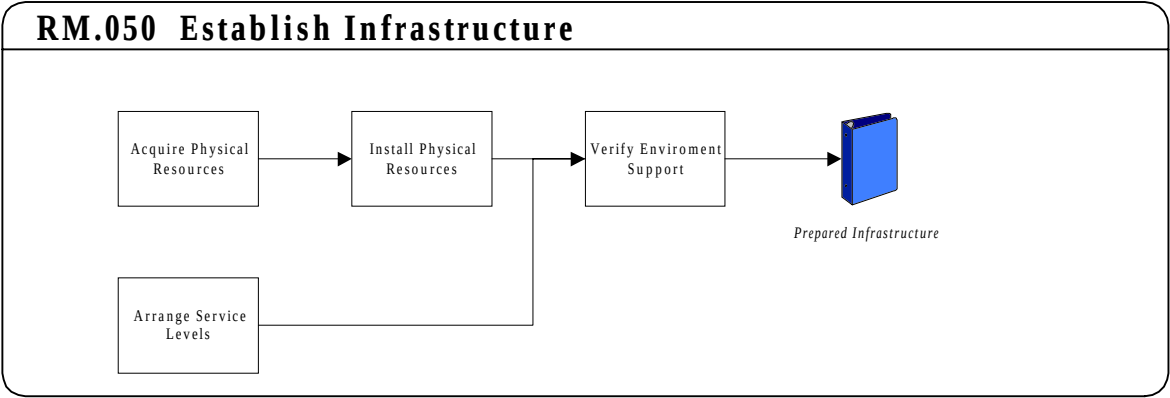


Figure 3-8 Task Flow for Establish Infrastructure

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Acquire physical resources.	Acquired Equipment, Software, and Facilities
2.	Install physical resources.	Installed Infrastructure
3.	Arrange service levels per service level agreements.	Verified Service Levels
4.	Verify environment support.	Prepared Infrastructure

Approach and Techniques

Acquire Physical Resources

- Order or request required resources.
- Arrange physical and technical installation.

Install Physical Resources

- Take delivery of resources. Use the Incoming Item Records form to record deliveries.
- Install items and establish equipment records. The supplier may take responsibility for installing equipment.

Arrange Service Levels

- Contact service provider and review schedule for initiating service.
- Review plans for service initiation.
- Pilot services, if availability of the service is critical.
- Raise client service issues to the project sponsor as soon as possible.

Verify Environment Support

- Test each environment to verify that resources meet compatibility, performance, and availability requirements.
- Communicate installation problems either to the client, as issues, or to each resource supplier, as equipment fault reports. Use the Issue form and the Equipment Fault report.
- Raise any risks or issues that are caused by the unavailability of resources to the client and consulting management.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Arrange for installation of resources and startup of services. Coordinate with suppliers, client service providers, and project members.	80
Project Manager	Ensure that the task is completed in a timely manner. Raise risks or issues caused by the non-availability of resources.	20
Client Project Manager	Direct client responsibilities, as agreed.	*

* Time not estimated.

Deliverable Components

The deliverable components for Prepared Infrastructure are:

- ☐ **Acquired Equipment, Software, and Facilities**
- ☐ **Installed Infrastructure**
- ☐ **Verified Service Levels**
- ☐ **Prepared Infrastructure**

Component Descriptions

This section describes each component.

Acquired Equipment, Software, and Facilities

Document by using incoming item records and logs. The Incoming Item Record can be used to check items into the project environment where there is a need to keep a close control, such as where items are used on a project before being delivered, with the solution, to the client. See the example Incoming Item Record form that follows.

Installed Infrastructure

Documented using installation records and test reports.

Verified Service Levels

Documented using the service level agreements previously prepared. Update after review and acceptance by service providers.

Prepared Infrastructure

Use equipment fault reports to document problems with environment support. Raise risks or issues caused by unavailability of resources to the client and consulting management.

The Fault Report can be used to document hardware, software, or tools faults for onward transmission to the manufacturer. For simplicity, use a Problem Report Form.

Equipment Fault details: List the details about the problem, including the item affected, if known. Describe the problem so that another reader not as familiar with the circumstances will understand and be able to resolve it.

Investigation: Describe the extent of the investigation, and the suggested or actual action to be taken. Where the problem is going to take more than half a work day to resolve, provide an estimate of required work.

Authorization: Normally authorized by the project manager or a nominee. It may also need to be authorized by the client, although usually client authorization will be obtained separately if any change requests are needed to resolve the problem.

[illegible]

Ref: ____ / ____ / IIR / ____

Supplier

Order Numbers:

Delivery Notes:

Release Notes:

Delivery Dates:

Qty

Description

Order Status

Spec'n / Version	
---------------------	--

Qty
Accept

Qty	Reject
-----	--------

Qty	Stock
-----	-------

Order status: P = PART
C = COMPLETE

Figure 3-9 Incoming Item Record

Audience, Distribution, and Usage

The Incoming Item Record is used to:

- record hardware, software, and tools, including their identifiers and versions, as a record of what was received
- enable release notes to be compiled whenever the environment is delivered to the client

The Fault Record is used to record the details relating to a fault for subsequent analysis by maintenance engineers or manufacturers

Distribute the deliverable to:

- project manager for review
- project library for control

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ All details, including version, part numbers, and serial numbers are present in incoming item records.
- ☐ Project responsibility for maintaining control of acquired physical resources is documented and delegated to staff members.

RM.060 - Staff Control

In this ongoing task you deal with staff management issues that arise during execution.

Prerequisites

Required

You need the following input for this task:

☐ **Staffing and Organization Plan (RM.020)**

The Staffing and Organization Plan defines the project organization and assignments of project staff within the organization to project roles.

☐ **Staff Records (RM.030, RM.060)**

Staff Records include information on staff assignments, skills, appraisals, and training.

☐ **Staff Roles and Organization Management Procedure (RM.010)**

The Staff Roles and Organization Management procedure provides specific instructions on how the project organization will be managed.

Task Steps

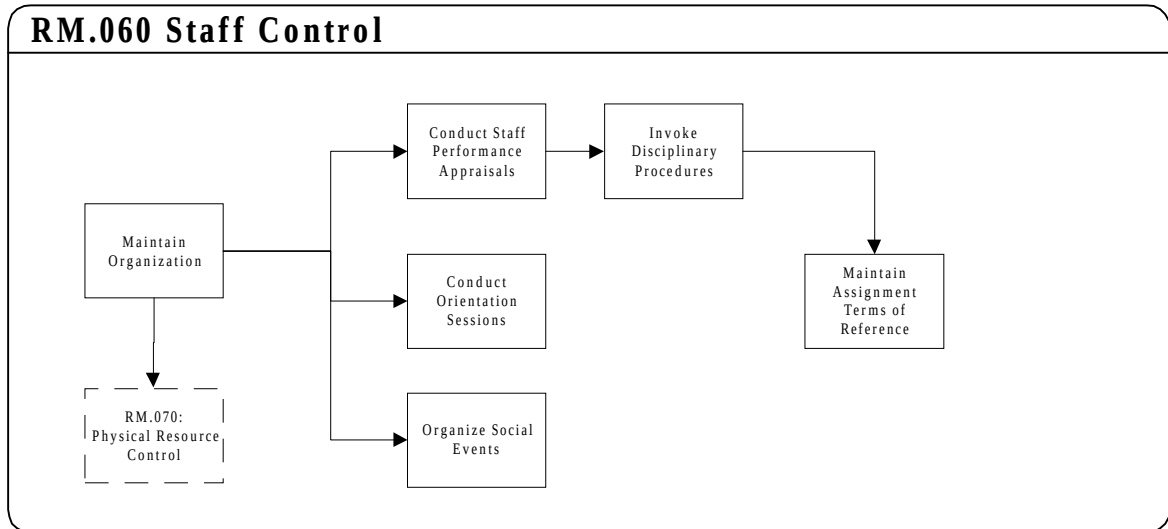


Figure 3-10 Task Flow for Staff Control

The steps involved in this ongoing task are:

No.	Task Step	Result or Output
1.	Maintain organization.	Adjusted Organization
2.	Conduct staff performance appraisals.	Performance Appraisals
3.	Invoke disciplinary procedures.	Disciplinary Reports
4.	Organize social events.	Social Calendar
5.	Conduct orientation sessions.	Oriented Project Staff
6.	Maintain assignment terms of reference.	Adjusted Assignment TOR

Approach and Techniques

Conduct Staff Performance Appraisals

- Arrange dates for individuals.
- Collect third party input.
- Prepare for appraisal (staff and manager).
- Conduct appraisal interview.
- Agree on future period objectives, actions, and results of appraisal.
- Update staff records.

Invoke Disciplinary Procedures

- Where there are disciplinary problems, (staff not following procedures, poor performance, high levels of absence), it is important to explore all the circumstances before invoking disciplinary procedures.
- Where formal disciplinary action is to be taken, it is essential to confirm practice procedures and to involve human resources staff from consulting.
- Always make a record of disciplinary interviews, formal or informal. Informal meetings do not necessarily have to be entered on a person's staff record, but minutes of the meeting must be kept so that a history is established in case of continued problems.

Organize Social Events

- Social events are essential to fostering and maintaining a good team spirit.
- When planning social events, consider such things as whether or not partners or children are invited. Is the function for consulting staff or the entire team? If free alcohol is provided, then will people be safe getting to where they are staying overnight?
- Plan a variety of social events and timings so that there is something for everyone.

Maintain Organization

- Maintain clear reporting lines within the organization.
- Over time, individuals will grow in their experience and capabilities and this should be reflected in the reporting lines.
- Reporting lines may become unstable as people enter and leave the organization.

Maintain Assignment Terms of Reference

- Assignment TOR will become out of date and need regular review and updating. This may be undertaken at any time, but should always be done as part of the performance appraisal.

Conduct Orientation Sessions

- When new staff members join, it is important to update them with the overall objectives, business climate, detailed procedures, and any other pertinent information.
- Initiate project records for new staff members, and arrange for physical resources.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Manage project staff.	70
Project Support Staff	Maintain records and orient new project staff.	30

RM.070 - Physical Resource Control

As the project progresses, enhancements and additions to the infrastructure will be required. In this ongoing task you implement these updates, as necessary, by managing facilities, technical resources, and service levels.

Prerequisites

Required

You need the following input for this task:

☐ **Physical Resource Plan (RM.040)**

The Physical Resource Plan defines the requirements and responsibility for all project physical resources and supporting services.

☐ **Environment Maintenance Procedure (RM.010)**

The Environment Maintenance procedure defines how physical resources will be ordered, installed, and maintained to support all project environments.

Task Steps

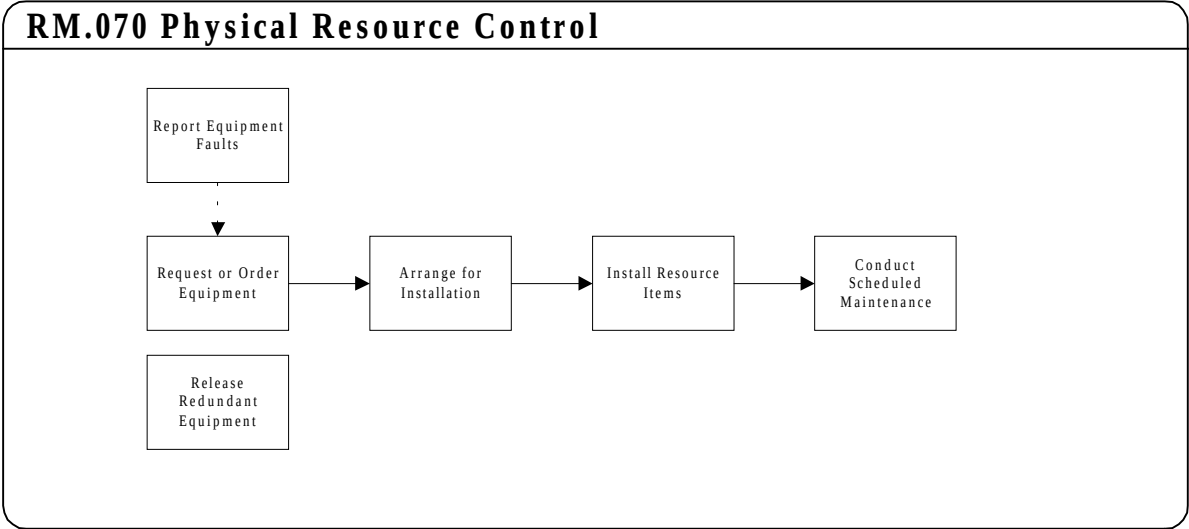


Figure 3-11 Task Flow for Physical Resource Control

The steps involved in this ongoing task follow:

No.	Task Step	Result or Output
1.	Request or order equipment.	Requests or Purchase Orders
2.	Arrange for installation.	None
3.	Install resource items.	Installation Reports, Incoming Item Records
4.	Conduct scheduled maintenance.	Updated Maintenance Schedule
5.	Report equipment faults.	Equipment Fault Reports
6.	Release redundant equipment.	Equipment Release Records

Approach and Techniques

Request or Order Equipment

- Complete agreed order documentation.
- Where equipment is not subject to formal orders, ensure that the request includes a date and time.
- Allow sufficient lead time for equipment to be delivered and installed.

Arrange for Installation

- Ensure that the space the equipment is to occupy is cleared, prior to delivery.
- Arrange installation in line with the service level agreements.

Install Resource Items

- Install the equipment. (Suppliers may undertake this task step.)
- Verify that installation is complete.

Conduct Scheduled Maintenance

- Maintain equipment in line with the maintenance schedule.
- Where maintenance is provided by a third party, ensure that it is carried out in a timely fashion.

Report Equipment Faults

- Report equipment faults to whoever is responsible for its maintenance.
- If necessary, arrange for replacement equipment.

Release Redundant Equipment

- When equipment is no longer needed (redundant), it should be released.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Maintain the infrastructure.	80
Project Manager	Ensure that resources are maintained.	20
Client Project Manager	Maintain client physical resources as agreed upon.	*

* Time not estimated.

RM.080 - Release Staff

In this task you release the staff resources whose services are not required for further project work. Release of a staff member from the project can be an emotional event for the project organization, the staff member, and even the client. Communicate your intentions clearly and carefully, to prevent misunderstandings and to make the release a positive experience for all concerned.

Deliverable

The deliverable for this task is **Released Staff**.

Prerequisites

Required

You need the following input for this task:

☐ **Staffing and Organization Plan (RM.020)**

The Staffing and Organization Plan identifies when staff are expected to be released.

☐ **Staff Records (RM.060)**

Staff records provide input for final performance appraisals.

☐ **Assignment Terms of Reference (RM.060)**

Assignment Terms of Reference provide an additional appraisal of some project staff.

Task Steps

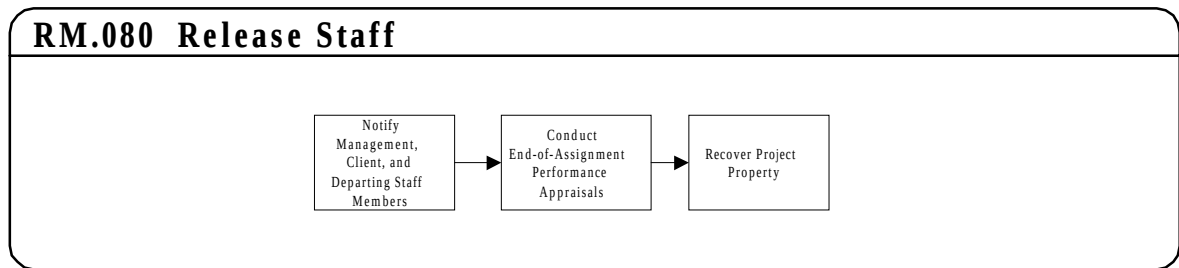


Figure 3-12 Task Flow for Release Staff

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Notify management, client, and departing staff members.	None
2.	Conduct end-of-assignment performance appraisals.	Performance Appraisals
3.	Recover project property.	Recovered Physical Resources

Approach and Techniques

Notify Management, Client, and Departing Staff Members

- Notify consulting managers of staff members of your plan to release them.
- Notify subcontractor management of plans to release subcontractors, and verify contractual compliance.
- Notify the client of staff release plans, and ensure that the client understands the rationale.
- Discuss release plans with affected staff members.

Conduct End-of-Assignment Performance Appraisals

- Use current performance management guidelines for consulting staff.
- Use subcontractor or client procedures for non-consulting staff, if these exist.
- Review the appraisal with the staff member.
- Forward and review the appraisal with the staff member’s manager.

Recover Project Property

- Collect any client property, such as passes, badges, keys, and office equipment from departing staff.
- Inventory physical resources, such as computer equipment and manuals belonging to the project.
- Take any standard security actions such as changing locks, passwords, or combinations to protect client or project property.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Manager	Perform coordination and performance appraisals.	80
Project Support Specialist	Recover project property.	20
Client Project Manager	Participate in release planning and performance appraisals of client staff.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Released Staff are:

- ☐ **Performance Appraisal**
- ☐ **Recovered Physical Resources**

Component Descriptions

This section describes each component.

Performance Appraisal

Give a final performance appraisal to each project member upon their departure from the project.

Recovered Physical Resources

Update equipment records to indicate the release of the departing staff member from responsibility for assigned project property.

RM.090 - Release Physical Resources

In this task you release control of the physical resources that are no longer needed to complete project execution.

Deliverable

The deliverable for this task is **Released Physical Resources**.

Prerequisites

Required

You need the following input for this task:

☐ **Physical Resource Plan (RM.040)**

The Physical Resource Plan identifies vendors, owners, support contracts, and license information for resources to be released.

☐ **Equipment Records (RM.070)**

Equipment records identify the current inventory, status, and maintenance of equipment to be released.

Task Steps

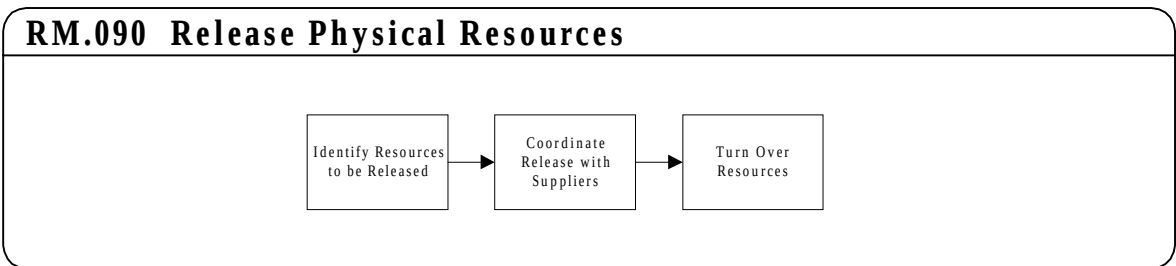


Figure 3-13 Task Flow for Release Physical Resources

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Identify the resources to be released.	None
2.	Coordinate the release with suppliers.	Coordinated Release Schedule
3.	Turn over the resources.	Released Physical Resources

Approach and Techniques

Identify Resources to be Released

- Compare ongoing environment requirements with the current infrastructure.
- Review the Staff Organization plan to determine the number of staff members requiring support.
- List all resources to be released from equipment records.

Coordinate Release with Suppliers

- Obtain turnover requirements for each resource item.
- Determine if there are any contractual obligations to be followed.
- Agree on a schedule for turnover.

Turn Over Resources

- Inventory and inspect items to be returned.
- Return items, or assist supplier with de-installation procedures.
- Obtain receipt and release from responsibility for all returned items.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Perform all task steps.	90
Project Manager	Review and approve release plans.	10

Quality Management (QM)

This chapter describes the Quality Management process.

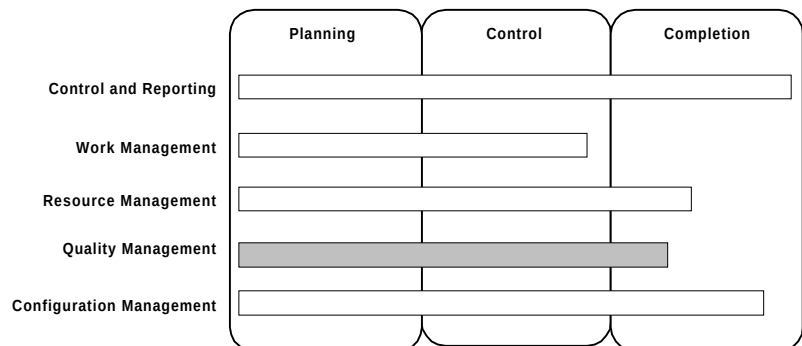


Figure 4-1 Quality Management Context

Process Flow

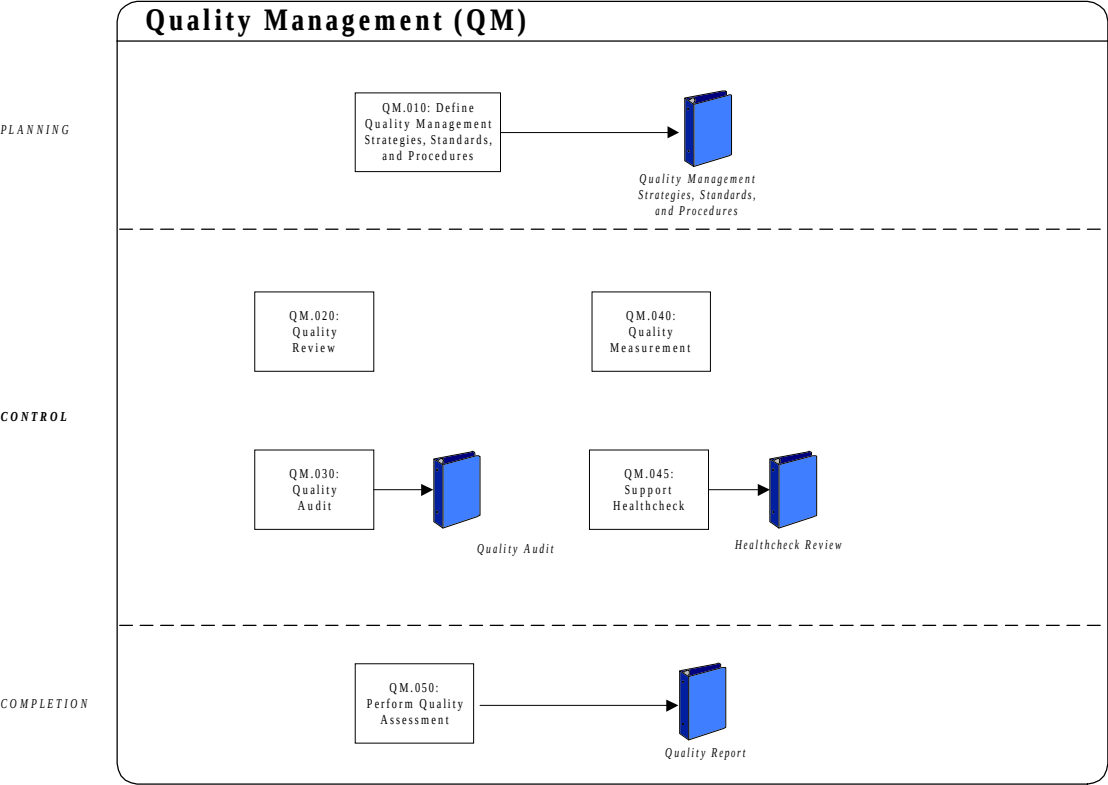


Figure 4-2 Quality Management Process Flow Diagram

Approach

A clear definition of Quality Management, including strategy and control mechanisms, in the Project Management Plan establishes a foundation for the project. Standards and procedures ensure consistency among team members and define quality assessment criteria. When the standards and procedures cannot be defined in the Project Management Plan, they can be developed as the project progresses, as long as they are ready when needed.

Use quality reviews and testing to assess and ensure the quality of the project deliverables, including the deliverable system. Testing is not covered in the Quality Management process but is a separate process included in execution methods, such as Custom Development Method (CDM) or Application Implementation Method (AIM). The testing strategy for a project is developed as part of the Project Management Plan.

During an Audit, you assess a project in terms of how execution has adhered to plans, standards, and procedures. Measurements of cost, time, and quality of the project enable the refinement of project estimating factors and enable reuse of the information for future projects.

The collection of measurement data is vital and often overlooked on projects. Quality Management attempts to focus the effort of measurement collection by concentrating on those measurements which are useful to the project or to consulting, through the preparation and implementation of the Project Management Plan.

Tasks and Deliverables

The tasks and deliverables for this process are:

ID	Task Name	Deliverable or Output Name	Type*
Planning			
QM.010	Define Quality Management Strategies, Standards, and Procedures	Quality Management Strategies, Standards, and Procedures	IT
Control			
QM.020	Quality Review	Quality Review	MI
QM.030	Quality Audit	Quality Audit	MI
QM.040	Quality Measurement	Metrics Report	O
QM.045	Support Healthcheck	Healthcheck Review	MI
Completion			
QM.050	Perform Quality Assessment	Quality Report	MO

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary in the *PJM Method Handbook*.

Objectives

The objectives of the Quality Management process are to:

- Provide definition of review, audit, and measurement processes.
- Monitor execution of these (and other) project processes.
- Ensure that comments from reviews and actions from audits are identified and managed through to resolution.
- Define and collect measurements (project metrics).

Key Deliverables

The key deliverables of this process are:

Deliverable	Description
Quality Management Strategies, Standards, and Procedures	Specific standards and procedures may be needed to amplify high-level statements in the Project Management Plan, such as a specific procedure for auditing.
Healthcheck Review	An evaluation of the status of a project, including progress against objectives, project management, and commercial control from both Oracle and client viewpoints.
Audit Report	Document that contains the findings from project audits.
Quality Report	Assessment of the completeness of quality arrangements on the project.

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Quality Auditor	Conduct audit, write audit report, and follow-up with appropriate action.
Project Support Specialist	Collect and compile documents and deliverables. Log and update action items. Schedule Healthchecks, reviews Healthcheck results, and distributes reports. Perform or arrange independent audits of the project. Participate in QM tasks. Administer the completion of documents.
Project Manager	Write standards and procedures. Arrange audits and reviews. Ensure that plans, standards, and procedures are maintained.
Project Staff	Participate in quality reviews and audits.
Reviewer	Conduct the review, write the review report, and follow-up with any appropriate actions.

Critical Success Factors

The critical success factors of the Quality Management process are:

- Standards are defined where needed, understood, and used.
- Deliverables are checked through the use of reviews.
- Processes are followed through the use of audits.
- Comments are resolved for reviews and actions from audits.
- Key measurements of processes are retained for repeatability and for use on future projects.

QM.010 - Define Quality Management Strategies, Standards, and Procedures

In this task you create or update the plans which define how the Quality Management process supports and implements the project scope, objectives, and approach.

You can define some project standards as the Project Management Plan is written, but generally, additional Quality Management standards need to be written as the project progresses. This task allows for the progressive writing and updating of Quality Management Standards and Procedures during the project.

Deliverable

The deliverable for this task is the set of **Quality Management Strategies, Standards, and Procedures**.

Prerequisites

Required

You need the following input for this task:

☐ **Scoping Project Management Plan (CR.010)**

The Scoping Project Management Plan sets the context for QM.

☐ **Policies**

Consider applicable consulting policies regarding QM in terms of how they apply to your project. The client may have their own policies regarding QM as well.

Optional

You may need the following input for this task:

☐ **Project Management Plan (CR.010)**

A Project Management Plan may exist from previous planning.

Task Steps

QM.010 Define Quality Management Strategies, Standards, and Procedures

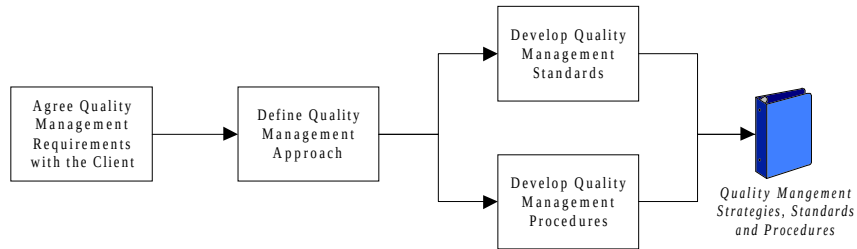


Figure 4-3 Task Flow for Define Quality Management Strategies, Standards, and Procedures

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Agree on the Quality Management requirements with the client.	Quality Management Requirements
2.	Define the Quality Management approach.	Quality Management Strategy
3.	Develop the Quality Management standards.	Quality Management Standards
4.	Develop the Quality Management procedures.	Quality Management Procedures

Approach and Techniques

The standards and procedures serves as a communication mechanism for project staff and subsequent assessment during audits. If the project team works together to define standards, it enhances consistency, eases maintenance, and improves productivity. To facilitate the completion of this task, reuse existing consulting standards to derive project-specific standards.

Agree on Quality Management Requirements with the Client

- Review the Project Management Plan.
- Identify any consulting policies affecting QM.
- Identify any client policies affecting QM.

Define Quality Management Approach

- Identify consulting best practices and standards.
- Document QM strategy using the Project Management Plan format.

Develop Quality Management Standards

- Identify any existing consulting standards.
- Customize standards to satisfy approach.
- Add additional standards to meet client requirements.
- Integrate, where necessary, with other technical standards.

Develop Quality Management Procedures

- Write detailed instructions for Quality Review.
- Write detailed instructions for Quality Audit.
- Write detailed instructions for Quality Measurement.



Suggestion: For small projects, the text in the Project Management Plan may be sufficient definition of Quality Management standards and procedures.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Prepare strategies, standards, and procedures. Note: If a project support specialist is not assigned to the project, then the responsibility rests with the project manager especially for small projects.	70
Project Manager	Review and approve all strategies, standards, and procedures.	30
Client Project Manager	Agree to QM requirements, and identify client policies affecting approach.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Quality Management Strategies, Standards, and Procedures are:

- ☐ **Quality Review**
- ☐ **Quality Audit**
- ☐ **Quality Measurement**

Component Descriptions

This section describes each component.

Quality Review

Review all project deliverables. Define techniques to be used and who is involved. Modify to suit the project and include client involvement.

Quality Audit

Audits focus on whether the project processes are functioning according to plans and procedures. Usually a project needs at least one audit, and large projects may need many. For example, you may need one audit per phase or one audit per major milestone. Audits use resources independent of the project team. Specify how auditing is to be conducted during the project.



Suggestion: If auditors from outside the project team are used, then they probably have their own procedure to follow, so this section may be omitted.

Quality Measurement

Maintain basic metrics in order to improve accuracy of the project estimates, both as the project progresses and for future. Specific metrics include time to design and build modules of all types, number of modules under development, number of problems during system test, number of problems at acceptance, and setup time for applications products. Define metrics and measurement procedures.

Audience, Distribution, and Usage

The QM Strategies, Standards, and Procedures are used:

- to amplify existing procedures in the Project Management Plan
- as detailed work instructions for team members

Distribute the documents to:

- project leader for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable, if appropriate

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are procedures clearly defined?
- ☐ Have you taken into account the impact of dependent tasks from other processes?
- ☐ Are the procedures consistent with the Project Management Plan?

QM.020 - Quality Review

In this task you conduct reviews of deliverables to ensure that they follow standards and fit the project needs (consistent, complete, correct, and concise).

Conduct Quality Reviews throughout the project, as defined in the Project Management Plan. Prepare a Review Comments List for each review.



Suggestion: You can either conduct the Quality Review internally, then offer the deliverable for client review, or you may want to conduct the review with the client present. If your relationship with the client is good and your confidence in the deliverable is high, then conduct the review with the client present.

Deliverable

The deliverable for this task is the **Quality Review**.

Prerequisites

Required

You need the following input for this task:

☐ **Project Management Plan (CR.010)**

The Project Management Plan contains the strategies for Quality Management.

☐ **Quality Review Procedure (QM.010)**

The Quality Review Procedure defines how the Quality Review task is to be conducted.

Task Steps

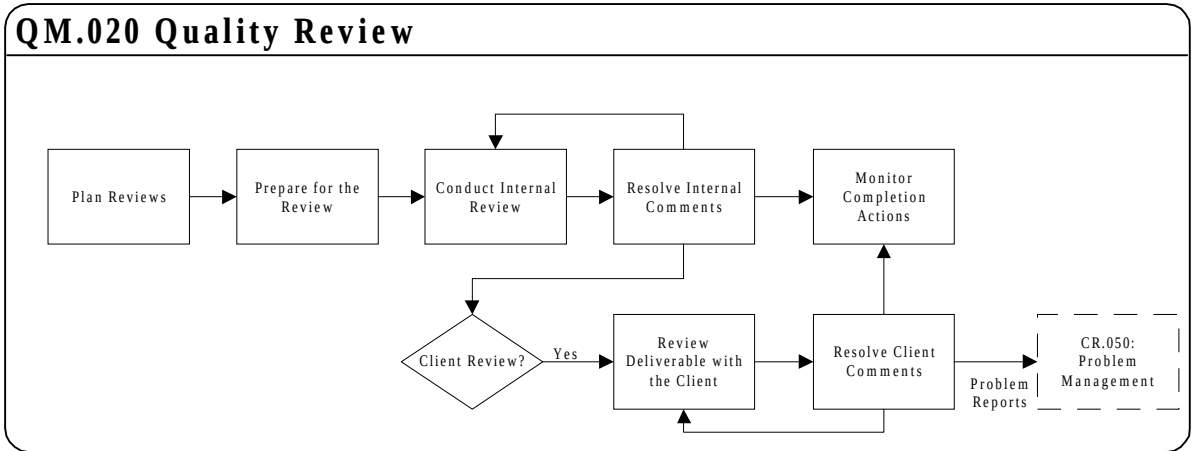


Figure 4-4 Task Flow for Quality Review

The steps involved in this task follow:

No.	Task Step	Deliverable Component
1.	Plan the reviews.	None
2.	Prepare for the review.	Review Checklist
3.	Conduct the internal review and assign review comments.	Review Comments
4.	Resolve internal comments.	Resolved Review Comments
5.	Review the deliverable with the client.	Review Comments
6.	Resolve any client comments and escalate unresolved comments.	Resolved Review Comments
7.	Monitor completion actions.	Problem Reports

Approach and Techniques

Plan reviews for each deliverable in the Project Management Plan. Select an appropriate technique based on the level of importance and complexity of the item under review. Use this task to conduct the review defined in the Project Management Plan. If the author is the only expert on the project team in the area under review, then independent staff should conduct the review.

Plan Reviews

- Plan review types (such as formal inspection, peer review).
- Document in the Workplan what review type is to be used for each deliverable, who is responsible for the review, and whether the customer is involved.
- Document review dates in the project plans.

Prepare for the Review

- Ensure that the review item is ready for review.
- Define a checklist appropriate to the review.
- Obtain project standards appropriate to the item under review.
- Get background material appropriate to the item under review.
- Nominate review staff and roles, and forward copies of the document under review to reviewers so that they can prepare.

Conduct Internal Review

- Review the item against the checklist, standards, and other input material. Note errors, omissions, and issues on the Review Comments List.
- List any external issues (those not directly associated with the item under review).
- Categorize the errors, omissions, and issues according to severity and priority to fix.
- Complete the Review Leader form.

- Give the completed forms to the member of the team who has been nominated by the team leader to resolve the issues.

Resolve Internal Comments

- Update the deliverable according to the comments where possible.
- Record on the Review Comments List the reason for not implementing any comments.

Review Deliverable with the Client

- Use Review Comments Lists or meeting minutes to document client issues with the deliverable.
- Ensure that all client issues are documented.

Resolve Client Comments

- Update the item under review, taking into account all client comments.
- Notify the client of any changes, and discuss any unresolved comments.

Monitor Completion Actions

- Ensure that all comments are resolved by either agreement not to take action or resolution of the issue. If comments are not resolved, then raise a Problem Report.
- Update the forms to show the outcome, completing the clearance box on each form.
- Retain the forms in the project library as a record of the review activities.

Roles and Responsibilities

The roles and associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Participate in reviews. Review and agree on findings.	60
Reviewer	Conduct the review, complete the review forms, and follow-up with any appropriate actions.	30
Project Manager	Arrange for the review, and agree on the findings.	10

Deliverable Components

The deliverable components for the Quality Review are:

- ☐ **Review Checklists**
- ☐ **Review Comments**
- ☐ **Problem Reports**

Component Descriptions

This section describes each component.

Review Checklists

Review checklists are used to help standardize the aspects reviewed for specific deliverables. They may be developed by the project or derived from Oracle Method standards.

Review Comments

This component is used to document the results of a quality review. Review comments should either be resolved and any changes made, or

held over by raising a Problem Report. Review comments can be documented using the Review Leader Form and the Review Comments List.

Review Leader Form: The Review Leader Form is a cover page for review comments, summarizing the results of a review. It can also be used to indicate those reviewers who wish to review the deliverable again after re-work and if a follow-up review is required. An example of this form is shown in Figure 4-5.

Review Comments List: Use the Review Comments List during reviews to register comments and issues with the item under review. In this way the author is clear on what needs to be investigated and potentially changed. An example Review Comments List is shown in Figure 4-6.

Indicate the outcome of the review by marking as either:

- ACCEPTED (Once comments have been acted upon), or
- NOT ACCEPTED (Want to review once comments have been acted upon)



Suggestion: Alternatively, use the Review Leader Form as a front page where there are a number of pages to the Review Comments List.

Comments are characterized by:

- a comment number for ease of reference
- a detailed description of the comment, with any referenced information such as page number or line number
- an action box for subsequent completion by the author after investigating the comment
- a closure box to indicate that the comment has been accepted.



Suggestion: If you accept the comment, just tick the closure box when it is incorporated. If you do not accept the comment, then write the reason why in the action box.

Problem Reports

Raise Problem Reports for any review comments which require in depth investigation or project management review. Reference the Problem Report identifier by the review comment so that it can be closed.

REVIEW LEADER FORM	
Ref: ____ / ____ / RLF / ____	
Associated Review Comments List:	
Attendees:	Were any actions assigned to them? Y or N
If any of the attendees wish to see the document again, after the rework has taken place then please circle their names in the above list	
If another review meeting is required after rework, please enter the date here:	
Comments:	

Figure 4-5 Review Leader Form

REVIEW COMMENTS LIST				
Ref: ____/____/ RCL/ ____				
Item Under Review				Author
Version:		Review Date:		
Reviewers Names OR Associated Review Leader Form Reference				
Outcome: (Circle One) ACCEPTED (Once comments have been acted upon) NOT ACCEPTED (Wish to re-review once comments have been acted upon)				
No	Reference	Comment	ACTION	CL
Categories: MA - MAJOR MI - MINOR I - INFORMATION O - OBSERVATION Problem Types: M - MISSING W - WRONG E - EXTRA/SUPERFLUOUS CL = Cleared (or tick)				

Figure 4-6 Review Comments List

Audience, Distribution, and Usage

The Review Leader Form and Review Comments List are used to:

- document comments that need addressing
- act as a quality record concerning the review and the monitoring and closure of the comments raised

Distribute the Review Leader Form with associated Review Comments Lists to:

- author
- project library
- quality review participants

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Checklists should be as complete as possible and based on standard Oracle Method standards, where possible.
- ☐ Make sure the Review Leader Form and Review Comments List are complete.
- ☐ Comments should be raised for issues with technical content and also consistency with any standards.
- ☐ Comments should lead to changes, no change for reasons stated, or the comment is held over as a Problem Report for later resolution.

QM.030 - Quality Audit

In this task you conduct audits of the project to assess the project performance against the project plans, and the adherence to standards and procedures.

The objective of this task is to confirm that procedures are defined, followed, and appropriate to the project.

Conduct Quality Audits throughout the project, as defined in the Project Management Plan. Prepare and complete an Audit Report and Audit Action Form for each audit to define issues needing attention.

Deliverable

The deliverable for this task is the **Quality Audit**.

Prerequisites

You need the following input for this task:

☐ **Quality Audit Procedure (QM.010)**

The Quality Audit Procedure defines how the Quality Audit task is to be conducted.

☐ **Project Management Plan (CR.010)**

The Project Management Plan defines requirements for quality audits.

Task Steps

QM.030 Quality Audit

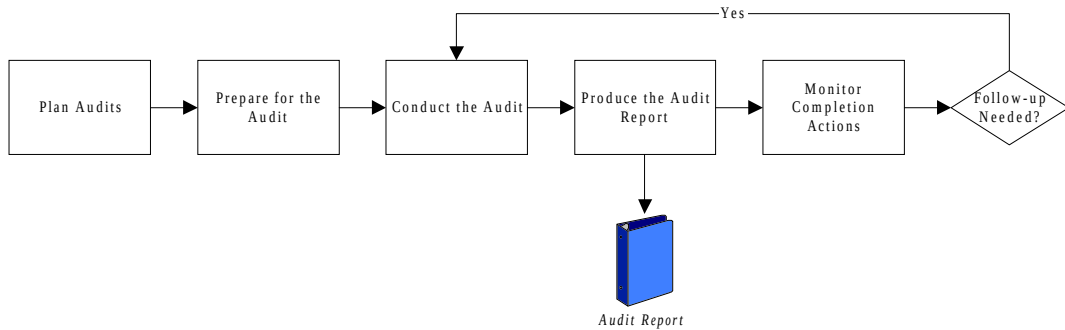


Figure 4-7 Task Flow for Quality Audit

The steps involved in this ongoing task are:

No.	Task Step	Deliverable Component
1.	Plan audits.	Project Management Plan and Workplan entries
2.	Prepare for the audit.	Checklist
3.	Conduct the audit.	Audit Actions
4.	Produce the audit report.	Audit Report
5.	Monitor completion actions.	None

Approach and Techniques

Audits are planned for particular points in the project and listed in the Project Management Plan. This task is then invoked to conduct the audit defined in the Project Management Plan. Audits are usually carried out by staff independent of the project team.

Clients may also want to participate in audits, or conduct their own audits. Clients usually appreciate a copy of the audit report as a comfort factor. Configuration audits have a different context and are explained in the Configuration Management chapter.



Attention: To get the most out of audits, training in auditing techniques is essential.

Plan Audits

- Plan project audits during preparation of the Project Management Plan.
- Identify your resource needs (trained auditors).

Prepare for the Audit

- Define the agenda (scope and scale of the audit).
- Collect any results or reports from previous internal and external audits.
- Define a checklist specifically for the audit.
- Collect any other evidence, such as plans and reports related to the area under audit, problem analysis, and risk lists.

Conduct the Audit

- Sample the processes under audit, and discuss plans with project staff.
- Record an audit action for each item where action is needed (and conceded to by the audited group) including the corrective action to be taken.
- Obtain agreement on action items.

Produce the Audit Report

- Complete the Audit Report based on Audit Action form details, and summarize the findings.
- Distribute to all staff involved in the audit.

Monitor Completion Actions

- Monitor that any necessary (corrective) action is taken.
- Conduct a follow-up audit (where many actions have been raised).

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Quality Auditor	Conduct audit, write Audit Report, and follow-up with any appropriate actions.	60
Project Support Specialist	Participate in the audit and carry out any agreed actions.	20
Project Manager	Arrange for audit and approve the findings of the audit.	20

Deliverable Components

The deliverable components for the Quality Audit are:

- ☐ **Audit Actions**
- ☐ **Audit Report**

Component Descriptions

This section describes each component.

Audit Actions

This component is used during audits to register actions at the time that the action is agreed to. In this way both the auditor and the person being audited are clear on what needs to be done. Actions are summarized in the Audit Report prepared after the audit is over. Figure 4-8 shows an example Audit Action Form which can be used to record audit actions.

Action Item: This factual description of the item requiring action includes enough detail to enable someone not involved with the audit to understand the circumstances of the action. Actions are raised where the project procedures and practices are not being followed.



Attention: Be clear that there is a need for action, so that unnecessary bureaucracy is avoided.

Background Information and Comments: This component provides any necessary qualification or background information about the circumstances surrounding the action, and identifies the procedure or practice that is being acted upon. Procedures and practices are those associated with your organization or as referenced in the Project Management Plan.

Proposed Corrective Action: This section lists the specific agreed upon actions and the names of the people who are assigned to the action item so that it is clear what they need to address. Think of corrective action in both the short term (to overcome any immediate problem) and long term (to get at any root causes). Usually actions will have to resolve both aspects, unless the action is a single lapse.

Audit Report

The Audit Report is the main deliverable from the Quality Audit. Before you leave an audit, you leave Action forms for those who are audited so they are clear on what actions have been agreed to. Then

you document the findings from an audit, actions, and responsibilities in Audit Reports. Figure 4-8 shows an example Audit Report form.

Overview: The overview section is your summary of the audit, noting anything you feel is worthy of highlighting, including good and bad points.



Attention: This is the section that managers will read, so what you say should be accurate but also either critical or offering praise, as appropriate.

The overview closes with the number of actions raised and an indication of whether a follow-up audit is required.

Action Summary: This lists the individual actions (forms) that were raised during the audit and left with the project team. The wording should just repeat what you filled out on the Action forms and not introduce new thoughts, unless these have been discussed further with the person being audited.

This also lists observations about the audit:

- where there may be opportunity for improvement
- as a record of the areas looked at during the audit
- to commend good work

Proposed Corrective Action (Summary): This section lists the actions and names of the people assigned action items so that it is clear to them what they need to address.

AUDIT ACTION FORM	
No: ___ / ___ / AAF / ___	
Business Unit	Project
Auditor	Date
Action Item	
Background Information / Comments	
Procedure reference:	
Proposed (Corrective) Action	
Agreed by: _____ Date: _____	
Person Taking Action	Proposed Completion Date
Follow-up Audit Required? YES/ NO _____	Follow-up Date
Action Category MA - Major Action A - Minor Action	Closure Signature & Date
Further Comments	

Figure 4-8 Audit Action Form

AUDIT REPORT			
No: ____/ ____/ AUR/ ____			
Business Unit	<Business Unit>	Project	<Project Name>
Auditor	<Auditor>	..	Date <Date of Audit>
Reason for Audit:	Scheduled / Follow-up *	Previous Audit Ref:	____/ ____/ AUR/ ____
Overview			
Action Summary			
1	A	Description	
X	O	Observation	
Proposed Corrective Action (Summary)			
Manager Signature		Proposed Completion Date	
.....			
Person Taking Action		Related Audit Report	
Follow-up Audit Required? Y/N		Date:	
Final Closure Sig & Date:			
Distribution	<Auditor>		
Categories of Action: MA - Major Audit Action A - Minor Audit Action O - Observation			

Figure 4-9 Audit Report

Audience, Distribution, and Usage

The Audit Action Form is used to:

- document actions that need addressing
- act as a quality record concerning the audit and the monitoring and closure of actions raised

Distribute the Audit Action Form to:

- person being audited
- auditor
- project library

The Audit Report is used to:

- summarize to people being audited the results of the audit
- make observations for improvement
- document actions that need addressing
- communicate quality issues or warm feelings to management

Distribute the Audit Report to:

- project manager
- business manager
- client, if agreed during preparation of the Project Management Plan
- project library

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Actions should only be raised for problems with procedure or working practice, not for routine project actions.
- ☐ Actions must be agreed to by the person being audited or their manager.
- ☐ The Audit Report should be reviewed by the project manager before distribution.
- ☐ The Audit Report should be checked for completeness and consistency by another person trained in the audit process.
- ☐ If there are political issues raised by the report, these should be discussed with business management before the report is published. Generally though, the audit report should be fact based.

QM.040 - Quality Measurement

In this ongoing task you collect measures of cost, time, and quality to help manage the development process or implementation project and implement improvements as progress occurs.

Specific metrics examples include problems found during testing at each level and risks and issues listed (open and closed) during a particular period. At a minimum, keep track of the actual time taken to perform a task, compared with the estimating metrics for that task. Relay recommended updates to these metrics to the party responsible for estimation.



Suggestion: Where possible, use available tools and easy to collect measurements, rather than spending time to create systems that collect the measurements.

Prerequisites

You need the following input for this ongoing task:

☐ **Quality Measurement Procedure (QM.010)**

The Quality Measurement Procedure defines how the Quality Measurement task is to be conducted.

☐ **Project Management Plan (CR.010)**

The Project Management Plan defines the strategy for Quality Measurement.

Task Steps

QM.040 Quality Measurement

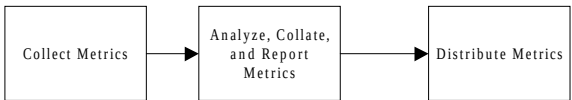


Figure 4-10 Task Flow for Quality Measurement

The steps involved in this ongoing task are:

No.	Task Step	Result or Output
1.	Collect metrics as execution progresses.	Project Metrics
2.	Analyze, collate, and report metrics to the project manager.	Metrics Reports
3.	Distribute metrics for use on other projects.	Distributed Metrics

Approach and Techniques

Collect Metrics

- Collect metrics as defined in the Project Management Plan or Measurement procedure.
- Collect metrics, where possible, from existing systems, rather than create project-specific measurement systems.

Analyze, Collate, and Report Metrics

- Analyze the metrics against any published ones, examining any discrepancies.
- Compile a metrics report, using the standard format.

Distribute Metrics

- Distribute within the project for use on future project work.
- Distribute to consulting management for use against standard measurements.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Collect measurements.	60
Project Manager	Nominate staff to provide and collate measurements, or arrange for a member of the project team to carry this out during the course of the project. Review collated measurements.	40

Outputs

The output for the Quality Measurement is:

☐ Metrics Report

Output Descriptions

This section describes each output for this ongoing task.

Metrics Report

The Metrics Report is a list of Project Metrics collected as defined in the Project Management Plan and published on a regular basis throughout the project for use by the project team.

The Metrics Report also contains data for use by consultants on other projects and to help refine estimating algorithms. To help with this data collection, send copies of the completed reports to your consulting practice.

Project Measurements: This table documents project performance against estimates. Modify the table sections to accommodate the phase you are in and the objectives for measurement set in the Project Management Plan.

Add any comments about the accuracy of the measurement and any actions that are needed.

Custom Development Evaluation: This table documents custom development module types, complexity, and estimates against actual performance of tasks for the range of tools you are using on the project.



Attention: Be clear as to whether your estimates include design and / or testing steps.

Tools, Techniques, and Approaches Evaluation: Track all tools, techniques, and approaches you have used during the project. It will be useful during the project if you have to look back at a previous phase, and to put the project measurements in the context of the environment and tools used.

Audience, Distribution, and Usage

The Metrics Report is used to:

- confirm actual figures against estimates
- predict and refine project completion dates
- quantify the quality of the development process
- feedback actual data to help future projects

Distribute the Metrics Report to:

- project manager
- project library
- consulting business manager

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Data has been checked for accuracy against project records.
- ☐ The report covers or exceeds what was defined in the Quality Plan.
- ☐ The report is consistent with any previous reports for the project, so that data can be compared for improvement.
- ☐ Get the report reviewed by a colleague or the project manager.

QM.045 - Support Healthcheck

The purpose of this task is to guide the project team in the performance of those project activities required by the Healthcheck process. The scope of the Healthcheck varies based upon many factors such as type of project, size, complexity, project status as determined by consulting policies and procedures, as well as other factors such the specific client, and visibility of the project.

The task steps described are intended to assist the project team in coordination with the external reviewer, in supplying the appropriate project information whether documents, deliverables, or process descriptions. In addition, guidance is provided to the consulting project team in compiling and using of the results of the Healthcheck and the presentation of those results to the client.

Deliverable

The deliverable for this task is the **Healthcheck Review**.

Prerequisites

Required

You need the following input for this task:

- ☐ **Quality Management Strategies, Standards and Procedures (QM.010)**

The Quality Management Strategies, Standards, and Procedures allow for the development of detailed quality management for the project. Specific guidance on the conduct of the Project Healthcheck will be derived from this document.

❑ Project Management Plan (CR.010)

The Project Management Plan defines the types of audits, reviews and Healthchecks to be conducted and delineates the requirements for each of these activities.

Optional

You may need the following input for this task:

❑ Consulting Policies and Procedures

The project manager and project team should consult the standard Consulting Policies and Procedures for specific information on the Healthcheck process.

❑ Healthcheck Review (QM.045)

Since Healthchecks may be conducted several times during the life of the project, previous results will provide valuable information and insights to the project team.

Task Steps

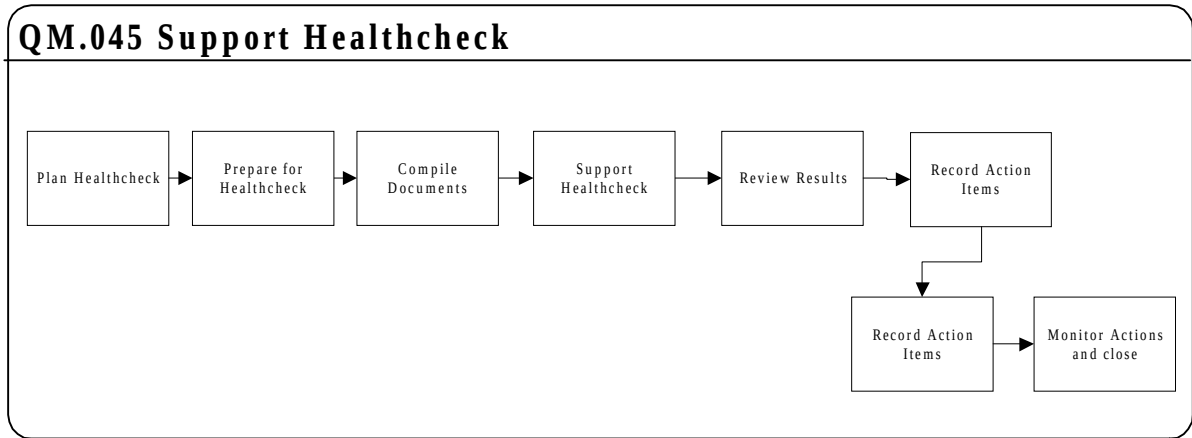


Figure 4-11 Task Flow for Support Healthcheck

The steps involved in this task follow:

No.	Task Step	Deliverable Components
1.	Plan Healthcheck.	Project Management Plan and Workplan Entries
2.	Prepare for Healthcheck.	Consulting Healthcheck Checklist
3.	Compile deliverables and documents for the Healthcheck.	Project Management Plan, Workplan, and Finance Plan
4.	Support the conduct of the Healthcheck.	Healthcheck Review Notes
5.	Review the Healthcheck results.	Healthcheck Report
6.	Record Action Items.	Action Items
7.	Review Healthcheck Report with client.	Record Client Action Items
8.	Monitor actions and close.	Resolved Healthcheck Review Notes

Approach and Techniques

This task provides guidance to the project manager and project team on the activities to be performed by the project team prior to, during, and after the Healthcheck is performed by external consultants. The scope and number of Healthchecks to be performed are determined by consulting policies and procedures and remain external to the project.

To further assist the project team, a preparatory checklist consisting of the areas to be covered by the Healthcheck as well as some of the deliverables reviewed during the Healthcheck has been added to the PJM templates.

Review results of the Healthcheck with the client after completing an initial review with the project staff.

Plan Healthchecks

- Plan the initial project Healthcheck during the preparation of the Project Management Plan.
- Identify the resource needs for the Healthcheck.

Prepare for Healthcheck

- Contact the external consultant to discuss Healthcheck review content.
- Determine the deliverables and documents required for review.
- Review the status of required deliverables and documentation.
- Develop interviewee list and schedule interviews.
- Obtain working space for reviewers.
- Prepare the documents and deliverables to be reviewed.
- Collect and compile any other required documentation.

Support Conduct of Healthcheck

- Participate in interviews, provide materials as required.
- Review interim results.

Review the Healthcheck Results

- Review the Healthcheck Review.
- Discuss any action items noted.
- Obtain agreement on the corrective action required.
- Distribute the Healthcheck review.

Assign Action Items

- Document the action items.
- Determine assignment of each action item.
- Set completion date or dates.

Review Healthcheck Report with Client

- Review the initial report with the external consultant.
- Note the recorded open and action items.
- Record any comments, recommendations from the client.
- Update the Issues Log with client notes.
- Distribute the updated Healthcheck review.

Monitor Completion Actions

- Perform weekly review of Issues log to monitor that all necessary correction actions are completed as scheduled.
- Conduct a follow up review with the client (if necessary, to clear action items).

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Collect and compile documents and deliverables, log and update action items, schedule Healthcheck, review Healthcheck results, and distribute reports.	60
Project Manager	Review Healthcheck schedule, meet with external consultants, review action items, assign action items, review results with client.	40
Client Project Manager	Meet with project manager to review Healthcheck results.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Healthcheck Review are:

- ☐ **Healthcheck Checklist**
- ☐ **Healthcheck Report**
- ☐ **Assigned Action Items**

Component Descriptions

This section describes each component.

Healthcheck Checklist

This component is used during the Healthcheck by external consultants to delineate the documents and deliverables to be reviewed for status, completeness, and quality.

Healthcheck Report

The Healthcheck Report is the narrative assessment of the overall status and inherent risk of the project. The report lists action items requiring correction by the project staff.

Assigned Action Items

After review with external consultants, action items noted in the Healthcheck report are recorded with assigned responsibility and due date. These Action Items are monitored for completion and required results.



Attention: Include scheduling the initial Healthcheck in your estimates.

Audience, Distribution, and Usage

The Healthcheck review is used to:

- Report the project progress and status to consulting management and the client.
- Communicate any deficiencies to the project team.

Distribute the Healthcheck Report to:

- client project manager
- project team
- project library

Distribute the Action Items to:

- client project manager
- project team
- project library

Distribute the Healthcheck Review to:

- client project manager
- project team
- project library

Quality Criteria

- ☐ Action items must be agreed to by the project team or their manager.
- ☐ Project manager must examine the Healthcheck Review prior to distribution.
- ☐ If there are political issues raised by the report, then resolve them with business management and the reviewer before the review is published.

QM.050 - Perform Quality Assessment

In this task you conduct an assessment of the completeness of Quality Control arrangements (reviews, audits, tests, and problem resolution) to assess the completion of the project to date.

A member of the project team may conduct the quality assessment, quality or it may be carried out by a quality consultant external to the project team.

Deliverable

The deliverable for this task is the **Quality Report**.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan (CR.010)**

The Project Management Plan defines the requirements for quality assessment.

☐ **Audit Action and Audit Reports (QM.030)**

The Audit Action and Audit Reports are used to provide evidence of the completeness of quality actions on the project.

☐ **Problem Log and Review Comments**

The Problem Log and Review Comments indicate the number of outstanding quality concerns about project deliverables at the time the assessment is conducted.

Task Steps

QM.050 Perform Quality Assessment



Figure 4-12 Task Flow for Perform Quality Assessment

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Assess the completeness of quality actions.	Completeness Assessment
2.	Document findings in a Quality Report.	Quality Report

Approach and Techniques

Keep the Quality Report as a stand-alone document, or incorporate it in the End Report. This decision depends on the scope of the project and whether either document will be distributed to the client.

Assess Completeness of Quality Actions

- Collect evidence from various quality records, complete quality related tasks and corrective actions, and document them in the Quality Report.
- Assess completeness of all deliverable reviews.
- Assess completeness of all audit actions.
- Assess closure of problem reports.

Document Findings in a Quality Report

- Complete the Quality Report using standard format, documenting the assessment findings.
- Ask the project manager to review and approve the report.
- Distribute the Quality Report.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Establish completeness of quality activities and document in a Quality Report.	95
Project Manager	Approve Quality Report.	5

Deliverable Components

The deliverable components for the Quality Report are:

- ☐ Executive Summary
- ☐ Quality Assessment Areas
- ☐ Summary of Action Items
- ☐ Appendices

Component Descriptions

This section describes each component.

Executive Summary

A summary and overall rating for the project, listing recommendations and client issues.

Quality Assessment Areas

A detailed report of the assessment topic (such as code development and testing) appropriate to the phase under assessment, including positive aspects, risk assessment, actions, long term actions, and the next quality review point.

Summary of Action Items

A summary of the recommendations and actions raised during the body of the report.

Appendices

Appendices contain supporting material relevant to the report, such as organization charts, workbench reports, and document copies.

Audience, Distribution, and Usage

The Quality Report is used:

- As an end-of-phase assessment of the quality (level) achieved
- As an end-of-project assessment of the quality achieved
- During spontaneous quality reviews or Healthchecks



Attention: Scheduled or follow-up audits are documented using the Audit Report.

Distribute the Quality Report to:

- project manager
- project library
- business manager

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Check the report for accuracy, completeness, and consistency (someone other than the author).
- ☐ Get the report approved by the project manager.

CHAPTER

5

Configuration Management (CM)

This chapter describes the Configuration Management process.

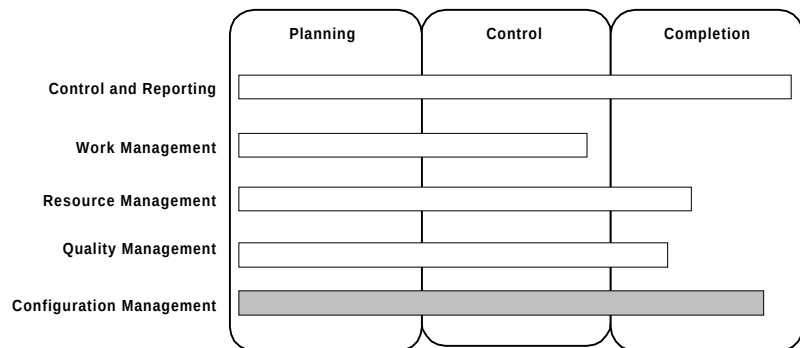


Figure 5-1 Configuration Management Context

Process Flow

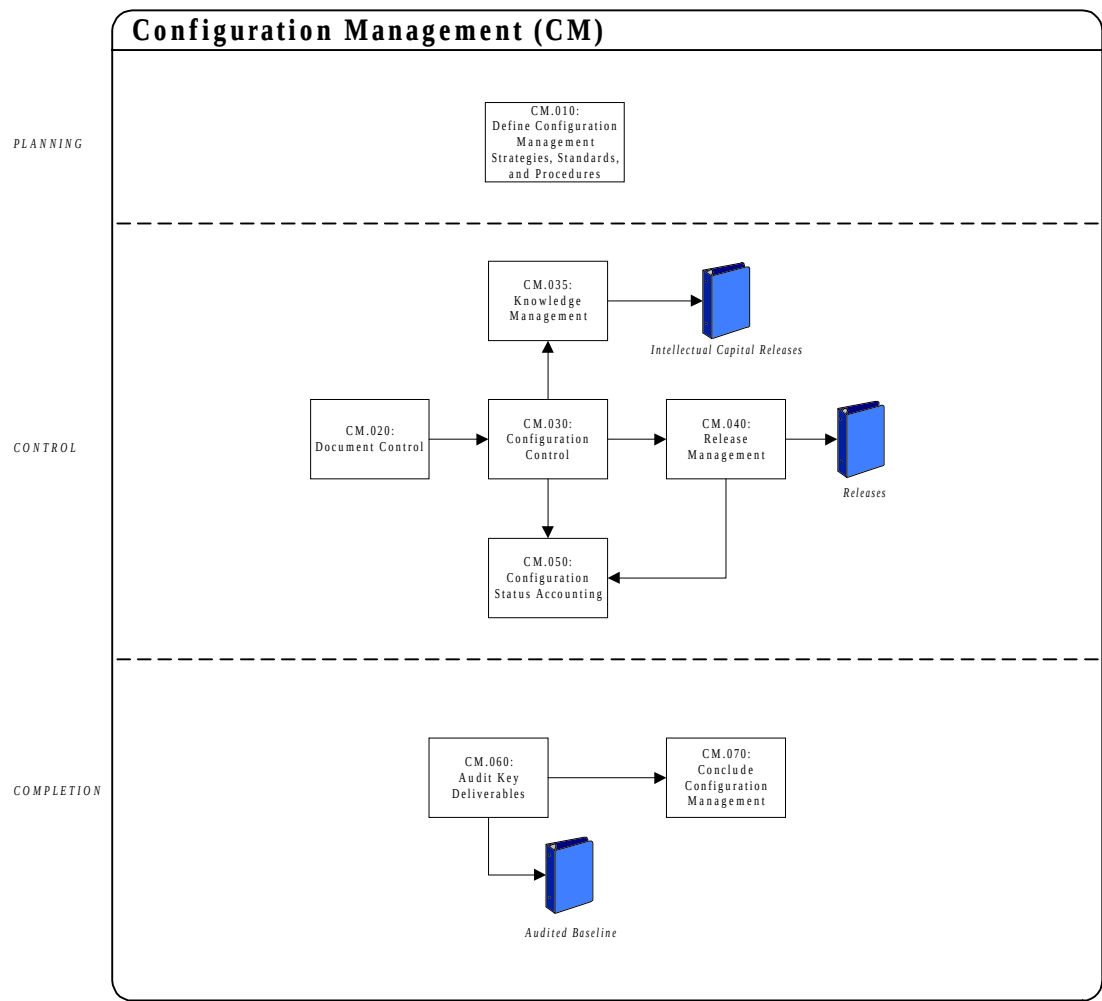


Figure 5-2 Configuration Management Process Flow Diagram

Approach

Configuration Management (CM) is the identification, control, and tracking of any item as it is produced during the development of a product. Think of Configuration Management as both a management service and a support service.

For the manager, the CM process represents the means by which the team incrementally assembles its deliverables into the final project products or releases. The manager relies on the CM process to provide reliable, complete, and reproducible releases of work to the rest of the project and to the client. CM ensures that only approved changes are made to project products.

For team members, the CM process represents a library system that permits them to function effectively as part of a team with common tasks and goals. The team members rely on the CM process to safeguard their work, give them recovery points, and to help organize workflow within the team.

Document Control manages project documents. Configuration Control supports the dynamic nature of these work products by maintaining changes. Knowledge Management facilitates the sharing of intellectual products through the capture of project knowledge contained in reusable project deliverables. Release Management provides Configuration Management products outside of CM. Configuration Status Accounting reports on the status and history of CM for management.

The Configuration Management process requires a CM Repository. The CM Repository contains a copy of all items under CM and also contains the information that relates those items to each other and to project deliverables.

Tasks and Deliverables

The tasks and deliverables for this process are:

ID	Task Name	Deliverable or Output Name	Type*
Planning			
CM.010	Define Configuration Management Strategies, Standards, and Procedures	Configuration Management Strategies, Standards, and Procedures	IT
Control			
CM.020	Document Control	Controlled Documents, Document Updates, Uncontrolled Documents	O
CM.030	Configuration Control	Configuration Items, Versions, Configurations, Promotions, Baselines, Configuration Changes	O
CM.035	Knowledge Management	Intellectual Capital Releases, Reuse Assets	O
CM.040	Release Management	Releases, Release Notes	O
CM.050	Configuration Status Accounting	Configuration Management Records, Progress Statements	O
Completion			
CM.060	Audit Key Deliverables	Audited Baseline	MO
CM.070	Conclude Configuration Management	Configuration Management Production Readiness	SI

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary in the *PJM Method Handbook*.

Objectives

The objectives of the Configuration Management process are:

- Identify and safeguard configuration items produced by project tasks.
- Ensure that configuration items can be traced to requirements.
- Includes the mechanism for reuse of deliverables and the preparation of deliverables into the Knowledge Repository.

- Ensure that changes to configuration items are performed in a controlled and agreed upon fashion.
- Ensure that the contents of each baseline are traceable to items under configuration control.
- Report the status of configuration items along with the reasons for any changes to those items.
- Record each release and its recipients.
- Ensure that project knowledge is shared within the consulting for use on future projects.

Key Deliverables

The key deliverables of this process are:

Deliverable	Description
Intellectual Capital Release	A release of project deliverables to the consulting Knowledge Repository.
Release	A baseline which is made available outside the CM process for a specific purpose, such as for testing, client delivery or capturing intellectual capital.
Audited Baseline	The baseline which confirms the functional and physical completeness of key deliverables.

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Client Project Manager	Acknowledge receipt of releases, and plan the transition of CM to the client. Approve deliverables for intellectual capital release.
Project Support Specialist	Develop and enforce CM procedures throughout the project, and manage the CM Repository and releases. Report CM status to the project manager. Manage the project library, and prepare CM documents and reports. Prepare, review, modify (if necessary), and submit deliverables to the Knowledge Repository as intellectual capital.
Project Manager	Ensure that all deliverables are available for reuse (when possible) and intellectual capital is captured and retained in the local knowledge repository. Authorize delivery of client releases and approve controlled documents.

Critical Success Factors

The critical success factors of the Configuration Management process are:

- All hardware, software, and documentation produced through, or delivered to, the project are under configuration management.
- Configuration Management procedures are simple, clear, and understood by all project members.
- The project infrastructure includes a tool set to support the CM Repository, and enforce CM procedures uniformly across all project activities.
- Changes to configuration items are controlled and documented.
- Releases to the client are timely, accurate, and acknowledged by the client.
- Controlled documents are reviewed and approved before delivery.

CM.010 - Define Configuration Management Strategies, Standards, and Procedures

In this task you create or update the plans which define how the Configuration Management process will support and implement the project's scope, objectives, and approach. You also plan how items submitted to CM will be identified, organized, promoted and controlled.

Deliverable

The deliverable for this task is the set of **Configuration Management Strategies, Standards, and Procedures**.

Prerequisites

Required

You need the following input for this task:

☐ **Scoping Project Management Plan (CR.010)**

The Scoping Project Management Plan is used to set the context for the CM Strategies, Standards, and Procedures.

☐ **Policies**

The consulting and client policies regarding CM define requirements for tools, repository management, auditing, archiving, and record keeping.

Optional

You may need the following input for this task:

☐ **Project Management Plan (CR.010)**

Use an existing Project Management Plan as the baseline for this task.

Task Steps

CM.010 Define Configuration Management Strategies, Standards, and Procedures

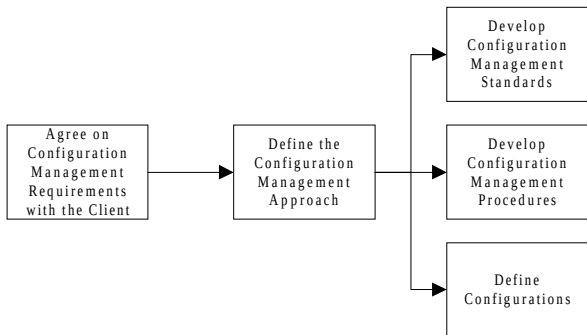


Figure 5-3 Task Flow for Define Configuration Management Strategies, Standards, and Procedures

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Agree on configuration management requirements with the client.	Configuration Management Requirements
2.	Define the configuration management approach.	Configuration Management Strategy
3.	Develop configuration management standards.	Configuration Management Standards
4.	Develop configuration management procedures.	Configuration Management Procedures
5.	Define configurations.	Configuration Definition

Approach and Techniques

The procedures you write in this task correspond to the CM ongoing tasks performed during project execution. Know how to determine the types of configuration items and configurations to be managed based on the method routes to be executed. Also, be familiar with both the project infrastructure and production environment.

Agree on Configuration Management Requirements with the Client

- Review the scope, objectives, and approach.
- Identify any consulting policies affecting configuration management.
- Identify any client policies affecting configuration management.

Define the Configuration Management Approach

- Identify consulting best practices and standards.
- Ensure that the project library structure supports the creation of the local consulting repository.
- Document configuration management strategy using the Project Management Plan format.

Develop Configuration Management Standards

- Identify any existing consulting standards.
- Customize standards to satisfy approach.
- Add additional standards to meet client requirements.

Develop Configuration Management Procedures



Suggestion: If support tools are expected to be used to support these procedures, but are not available at the time this task is executed, then write the procedures initially without reference to the tools. Next, revise the procedures during subsequent planning to incorporate tool-specific provisions.

- Write detailed instructions for document control, configuration, control configuration status accounting, and configuration audit.
- Schedule appropriate Knowledge Management Reviews of deliverables at the end of phases and upon project completion.

Define Configurations

- Define configuration item types. All items of significance to a project should be under configuration management, unless they are intended to be discarded after a short time (such as working notes).
- Identify configurations. A configuration is a logical grouping of configuration items. Name each expected configuration and its member configuration items (or item types) identified.
- Define the interrelationships between item types. There are a number of reasons why items are associated. For example, a traceability relationship describes the derivation of one item from another. A dependency relationship indicates that an item type is dependent on another item type, in whole or in part, for its definition or use.
- Define a life-cycle for each item type. Identify an hierarchy of possible states for the item type. For each state transition (promotion), document a quality criteria and authorization requirement to be enforced.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Prepare strategies, standards, and procedures. Define configurations.	80
	Ensure the scheduling of appropriate Knowledge Management reviews.	
Project Manager	Review and approve all strategies, standards, and procedures.	20
Client Project Manager	Agree to configuration management requirements, and identify client policies affecting approach.	*

* Time not estimated.

Deliverable Components

The deliverable components for the Configuration Management Strategies, Standards, and Procedures are:

- ☐ **Document Control**
- ☐ **Configuration Definition**
- ☐ **Configuration Control**
- ☐ **Release Management**
- ☐ **Configuration Status Accounting**
- ☐ **Configuration Audit**

Component Descriptions

This section describes each component.

Document Control

This component defines standards and procedures for creating, distributing, maintaining, and safeguarding project documents. The project library, in physical terms, is usually part paper and part electronic, so the key aspect of this component is delimiting the paper from the electronic items, while defining control mechanisms for both.

Configuration Definition

Describe any project-specific naming conventions for each type of item to be produced and documentation naming conventions. Describe directory names and storage arrangements, including versioning, if different from item naming. Define the following:

- Configuration item types of significance to a project.
- Interrelationships between item types.
- A life-cycle for each item type. Identify a hierarchy of possible states for the item type. For each state transition (promotion), document a quality criteria and authorization requirement to be enforced.

Configuration Control

Describe arrangements for the control of configuration items, such as placing in a controlled area when tested and the mechanisms to be employed for managing technical changes, including approval for the change and correction due to problems being found. Describe when baselines will be created and who will create it.

Release Management

Describe the procedure for preparing a release, including responsibilities and deliverables. Describe the arrangements for baselining and releasing software to the client (for acceptance or final delivery), including how a copy will be maintained in the project environment to support acceptance testing and problem solving.

Configuration Status Accounting

Define any item completion reporting to record progress and track problems.

Configuration Audit

Define arrangements for configuration auditing or refer to a separate procedure.

Audience, Distribution, and Usage

The Configuration Management Strategies, Standards, and Procedures are used:

- to detail arrangements for configuration management beyond the high level definition in the Project Orientation Guide
- to task individual team members for configuration management tasks

Distribute the deliverable to:

- project leader for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable, where appropriate

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are the procedures clearly defined?
- ☐ Are the components complete and consistent, especially amongst themselves?

CM.020 - Document Control

In this ongoing task you create and maintain documents for use within the project and for delivery to the client.

The objectives of this process are to verify that:

- Documents are easily accessed for reading and protected from unauthorized change.
- Formal documents have a controlled circulation.
- Document recipients are aware of the document status.
- Appropriate project members review and approve documents before documents are issued.

Prerequisites

Required

You need the following input for this task:

☐ **Document Control Procedure (CM.010)**

The Document Control Procedure specifies how this task is conducted on the project.

☐ **Project Library**

The Project Library is the repository for all project documentation and document archives managed by this task. The library is part of the CM Repository.

Task Steps

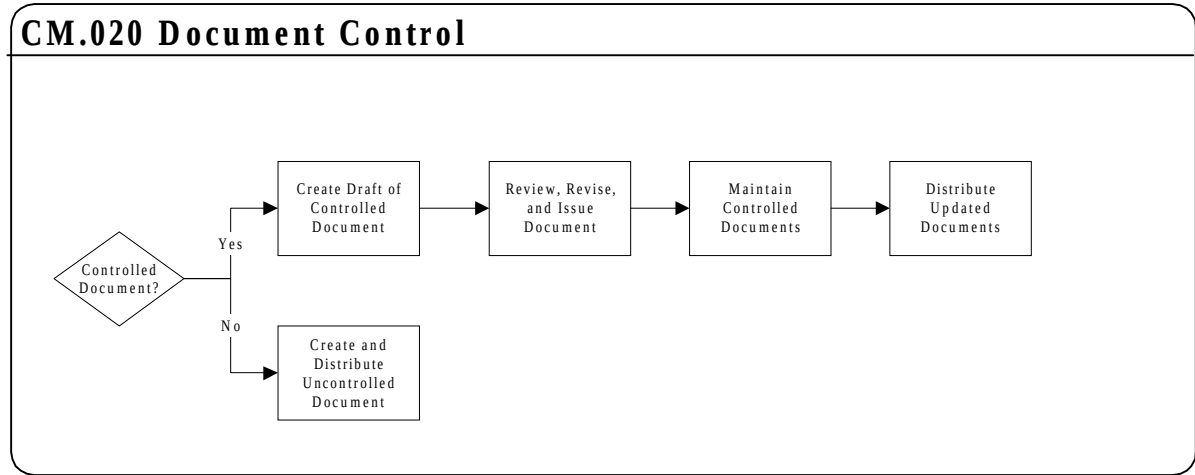


Figure 5-4 Task Flow for Document Control

The steps of this ongoing task are:

No.	Task Step	Result or Output
1.	Create draft of a controlled document.	Initial Draft Document
2.	Review, revise, and issue document. Log and store the master in the project library.	Approved Document
3.	Maintain controlled documents.	Updated Documents
4.	Distribute updated documents with notification of update.	Document Updates
5.	Create and distribute an uncontrolled document.	Uncontrolled Document

Approach and Techniques

Determine the level of control of each document produced in your project. As a rule, **control** any document that influences project scope. These documents usually represent, or are themselves, the key deliverables of your project, and undergo a quality review and approval. Other documents may contain interim, internal project information, or are working documents, such as meeting minutes. These may be issued as **uncontrolled** documents.

Create Draft of Controlled Document

- Assign a version number to the document. Place the word *Draft* in front of the version number to indicate an unapproved document. Assign a control number to the document, and complete the cover page.
- Create the change record which is used to give a history of changes to the document.
- Create the reviewer list which indicates those who initially reviewed the document.
- Create the distribution list, which shows the recipients of the document, including the project library and other possible stores.
- Create the table of contents, which lists the first two levels of headings, as a rule.
- Create the introduction. Most deliverable documents have an introduction component that introduces the reader to the deliverable, indicates why they are reading it, and tells them what they are expected to do with it.
- Create the body of the document.
- Log the document and version in the Document Index.

Review, Revise, and Issue Document

Review every controlled document you produce. Determine the specific number and type of reviews each document requires. If the document is to be delivered to the client, then plan to review the document internally before releasing it to the client for their review.

- Determine a review technique, the participating reviewers, and complete the reviewer list.
- Distribute the document for review, and ensure that one copy is stored in the Project Library.
- Conduct the review of the document using the quality review procedure. Capture review comments using the Review Comments List form. Use a new form for each review.
- Assign each review comment to a project member for resolution. Some comments require more extensive investigation as a problem or an issue.
- Resolve review comments, and record the resolution action on the review comments list.
- Publish and distribute the revised document, logging the new version number in the Document Index. If the document is still in draft form, then attach the annotated Review Comments List to it.

Maintain Controlled Documents

Control updates to all controlled documents. On a project, this will usually be done by using a controlled project library (one person controls access to the documents) or controlled electronic directories. For each controlled document update:

- Make the required changes to the document, and ensure that the version, date, and change record of the document are updated.
- If the document is approved, then indicate approval on the cover page of the document, and complete the distribution list. Otherwise, update the reviewer list.
- Complete a Document Update Notice, and place it in front of the document cover page.
- If an unapproved document is being updated as a result of a document review, then place the Review Comments List behind the Document Update Notice.

Distribute Updated Documents

For each document update to be distributed:

- Distribute the document update, and ensure that a copy of the complete updated document is stored in the project library.
- If the document is copy-controlled, then request that superseded copies are returned or destroyed.
- Log the document version in the project Document Index.

Create and Distribute Uncontrolled Document

A document may be uncontrolled, because it will not be updated or will be shortly replaced by a controlled document. Review the document, and retain a copy of it in the project library.

- Assign a control number to the document and complete the cover page.
- Create the table of contents, introduction, and body of the document, as needed.
- Distribute the document, and store a copy of the document in the Project Library. Log the document in the project Document Index.

Roles and Responsibilities

The roles and their associated responsibilities for this task follow:

Role	Responsibility	%
Project Support Specialist	Enforce document control procedures and standards. Ensure documents that are configuration items are also controlled according to the Configuration Control Procedure.	90
	Establish and maintain the Project Library per document control procedures and standards. Issue new and updated documents and prepare Document Update Notices.	
Project Manager	Determine key deliverables which require controlled documents. Approve all controlled documents.	10
Client Project Manager	Approve all controlled documents requiring client approval.	*

* Time not estimated.

Outputs

The outputs for the Document Control are:

- ☐ **Controlled Document**
- ☐ **Document Update**
- ☐ **Uncontrolled Document**

Output Descriptions

This section describes each output for this ongoing task.

Controlled Document

A controlled document is a document which constitutes or represents a project deliverable for approval internally, or by the client. This document is subject to change. A Document Index provides a list of controlled project documents for reference by team members. See the Document Index form for small and medium-sized projects.

Document Update

A controlled document is periodically updated. All updates to a controlled document should also be controlled. The Document Update Notice (DUN) can be used to record the change activity when a document is updated. Information on the DUN should provide the reader with a clear view of what changes have been made in updating the document. See the Document Update Notice.

Uncontrolled Document

An uncontrolled document is produced once, for information only, and is not subject to formal approval or change control. These documents are not tracked; however, master copies are stored in the Project Library.

DOCUMENT INDEX

Ref: ____/____/IND/____

Document Title	Document Reference	Document Version & Date			Storage Location
		*	*	*	

* Record Latest Issued (non-Draft) Version & Date

Figure 5-5 Index and Master List

DOCUMENT UPDATE NOTICE

Ref: ____/ ____/ DUN/ ____

Date: _____

The document attached (document reference : ____/ ____/ ____/ ____) has been updated as follows:

Previous Version Number: Issue ____

New Version Number: Issue ____

Summary of Update

Instructions to Document Holder

Figure 5-6 Document Update Notice

Audience, Distribution, and Usage

The Index and Master List is used in the following ways:

- to maintain a record of the latest documents and their versions

The Document Update Notice is used in the following ways:

- to explain update details to document owners

Distribute the outputs as follows:

- to the project leader for review
- to the Project Library for control
- to the document distribution list

Quality Criteria

Use the following criteria to ensure the quality of these outputs:

- ☐ Ensure that each controlled document is maintained up-to-date.
- ☐ Ensure that a copy of the complete updated document is stored in the Project Library.
- ☐ If a document is copy-controlled, have superseded copies returned or destroyed.
- ☐ Ensure that DUN instructions are precise.
- ☐ Check the details of the DUN instructions.

CM.030 - Configuration Control

In this ongoing task you manage the dynamic aspects of configurations.

When a new project deliverable is created, a new configuration item is usually also created. When you accept an authorized revision to a configuration item, a new version of that item is created. The degree of control you exercise over changes to any configuration item is based on the state of the item in its life-cycle. Promotion of an item reflects increasing value and confidence in the item and, hence, a need for stricter control of changes to the item.

During the project you will find it necessary to baseline the contents of a configuration by designating a version of each item in the configuration. Once a baseline has been designated, it is logically *frozen*, and the versions represented in that baseline are fixed.

Prerequisites

Required

You need the following input for this task:

☐ **Configuration Control Procedure (CM.010)**

The Configuration Control Procedure specifies how this task is conducted on the project.

☐ **Configuration Definition (CM.010)**

The Configuration Definitions contain the specifications of how each configuration on the project is to be constructed during this task.

☐ **CM Repository**

The CM Repository stores and organizes all items, and information about them, created by this task.

Task Steps

CM.030 Configuration Control

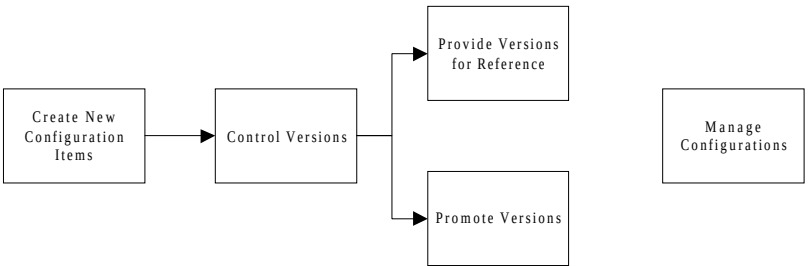


Figure 5-7 Task Flow for Configuration Control

The steps of this ongoing task are:

No.	Task Step	Result or Output
1.	Create new configuration items and assign them to configurations.	Configuration Items, Configurations
2.	Control the creation of new versions of configuration items.	Versions
3.	Provide copies of versions for reference.	Reference Versions
4.	Promote versions to a higher level of quality and change control.	Promotions
5.	Manage the content and baselines for configurations.	Configurations, Baselines

Approach and Techniques

Manage Configurations

Configurations require routine maintenance. Update the Configuration Definition when any of the following actions is taken.

- Add or remove items in a configuration as agreed to by project management.
- Create or delete configurations as agreed to by project management.
- Create a baseline for a configuration by designating a member version for each item in the configuration. Designate a baseline identifier, and indicate whether it is an incremental or full baseline.
- Protect (freeze) versions in a configuration from change, as agreed to by project management.
- Remove protection (unfreeze) versions in a configuration as agreed to by project management.

Create New Configuration Items

As deliverables are produced on the project, identify within each deliverable the products that need to be controlled as unique configuration items according to your established CM strategy, standards, and procedures.

- Assign an item type and unique identifier to the item, and log the item in the Configuration Item Index.
- Identify the configuration (or configurations) to which the item belongs.
- Identify any relationships between the item and other configuration items.

Control Versions

When you accept an authorized revision to an item, you create a new version for it. Record the following actions in the Configuration Item Index:

- Ensure that the need for revision to an item is valid. Use the established Configuration Control Procedure.
- Provide, or *check out*, a working copy of the version to the designated project member for use within a project task.
- *Lock* the version from change by others, unless specifically authorized in your procedures.
- Store, or *check in*, the working copy from the authorized project member on demand and associate it with the predefined version identifier.
- Remove protection from, or *unlock* the changed version.

Provide Versions for Reference

Give project members the ability to examine the contents of any version of a configuration item by providing a reference, or *read-only* copy of the version.

- Identify the version of a configuration item requested for reference, the target format, and the destination of the copy.
- Create a copy of the version, and provide it to the requesting project member in the requested format and location.
- Do not allow the reference copy to be returned as a new version of the item.

Promote Versions

Promote a version when it has passed a quality requirement or when an authorized manager has requested it. Use the Project Management Plan and the Configuration Control Procedure as the basis for the promotion. Note the following actions in the Configuration Item Status Record:

- Promote an item when it meets a quality requirement defined in the Configuration Control Procedure.
- Promote or demote an item when requested by an authorized project member.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Perform all steps, unless delegated to other project members, as specified in the Configuration Control Procedure.	80
Project Manager	Perform all steps and actions which are delegated to them by the project manager. Provide management input and decisions.	20
Client Project Manager	Perform all steps and actions which are delegated by the project manager. Provide management input and decisions.	*

* Time not estimated.

Outputs

The outputs for Configuration Control are:

- ☐ **Configuration Item**
- ☐ **Version**
- ☐ **Configuration**
- ☐ **Baseline**
- ☐ **Promotion**

Output Descriptions

This section describes the outputs for this ongoing task.

Configuration Item

A configuration item is a product of a task, or a receivable of the project, which is placed under configuration control. There may be multiple configuration items for a given deliverable, such as when multiple modules are represented in the same deliverable. Configuration items are tracked using a unique identifier, according to the project configuration control procedure.

Versions

A version of a configuration item represents the state of that item at a given point in time. Versions are derived from one another by revision, starting from a common initial version.

Configuration

A configuration represents the aggregation of multiple configuration items into logical groups for tracking and manipulation as a whole. For example, a subsystem or work package might be given a named configuration. All configuration items must be members of at least one configuration.

Baseline

A baseline is a named set of versions within one or more configurations, fixed at a specific point in time. Baselines record the state of project deliverables at key points in the project or to prepare project deliverables for release.

Promotion

A promotion of a configuration item indicates that it has added value to the project. Promotions are made within a predefined life-cycle for the relevant type of configuration item.

CM.035 - Knowledge Management

In this ongoing task you manage project deliverables for reuse within the consulting organization and the project. This task facilitates the sharing of intellectual capital through the capture of project knowledge.



Attention: By contractual definition, consulting organizations may have intellectual property rights to the intellectual capital amassed during a project. Consideration must be given to the nature of any site or client-specific information incorporated in the documentation of intellectual capital prior to its distribution or reuse.

Prerequisites

Required

You need the following inputs for this task:

☐ **Project Management Plan (CR.010)**

The Project Management Plan defines the names and types of the key deliverables to be produced. It also contains the project-level plans and policies regarding Knowledge Management.

☐ **Configuration Management Strategies, Standards and Procedures (CM.010)**

The Knowledge Management Procedure specifies how this task is conducted on the project, using the tools and organization of the project.

☐ **Consulting Knowledge Management Policies and Procedures**

These define how to prepare and submit the project deliverables by the consultant as intellectual capital to the consulting Knowledge Repository. They include standards for assigning attributes to each deliverable to ensure consistency in cataloging submissions.

❑ CM Repository

The CM Repository is the project deliverable library, containing all project deliverables which are candidates for submission to the Knowledge Repository as intellectual capital.

❑ Knowledge Repository

The Knowledge Repository is the system employed by consulting for storing, cataloging, and retrieving intellectual capital for reuse within the consulting organization. It may include project-level components which are used locally by the project team in conjunction with the project CM Repository.

Task Steps

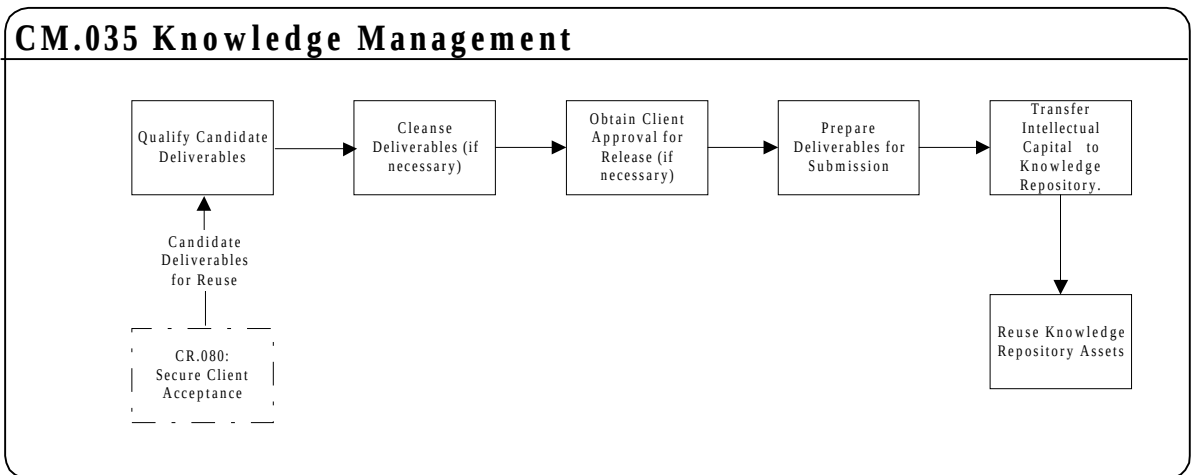


Figure 5-8 Task Flow for Knowledge Management

The steps of this ongoing task are:

No.	Task Step	Deliverable Component
1.	Qualify candidate deliverables.	Qualified Deliverables
2.	Cleanse deliverables (if necessary).	Cleansed Deliverables

No.	Task Step	Deliverable Component
3.	Obtain client approval for release (if necessary).	Deliverables Approved for Release
4.	Prepare deliverables for submission to the Knowledge Repository.	Intellectual Capital Baseline
5.	Transfer intellectual capital to the Knowledge Repository.	Intellectual Capital Release
6.	Reuse Knowledge Repository assets.	Reuse Assets

Approach and Techniques

Qualify Candidate Deliverables

- Verify project deliverables that are candidates for submission to the Knowledge Repository, either from CR.080 (Secure Client Acceptance) after client acceptance of key deliverables or at project milestones for other deliverables.
- Verify that the deliverables have been reviewed by the project team for quality and completeness.
- Perform any additional quality checks on the deliverables required by consulting policy.

Cleanse Deliverables (if necessary)

- Check out deliverables from the CM Repository (see CM.030, Configuration Control).
- Cleanse sensitive, proprietary, and confidential client information from the deliverables, as required by the Project Management Plan and Knowledge Management Procedure.
- Check in new revisions of the deliverables to the CM Repository, indicating that they are configured for intellectual capital release.

Obtain Client Approval for Release (if necessary)

Perform the following only when consulting has qualified intellectual property rights to the project deliverables:

- Prepare a review copy of all deliverables to be released.
- Schedule a review meeting with the client project manager.
- Review the deliverables to be released.
- Document any comments or changes required by the client, and modify the deliverables in the CM Repository as required.
- Obtain written approval for release of the deliverables from the client project manager.

Prepare Deliverables for Submission

- Using consulting Knowledge Management policies and procedures:
 - Verify that all deliverables meet the designated consulting quality criteria, and remedy deficiencies as required.
 - Properly catalog (tag) all deliverables (for example, type of document, industry, service line, product designations if applicable, method task, and deliverable).
 - Request support from the consulting Knowledge Management staff for review and assistance as needed.
- Create a baseline for the prepared deliverables in the CM Repository, identifying them as intellectual capital ready for release.

Transfer Intellectual Capital to Knowledge Repository

- Use CM.040, Release Management, to construct a release of the intellectual capital baseline from the CM Repository.
- Follow Knowledge Management policies and procedures.
- Place the deliverables from the intellectual capital release into the designated project or consulting organization location for incorporation into the Knowledge Repository.
- Document release contents in the Knowledge Repository.

Reuse Knowledge Repository Assets

- Research the consulting Knowledge Repository for assets that could be reused by the project team.
- Download reuse assets into the project CM Repository for project team access.
- Track project modifications to the reuse assets that made them more suitable to the project.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Identify, qualify, prepare, review, modify (if necessary), and submit deliverables into the Knowledge Repository as intellectual capital.	80
	Research Knowledge Repository for potential reusable components for the project team and make them available to the project team.	
Project Manager	Review the deliverables for adherence to quality standards and completeness.	20
	Recommend modifications and review the deliverables with the client for approval.	
Client Project Manager	Approve the deliverables for release.	*

Role	Responsibility	%
Consulting Business Manager	Ensure that candidate project deliverables are reviewed and submitted into the consulting Knowledge Repository according to consulting policies and procedures.	*

* Time not estimated.

Outputs

The outputs for the Intellectual Capital Release are:

- ☐ **Intellectual Capital Release**
- ☐ **Reuse Assets**

Output Descriptions

This section describes each output for this ongoing task.

Intellectual Capital Release

This component contains a release of project deliverables constructed from a baseline in the project CM Repository according to Release Management procedures, after being prepared for submission to the Knowledge Repository.

Each project deliverable in this baseline is qualified, reviewed, and cleansed of sensitive, confidential, or proprietary client data if necessary. Each deliverable is cataloged according to consulting and project policies, standards, and procedures. The resulting deliverable revision in this baseline may therefore be specifically intended for Knowledge Management reuse and not intended for further use on the project. If this is the case, the revision is clearly identified by a branch or similar mechanism in the CM Repository.

The Release Receipt documents information about the contents of the release and records acknowledgment of their receipt. A receipt is used

when deliverables are provided to client satisfaction, in accordance with project requirements, or when cleansed deliverables are submitted to the Knowledge Repository. An example of a Release Receipt is found in CM.040 Issue/Risk Management.

Reuse Assets

Reuse Assets are information components from the consulting Knowledge Repository that provide value to the project in terms of quality, cost, or schedule. Once identified, they are downloaded from the Knowledge Repository and stored in the CM Repository for assessment by the project team according to the Knowledge Management Procedure.

Audience, Distribution, and Usage

The Intellectual Capital Release is used:

- as potential reuse assets within consulting
- as a basis for future knowledge reviews of deliverables on the project
- to document the approved release of project deliverables outside of the project

Reuse Assets are used:

- to improve the quality of project deliverables
- to reduce the cost or schedule required to produce project deliverables
- to demonstrate consulting best practices to the client

Distribute the Intellectual Capital Release to the designated Knowledge Repository uptake point. Distribute the Intellectual Capital release receipt to:

- project manager for review
- consulting business unit responsible for the project
- project library
- client project manager to agree to the release, if required

- Knowledge Repository according to consulting procedures

Distribute Reuse Assets to the CM Repository for assessment and use by the project team.

Quality Criteria

Use the following criteria to ensure the quality of this deliverable:

- ☐ Are the procedures clearly defined?
- ☐ Are the components complete and consistent, especially among themselves?

CM.040 - Release Management

In this ongoing task you construct releases, and deliver them to designated project locations or to the client, in an arranged format or medium.

A release may issue items from the CM Repository in a wide variety of forms, such as documentation, source code, executables, and data files. Releases can occur from the CM Repository as often as necessary to support the internal needs of a project or the delivery requirements of the client.

Document a procedure for any release which is complex or requires repetitive execution. Consider tool support for building releases if build procedures are complex and can be automated. Consider automation support for distributing releases if multiple sites are involved, releases are frequent, or release installation is complex.

Acknowledge releases to the client with a Release Note. This does not constitute client acceptance but rather acknowledgment of receipt of the release.

Prerequisites

Required

You need the following input for this task:

☐ **Release Management Procedure (CM.010)**

The Release Management Procedure specifies how this task is conducted on the project.

☐ **CM Repository**

The CM Repository contains the configurations which are used to establish baselines and releases.

Task Steps

CM.040 Release Management



Figure 5-9 Task Flow for Release Management

The steps of this ongoing task are:

No.	Task Step	Result or Output
1.	Plan releases.	Build Procedures
2.	Construct releases.	Constructed Releases
3.	Deliver releases.	Delivered Releases
4.	Secure client receipt.	Release Receipts

Approach and Techniques

Plan Releases

- Plan the content, media, and destination for a release. If it is a client release, then agree on the purpose and recipient. Assign an identifier to the release.
- Create or document a procedure to construct the release from the CM Repository.

Construct Releases

- Execute the build procedure for the release.
- Verify the release contents by inspection or test.
- Revise and repeat the release procedure if necessary.
- Archive the build procedure for later execution if needed.

Deliver Releases

- Verify that the destination is prepared for release.
- If multiple destination sites are involved, then verify the procedure for release distribution.
- Deliver the release to each destination in the specified media.
- Verify that the release has been properly received and installed at the destination.
- Resolve any release distribution problems that are encountered.

Secure Client Receipt

- Prepare the Release Note for the release.
- Meet with the client, and verify that the release contents were received and are acceptable.
- Ask the client to acknowledge receipt of the release by signing the Release Note.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Determine contents of releases and prepare release notes. Construct and execute build procedures.	90
Project Manager	Authorize delivery of client releases.	10

Role	Responsibility	%
Client Project Manager	Acknowledge receipt of client releases.	*

* Time not estimated.

Outputs

The outputs for Release Management are:

- ☐ **Release**
- ☐ **Release Receipt**

Output Descriptions

This section describes each output for this ongoing task.

Release

This component represents a baseline constructed from Configuration Management according to a procedure. The release is then delivered to a destination or target. A release may contain items in a variety of formats, such as executable programs, source code, and documentation.

Release Receipt

This component documents information about the contents of a release and records acknowledgment of their receipt. Always use a when deliverables are provided to client satisfaction, in accordance with project requirements. See the example Release Note form which can be used to document release receipt.

RELEASE NOTE	
No: ____ / ____ / RLN / ____	
Project Code:	
Contract Number:	
Name of Release	
Baseline for Release	
Reason for Release:	
Observations from testing: (Specific problem reports are referenced later in this document)	
Comments:	
Date:	
The deliverables specified in this document are agreed to be a complete list of items necessary to hand over from the project. Receipt Please complete this section if all the items specified in the attached documentation have been successfully received and are believed to be present and correct.	
For Consultant	For Customer:
Agreed by:	Agreed by:
Position:	Position:
Date:	Date:

Figure 5-10 Release Note

Audience, Distribution, and Usage

The Release Note is used:

- for any release outside of the project
- for internal releases, such as for system testing, where an accurate record of the release is needed

Distribute the Release Note to:

- project leader for review
- project library for control
- client project manager to accept the components and agree on the scope, detail, and quality of the deliverable
- anyone receiving the release

Quality Criteria

Use the following criterion to ensure the quality of these outputs:

- ☐ Are all tests and reviews complete?

CM.050 - Configuration Status Accounting

In this ongoing task you monitor and report on the status and history of configuration items and configurations.

The purpose of Configuration Status Accounting is to maintain a continuous record of the status of all items in the CM Repository. The information required for comprehensive status accounting includes:

- a description of each configuration item
- the configuration which represents each baseline
- the time at which each baseline was established
- when each configuration version and change was included in a baseline
- the status and item impacts of each configuration change
- the reason for each revision to a configuration item

Although status information may be routinely prepared for management review, make it easily accessible to all members of the project. Make predefined reports available from the tools which automate the CM Repository.

Prerequisites

Required

You need the following input for this task:

☐ **Configuration Status Accounting Procedure (CM.010)**

The Configuration Status Accounting Procedure specifies how this task is conducted on the project.

☐ **CM Repository**

The CM Repository is the organizational structure which stores information and records about the configurations examined by this task.

Task Steps

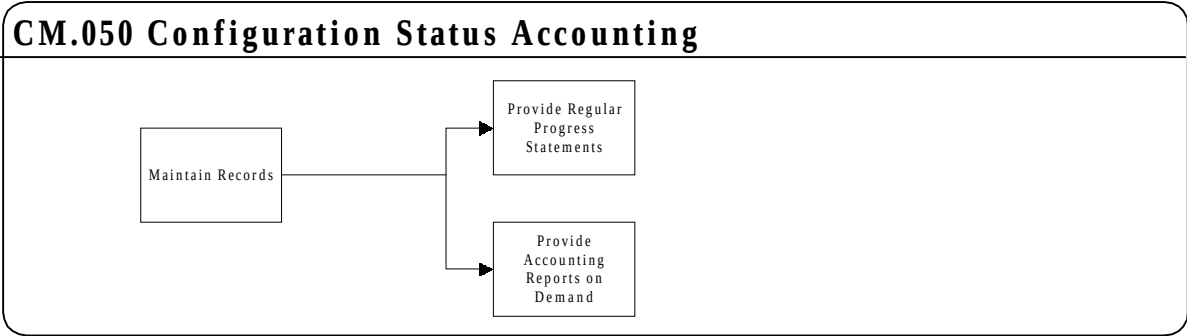


Figure 5-11 Task Flow for Configuration Status Accounting

The steps of this ongoing task are:

No.	Task Step	Result or Output
1.	Maintain records of configuration management activity and the CM Repository.	Configuration Management Records
2.	Provide regular progress statements on configuration management.	Configuration Management Progress Statements
3.	Provide accounting reports on demand.	Configuration Management Accounting Reports

Approach and Techniques

Maintain Records

- Ensure that records of all configuration items, configurations, baselines, and releases are maintained.
- Ensure that accounting records of all CM process transactions are maintained for traceability.

Provide Regular Progress Statements

- Summarize new configuration items created during the period.
- Summarize configuration changes and baselines created during the period.
- Summarize change and revision activity to configurations during the period.
- Summarize release activity during the period.

Provide Accounting Reports on Demand

- Report on the relationships between configuration items.
- Report on the promotion history of versions.
- Report on the revision history of configuration items.
- Report pair-wise comparisons of versions.
- Report on the status and change history of configuration items.
- Report on the history of configurations and their baselines.
- Report on the history and content of all project releases.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Produce and distribute progress statements. Review reports with project leaders. Review information required for progress statements.	80
Project Manager	Obtain accounting information on demand from the CM Repository.	20
Client Project Manager	Obtain accounting information on demand from the CM Repository.	*

* Time not estimated.

Outputs

The outputs for Configuration Status Accounting are:

- ☐ **Configuration Management Records**
- ☐ **Configuration Management Progress Statements**
- ☐ **Configuration Management Accounting Reports**

Output Descriptions

This section describes each output for this ongoing task.

Configuration Management Records

This component represents all information about the status and activity history of the Configuration Management (CM) process on the project. It is strongly recommended that all information technology projects invest in an automated configuration management support tool. A CM tool is relatively inexpensive and can maintain CM records transparently while supporting the CM process. If a CM tool is not employed, then a paper system can be established using forms such as the Configuration Item Status Record.

Configuration Management Progress Statements

This component represents a group of summary reports which are provided to project management for incorporation into the Progress Report or produced on demand. Base the reports on a time period covered by the progress being reported to management, and as a minimum cover:

- New configurations and configuration items created during the period
- Configuration changes implemented during the period
- Baselines created during the period
- Releases delivered outside of the project during the period
- Promotions made during the period by item type or configuration

Configuration Management Accounting Reports

This component consists of a family of reports or information which can be obtained as required from CM records. The following information is normally provided:

- relationships between configuration items
- promotion history of versions
- revision history of configuration items
- pair-wise comparisons of versions
- status and change history of configuration items
- history of configurations and their baselines
- history and content of all project releases

CONFIGURATION ITEM STATUS RECORD

(To support version numbering)

Item Reference:

Item name:

Item type:

Ref No: ____/____/ISR/____

	Version	Authorization	Version	Authorization
Reason for change				
State 1				
State 2				
State 3				
State 4				
State 5				
State 6				
	Version	Authorization	Version	Authorization
Reason for change				
State 1				
State 2				
State 3				
State 4				
State 5				
State 6				

Figure 5-12 Configuration Item Status Record

Audience, Distribution, and Usage

Configuration Status Accounting outputs are used:

- For maintaining accurate records of all CM transactions to support audits and management reviews
- To provide management and the client with an accurate representation of the state of deliverables produced to date on the project
- To provide input to the project Progress Report

Distribute Configuration Status Accounting outputs as follows:

- Progress Statements to the project manager for progress reporting
- Accounting Reports to team leaders and project management on demand
- Copies of records to project members, as needed to support project tasks

Quality Criteria

Use the following criterion to ensure the quality of these outputs:

- ☐ Ensure that accounting records of all CM process transactions are maintained for traceability.

CM.060 - Audit Key Deliverables

In this task you demonstrate to the project manager and to the client that the deliverables produced were those which were intended and that adequate control over development of those deliverables was exercised. The results of this task are usually used by the project manager in the Control and Reporting process to obtain approval by the client for a key project milestone. However, this task can coincide with any milestone or client release.

A physical configuration audit verifies that all intended key deliverables are baselined as configuration items. A functional configuration audit demonstrates that configuration items in the baseline can be traced backward in time to their requirements.

Deliverable

The deliverable for this task is an **Audited Baseline**.

Prerequisites

Required

You need the following input for this task:

☐ **CM Repository**

The CM Repository is the organizational structure which stores all items and any information about them which is necessary for performing the audits.

Task Steps

CM.060 Audit Key Deliverables

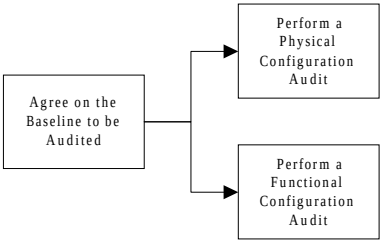


Figure 5-13 Task Flow for Audit Key Deliverables

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Agree with the client and the project manager on the baseline to be audited.	Agreed Baseline
2.	Perform a physical configuration audit of the baseline.	Physical Configuration Audit Report
3.	Perform a functional configuration audit of the baseline.	Functional Configuration Audit Report

Approach and Techniques

Agree on the Baseline to be Audited

- Identify key deliverables for the milestone being confirmed with the project manager and the client.
- Review audit procedures with the client and resolve any issues that are raised.

Perform a Physical Configuration Audit

- Verify that all deliverables are present in the baseline and are properly identified in the CM Repository as Configuration Items.
- Verify that the deliverables are physically complete and current.
- Document audit results using the Audit Report.

Perform a Functional Configuration Audit

- Verify that the project records demonstrate that the deliverables meet their requirements.
- Verify that changes to requirements have been adequately documented, and that configuration changes can be traced to approved requests for change.
- Document audit results using the Audit Report.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Secure agreement on the audit requirements and key deliverables. Produce status accounting reports needed for the audits. Perform audits	85
Project Manager	Review and approve audit results.	15

Deliverable Components

The deliverable components for the Audited Baseline are:

- ☐ **Physical Configuration Audit Report**
- ☐ **Functional Configuration Audit Report**

Component Descriptions

This section describes each component.

Physical Configuration Audit Report

This component verifies that all intended key deliverables are baselined as configuration items. The identity, completeness, and relationships of configuration item versions in the baseline is inspected and verified.

Functional Configuration Audit Report

This component documents the traceability of items in the baseline back to their requirements, accounting for all change control adjustments made during the period.

Audience, Distribution, and Usage

CM Audit Reports are used to demonstrate to project management product integrity for a key project milestone.

Distribute CM Audit Reports outputs to:

- project manager for review
- client during acceptance

Quality Criteria

Use the following criterion to ensure the quality of this deliverable:

- ☐ Use the Configuration Audit procedure to prepare these outputs.

CM.070 - Conclude Configuration Management

In this task you transfer the responsibility for Configuration Management to the client, and archive the project deliverables in accordance with consulting policy.

It is important to perform an orderly transition of Configuration Management products, methods, and tools produced and used during the project from consulting to the client. The amount of transfer required is project-specific and highly dependent on the client. The client might request all or part of the CM Repository or only a final release for transfer to their own CM environment. If the client does not plan to maintain the final deliverables of the project, then it is possible that nothing may be transferred.

Deliverable

The deliverable for this task is **Configuration Management Production Readiness**.

Prerequisites

Required

You need the following input for this task:

☐ **CM Strategies, Standards, and Procedures (CM.010)**

The CM Strategies, Standards, and Procedures are documents which define how to conduct the CM process.

☐ **CM Repository**

The CM Repository contains all items, information, and records about configurations which will be transferred to the client.

☐ **CM Environment (RM.070)**

The CM Environment is the set of software, data, and automated procedures which permits the client to manage the CM Repository and the Project Library.

Task Steps

CM.070 Conclude Configuration Management

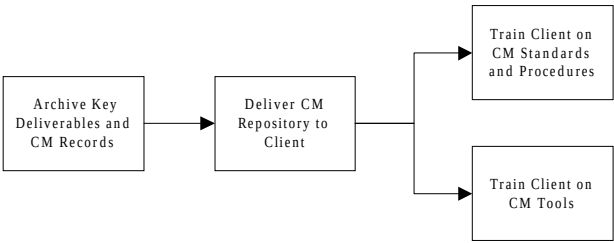


Figure 5-14 Task Flow for Conclude Configuration Management

The steps of this task are:

No.	Task Step	Deliverable Component
1.	Archive key deliverables and CM records according to consulting policy.	Archived CM Repository
2.	Prepare and deliver the CM Repository or the contents, to the client according to the agreement.	Transferred CM Repository
3.	Deliver the CM procedures and standards, and train the client in their use according to the agreement.	Transferred CM Procedures and Standards
4.	Deliver the CM tools, and train the client in their use according to the agreement.	Transferred CM Tools

Approach and Techniques

Archive Key Deliverables and CM Records

- Produce an internal release which contains all the key deliverables of the project.
- Extract or produce a complete archive record of the CM Repository and records.
- Place the Internal Release and CM Archive in the business unit archives according to consulting practice policy.

Deliver CM Repository to Client

- Determine the format and content of the CM Repository to be delivered.
- Execute the extraction of data or records, if necessary, from the CM Repository.
- Install, load, or administratively transfer the CM Repository contents to the client.
- Produce a final release and release note, if a record of items delivered to the client is deemed to be prudent.

Train Client on CM Standards and Procedures

- Obtain a list of the personnel who will comprise the client CM team and require training. Assess their knowledge and skill levels.
- Plan delivery and familiarization training.
- Organize a training session for the client personnel.
- Conduct a training session.

Train Client on CM Tools

- Assess the knowledge and skill level of the client personnel who will require training.
- Organize a training session for the client personnel.

- Conduct a training session.
- Ensure that the resource management process transfers any product licenses the client will need to be able to use the CM tools.

Roles and Responsibilities

The roles and their associated responsibilities for this task are:

Role	Responsibility	%
Project Support Specialist	Manage the transfer of the CM process to the client. Archive CM records.	80
Project Manager	Agree on transfer steps with the client.	20
Client Project Manager	Determine Configuration Management Production Readiness requirements and agree on transfer steps.	*

* Time not estimated

Deliverable Components

The deliverable components for Configuration Management Production Readiness are:

- ☐ **Archived CM Repository**
- ☐ **Transferred CM Repository**
- ☐ **Transferred CM Procedures and Standards**
- ☐ **Transferred CM Tools**

Component Descriptions

This section describes each component.

Archived CM Repository

This component contains all key project deliverables for consulting archives, if the consulting agreement permits. The archive is in a format and media as defined by the consulting practice

Transferred CM Repository

The CM Repository contains all items, information, and records of the CM process on the project. The transfer of this repository to the client is done as agreed upon with the client. The client may simply adopt the repository in total, or the client may request extracts of the final versions of the project deliverables in a specified format. Use a Release Note to record the transfer of the repository to the client.

Transferred CM Standards and Procedures

This component exists if the client wishes to adopt the project CM standards and procedures for use during production and maintenance of the project deliverables.

Transferred CM Tools

The client may adopt the project CM tools as agreed. This is normally done in order to preserve the contents of the CM Repository. If the client intends to actively use the tools to perform CM, then training may also be involved. In this case, complete training records.

A

Tasks & Deliverables

This appendix indexes the PJM tasks and deliverables. The sections of this appendix list tasks and deliverables in the following ways:

- alphabetically by deliverable name
- alphabetically by task name

Index by Deliverable Name

Deliverable Name	Task ID	Task Name	Project Management Life-Cycle Category
Adjusted Finance Plan, Financial Progress Statements	WM.050	Financial Control	Control
Adjusted Workplan, Work Progress Statements	WM.040	Workplan Control	Control
Quality Audit	QM.030	Quality Audit	Control
Audited Baseline	CM.060	Audit Key Deliverables	Completion
Client Acceptance	CR.080	Secure Client Acceptance	Completion
Client Progress Reports, Consulting Progress Reports, Action Items, Issues/Risks	CR.070	Status Monitoring and Reporting	Control
Configuration Items, Versions, Configurations, Promotions, Baselines, Configuration Changes	CM.030	Configuration Control	Control
Configuration Management Production Readiness	CM.070	Conclude Configuration Management	Completion
Configuration Management Records, Progress Statements	CM.050	Configuration Status Accounting	Control
Configuration Management Strategies, Standards, and Procedures	CM.010	Define Configuration Management Strategies, Standards, and Procedures	Planning
Control and Reporting Strategies, Standards, and Procedures	CR.020	Define Control and Reporting Strategies, Standards, and Procedures	Planning
Controlled Documents, Document Updates, Uncontrolled Documents	CM.020	Document Control	Control
Finance Plan	WM.030	Establish Finance Plan	Planning
Healthcheck Review	QM.045	Support Healthcheck	Planning
Intellectual Capital Releases, Reuse Assets	CM.035	Knowledge Management	Control
Metrics Report	QM.040	Quality Measurement	Control
Performance Appraisals, Adjusted Organization, Adjusted Assignment Terms of Reference	RM.060	Staff Control	Control
Physical Resource Plan	RM.040	Establish Physical Resource Plan	Planning
Prepared Infrastructure	RM.050	Establish Infrastructure	Planning
Prepared Organization	RM.030	Implement Organization	Planning
Project Orientation Guide	RM.025	Create Project Orientation Guide	Planning
Quality Management Strategies, Standards, and Procedures	QM.010	Define Quality Management Strategies, Standards, and Procedures	Planning
Project Management Plan [CR.010]	CR.030	Establish Management Plans	Planning
Quality Report	QM.050	Perform Quality Assessment	Completion
Released Physical Resources	RM.090	Release Physical Resources	Completion
Released Staff	RM.080	Release Staff	Completion
Releases, Release Notes	CM.040	Release Management	Control

Deliverable Name	Task ID	Task Name	Project Management Life-Cycle Category
Installation Reports, Incoming Item Records, Equipment Release Records, Fault Reports	RM.070	Physical Resource Control	Control
Resolved Issues, Risk Containment Measures, Change Requests	CR.040	Issue/Risk Management	Control
Resolved Problems, Issues/Risks, Change Requests	CR.050	Problem Management	Control
Resource Management Strategies, Standards, and Procedures	RM.010	Define Resource Management Strategies, Standards, and Procedures	Planning
Quality Review	QM.020	Quality Review	Control
Scope Changes, Approved Change Requests	CR.060	Change Control	Control
Scoping Project Management Plan	CR.010	Establish Scope, Objectives, and Approach	Planning
Staffing and Organization Plan	RM.020	Establish Staffing and Organization Plan	Planning
Work Management Strategies, Standards, and Procedures	WM.010	Define Work Management Strategies, Standards, and Procedures	Planning
Workplan	WM.020	Establish Workplan	Planning

Index by Task Name

Task Name	Task ID	Deliverable Name	Project Management Life-Cycle Category
Audit Key Deliverables	CM.060	Audited Baseline	Completion
Change Control	CR.060	Scope Changes, Approved Change Requests	Control
Conclude Configuration Management	CM.070	Configuration Management Production Readiness	Completion
Configuration Control	CM.030	Configuration Items, Versions, Configurations, Promotions, Baselines, Configuration Changes	Control
Configuration Status Accounting	CM.050	Configuration Management Records, Progress Statements	Control
Create Project Orientation Guide	RM.025	Project Orientation Guide	Planning
Define Configuration Management Strategies, Standards, and Procedures	CM.010	Configuration Management Strategies, Standards, and Procedures	Planning
Define Control and Reporting Strategies, Standards, and Procedures	CR.020	Control and Reporting Strategies, Standards, and Procedures	Planning
Define Quality Management Strategies, Standards, and Procedures	QM.010	Quality Management Strategies, Standards, and Procedures	Planning
Define Resource Management Strategies, Standards, and Procedures	RM.010	Resource Management Strategies, Standards, and Procedures	Planning
Define Work Management Strategies, Standards, and Procedures	WM.010	Work Management Strategies, Standards, and Procedures	Planning
Document Control	CM.020	Controlled Documents, Document Updates, Uncontrolled Documents	Control
Establish Finance Plan	WM.030	Finance Plan	Planning
Establish Infrastructure	RM.050	Prepared Infrastructure	Planning
Establish Management Plans	CR.030	Project Management Plan [CR.010]	Planning
Establish Physical Resource Plan	RM.040	Physical Resource Plan	Planning
Establish Scope, Objectives, and Approach	CR.010	Scoping Project Management Plan	Planning
Establish Staffing and Organization Plan	RM.020	Staffing and Organization Plan	Planning
Establish Workplan	WM.020	Workplan	Planning
Financial Control	WM.050	Adjusted Finance Plan, Financial Progress Statements	Control
Implement Organization	RM.030	Prepared Organization	Planning
Issue/Risk Management	CR.040	Resolved Issues, Risk Containment Measures, Change Requests	Control
Knowledge Management	CM.035	Intellectual Capital Releases, Reuse Assets	Control
Perform Quality Assessment	QM.050	Quality Report	Completion
Physical Resource Control	RM.070	Installation Reports, Incoming Item Records, Equipment Release Records, Fault Reports	Control

Task Name	Task ID	Deliverable Name	Project Management Life-Cycle Category
Problem Management	CR.050	Resolved Problems, Issues / Risks, Change Requests	Control
Quality Audit	QM.030	Quality Audit	Control
Quality Measurement	QM.040	Metrics Report	Control
Quality Review	QM.020	Quality Review	Control
Release Management	CM.040	Releases, Release Notes	Control
Release Physical Resources	RM.090	Released Physical Resources	Completion
Release Staff	RM.080	Released Staff	Completion
Secure Client Acceptance	CR.080	Client Acceptance	Completion
Staff Control	RM.060	Performance Appraisals, Adjusted Organization, Adjusted Assignment Terms of Reference	Control
Status Monitoring and Reporting	CR.070	Client Progress Reports, Consulting Progress Reports, Action Items, Issues / Risks	Control
Support Healthcheck	QM.045	Healthcheck Review	Planning
Workplan Control	WM.040	Adjusted Workplan, Work Progress Statements	Control

APPENDIX

B

PJM Roles

This appendix defines roles used in PJM.

Role Descriptions

Bid Manager

The bid manager is the role responsible for preparation of the bid, negotiation, and award. This role assists in the hand over of materials and information accumulated during the bid to the project manager at the start of the project.

Client Project Manager

The client project manager is responsible for the daily management of the client's contractual commitments to the project. This role must understand the client's business objectives for the project so that these can form the basis for resolving problems, conflicts of interest, and making compromises.

The client project manager obtains physical resources such as accommodation space, office equipment, computer equipment, materials and also arranges for client staff to make time available to the project. The client project manager introduces the consulting staff to other client staff. This role is responsible for monitoring the project's performance: progress against milestones, appropriateness of work, quality of work, and seeks to resolve any problems with work or relationships between development and business staff. The client project manager usually performs intermediate and phase-end acceptance.

Consulting Business Manager

The consulting business manager is the practice manager in the consulting organization who is responsible for the successful execution of the project. This role manages the practice which makes consulting staff available, and coordinates resources needed by the project with other practices. The consulting business manager represents the consulting organization in the contractual agreement with the client, and resolves business issues with the project sponsor. This role participates in project reviews with the client as well as internal practice reviews of project progress.

Project Manager

The project manager is ultimately held responsible for the success or failure of the project. This role must understand the project business objectives of both the client and consulting, and have a clear vision of how to achieve those objectives. The project manager agrees on the scope of the project with the client and resolves any conflicts among the various objectives of its stakeholders.

The project manager is responsible for planning the project, resourcing that plan, and monitoring and reporting the project's progress against the plan. This role obtains any physical resources required for the project, recruits staff, and, if necessary, dismisses staff. The project manager is responsible for ensuring that quality activities are performed in accordance with the Project Management Plan.

Internal responsibilities of the project manager role should be delegated to subordinate team leaders, as documented in the project organization plan.

Project Sponsor

The project sponsor holds the budget and pays for the project. A role at senior management level usually performs this role, and on large, cross-functional projects may be at the board level. This role must have clear views of the objectives of the project, particularly concerning delivery of the business benefits. The project sponsor is the ultimate arbiter on conflicting business requirements, if these cannot be resolved at lower management level, and on scope changes. The project sponsor expects the project to be delivered on time and within budget.

The project sponsor is responsible for ensuring other members of the management share his or her commitment to the project. This role may provide the resources, particularly staff time, required to make the project a success. The project sponsor usually performs the final approval.

Project Support Specialist

The project support specialist is responsible for assisting the project manager in the daily management of the project. On larger projects, this role may be performed by project office staff.

Specific duties are delegated by the project manager but will normally include some or all of the following:

- establish and maintain the Quality Plan, project standards, and project procedures
- coordinate and execute effective phase management
- establish and maintain the Workplan and Finance Plan
- procure staff or physical resources for the project
- monitor and perform analysis of risks, issues, and problems for trends requiring project manager corrective action
- perform coordination and communication functions within the project organization
- organize the Project Library, assign documents into the library, and maintain control of documents in the library
- orient new project members to the project environment, policies and procedures
- coordinate with administrators in client and subcontractor organizations
- prepare project progress reports
- record and distribute minutes, decisions and actions from management meetings
- generate routine status information from project records
- maintain information on project staff such as grade, qualifications, training, parent business unit or subcontractor, telephone and address, project assignment history, and other pertinent information
- develop, document, and implement Configuration Management plans and procedures
- establish project baselines and determine the content of project releases
- ensure that no unauthorized changes are made to a project baseline
- enforce Configuration Management procedures across all project processes

- establish the Configuration Management Repository and assist in the maintenance and protection against damage or loss
- ensure that the standards and procedures which have been defined for the project in the Project Management Plan are implemented
- ensure that quality reviews and quality audits are conducted as required

Quality Auditor

The quality auditor is responsible for conducting quality audits of the project to include a review of the Project Management Plan. This role should be filled by a role independent of the project staff in the consulting organization. The quality auditor needs training in the audit process. This role prepares for, conducts, and reports on the quality audit or audits undertaken, following up any actions raised.

Reviewer

The reviewer plans, conducts, and reports on the results of a quality review. The reviewer also assists in the resolution of review comments and monitors compliance actions by project staff after the review.